

NOT A POTATO

# THE UNKNOWN CORNER



ANDRIES J. GREYLING



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# Chapter 1 — The Arrival at Altitude

The bus door folded open and the air came in like a thing with opinions.

Naila felt it in her teeth first — that thin, scoured cold that doesn't bother to warm up before it gets to you. Twelve thousand feet will do that. The atmosphere up here has packed its bags and left, taking a third of the oxygen with it, and your body spends the next three days writing complaint letters nobody answers. She stepped down onto gravel, and her pulse did a little flutter of protest, and she ignored it the way she ignored most of what her body had to say when there was something to look at.

There was something to look at.

The plateau ran out flat in every direction, gold-brown grass cut low by wind that never quite stopped, and at the rim of it the Cordillera held up a line of snow against a sky so blue it looked overdone. Lake Titicaca was a hard blue blade off to the north, too bright to study. And in the middle distance, low and broken and stubborn, the stones.

"Okay," said Specs, behind her, dragging the rig case down the steps one bounce at a time. "Okay. That's — yeah. That's a lot of stone for a place with no trees and no road and no air."

"There's a road," said Rani, not looking up from the printout in their hands, the corner of it snapping in the wind. "Route four, paved nineteen-seventies, but the original transport question is — theandesite quarries are across the lake. Copacabana side. Roughly ninety kilometres if you go around. Less if you don't." A pause. "You can't

not go around. It's a lake."

"Unless you go on it," Specs said.

"That's the debate," Rani agreed, pleased, and went back to the page.

Naila let them. She was watching the stones.

From here you couldn't see the precision, only the wreckage of it — great red sandstone slabs tipped and scattered like something enormous had been dropped and shattered and the pieces left exactly where they fell, which, more or less, was the truth of it. A few stood. Most lay. Even broken, even ruined, the scale did something to your sense of proportion. Those long blocks were the size of shipping containers. The biggest one on the site, she knew from the file, ran past a hundred and thirty tonnes. You don't move a hundred and thirty tonnes by accident. You don't move it at all without wanting it very badly.

The Lens was already out of the bus and already working, the camera up, that particular stillness in the shoulders that meant the world had narrowed to a frame. Naila had stopped finding it strange. The camera was how the Lens stood in a place without being swallowed by it. A pane of glass between the eye and whatever the eye couldn't afford to feel directly. She let that be too.

"Right," Naila said. "Where's Jakobus."

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He was at the fence.

Of course he was at the fence — he was always already wherever the made thing was, the way water is always already at the lowest point in the room. By the time the crew had hauled their cases across the lot he hadn't moved, just stood with his weathered hands loose at his sides and his eyes going along a row of fallen blocks like a man reading a sentence in a language he half-knew. He didn't greet them. He tipped his chin at the stone in front of him.

"Look at the cut," he said. To no one. To the stone, maybe.

Naila looked.

It was an H-block — one of the famous ones, the shape that gets printed on every wild website about this place, the squared-off recess

like a letter H lying down. And the cut. God, the cut. The inside angles were *clean*. Not weathered-clean, not approximately-clean — clean the way a machinist means it, a right angle that stayed a right angle all the way into the corner where most stone-cutters give up and leave a fillet of rubble. The face was so flat it shamed the daylight, throwing back an even sheen with no waver in it.

“There it is,” said Specs, crouching, holding a hand near the surface like you’d hold it near a hot pan. “That’s the thing. That’s the thing everyone loses their minds about. You ever try to make granite do that? Andesite’s harder than granite. Mohs six, six and a half, near enough. You’d burn through steel.”

“They didn’t have steel,” Rani said. “No iron, no steel anywhere on the site. Stone hammers, bronze tools. Pecking and grinding and abrasive — sand and water and time. Protzen and Nair worked out the toolmarks. It’s all hand work.” They flipped a page. “It’s *achievable*. That’s the consensus. Achievable. Given skill. Given time.”

“Given a lot of time,” Naila said.

“Given a *civilisation*,” said Rani, and there was something almost fierce in it, a footnote with a fist. “Tiwanaku. Five hundred to nine-hundred of the common era, give or take. A real polity, a capital, a population, organised labour, generations of carvers who learned it from carvers who learned it from carvers. Ancestral to the Aymara who live right here. Right now. This is theirs.”

Specs straightened up, looked along the broken rows, at the snow, at the lake, at the immense scattered patience of the place. “Yeah,” they said, quieter. “No little grey men.”

“No little grey men,” Naila agreed.

It needed saying, plainly, every time, because the place had been so thoroughly colonised by little grey men in the popular imagination that you almost had to apologise to the stone for them. *Impossible precision. Machine-flat. Lost technology*. You heard it everywhere except in the actual ground, where the actual ground said, in its dry, patient, gritty way: *hand tools, friend. Sand and water and a thousand pairs of hands and a reason that mattered enough to spend them on*.

The reason being the part nobody had the manners to ask about.

Jakobus had moved. He was three blocks down now, hunkered at a stone Naila hadn't looked at yet, and there was a particular quality to his quiet that pulled her after him. She crossed the wind-flattened grass.

The stone had a groove in it. A channel, cut clean into the surface, with two squared lobes at either end — a long shallow keyhole, an *I* lying on its side, or maybe two *T*s kissing. She knew exactly what it was before he said anything. She'd been thinking about it for three weeks. She'd dreamed about it twice, which she'd told no one.

"They poured metal in here," Jakobus said. He laid one finger an inch above the channel, not touching, the way you don't touch what isn't yours. "Molten. Into the stone. To bind one block to the next." He looked up at her, and the question was already in his face before his mouth got to it, the only question he ever really asked. "What's it *for*, this?"

"To hold," Naila said. The metallurgist in her had stepped to the front and taken the wheel; it always did, around metal. "Cramp. U-shape or I-shape, you cut a matching channel across the seam of two stones, you pour the bronze in molten, it cools and locks them. Earthquake comes —" she made a small shudder with her hand — "the seam wants to open. The cramp says no."

"Bronze," Jakobus said.

"Arsenical bronze, here. They've assayed it. Copper with arsenic, not tin — gives you a good hard pour, low melt." She crouched, getting her eye down level with the channel, reading the way the lobes fed the long run. Whoever cut this had thought about *flow*. The channel widened where it needed to fill and the corners were eased just so. This wasn't a slot. This was a *gating system*. This was someone who understood what liquid metal does when it meets cold stone — that it wants to freeze before it's finished, that it grabs the walls and chills and goes solid in the wrong place and leaves you a cramp full of voids and cold-shuts, a casting that looks like a casting and holds like wet cardboard.

"You poured it *here*," she said, half to Jakobus, half to the channel. "On the stone. In situ. You didn't cast a bar in a workshop and carry it up. The fit's too good. You poured it on the seam so it took the exact shape of both blocks." She sat back on her heels. The wind



found the gap at her collar. “Which means you melted bronze on site. At this altitude. On a platform with no trees.”

“No trees,” Jakobus said, agreeably, and let the silence do the rest.

Because that was it, wasn’t it. That was the small cold draught under the door of the whole magnificent official story, and you only felt it if you’d ever stood in front of a crucible. Bronze pours around a thousand degrees. To melt it you need a furnace, and to run a furnace you need fuel, and to run it *hot enough at four kilometres up*, where the thin air starves your fire and steals your heat, you need a *lot* of fuel and forced air and someone who knows exactly what they’re doing — and then you need to do it *next to the stone*, repeatedly, consistently, channel after channel after channel, every pour filling clean, every cramp gripping, across a platform of hundreds of blocks. Not once as a stunt. As a *method*.

And the thing was — and Naila said this part out loud, because it was the honest part — “They obviously did it. The cramps are right there. It worked. It’s not a question of whether.”

“No,” said Jakobus.

“It’s a question of *how, exactly*. The whole run of it. Smelt to pour to fit.” She rubbed grit between her fingers. “And I don’t think anyone’s actually done it. Full scale. Up here. With what they had, start to finish, and published the workflow.” She looked at the channel again, at its unhurried competence. “Rani would know.”

“Nobody’s published it,” Rani said, from directly behind her, which was a thing Rani did. “I checked. People have studied the cramps — alloy, shape, the moulds. Lechtman on the metallurgy. There’s good work. But the full-scale, on-site, period-method *replication*, the whole chain at altitude —” they stopped, the way they stopped when the source ran out and they refused to walk past the end of it. “It isn’t there. The literature has a gap. The cramp itself works. The *recipe for making the cramp the way they made it* — that part’s open.”

For a moment nobody said anything. The wind went over the grass.

“Huh,” Specs said, finally, and there was a light coming on behind their eyes, the dangerous one, the one that meant a test rig was already being built in some back room of their skull. “So we could just. Do it.”

"It is never *just*," Naila said.

"We could *try* to just do it."

"That," said Jakobus, standing, his knees clicking, "is the only honest way to ask the question." He didn't smile, exactly. The lines at his eyes deepened. "What's it for. You can't read it from the chair. You have to put your hand to it and find out where it fights you." He looked at the channel one last time, the long competent keyhole of it. "Then it tells you what it knows."

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They'd been seen.

Naila had felt it for a few minutes without naming it — that prickle of being read while you're busy reading. When she turned, the woman was thirty metres off and in no hurry, coming along the fence line in a thick jacket and a wide hat against the glare, hands in her pockets, watching them with the unbothered patience of someone who has watched a great many people get very excited about her stones.

She stopped a comfortable distance away. She did not offer a hand. She let the wind fill the gap a moment, the way Jakobus did, and Naila revised her sense of who in this conversation held the most camera.

"You're the makers," the woman said. Not a question. Her Spanish-accented English was precise and unhurried, with something dry folded into it. "The ones who write the books with the question marks."

"Some of us write the question marks," Naila said. "I melt things."

A flicker, there and gone, that might have been amusement. "Dr. Lucía Quispe." She tipped her head a degree toward the broken rows behind her, claiming nothing, owning everything. "I look after this. Among other things." Her eyes went over them — over Specs and the rig case, over Rani and the snapping printout, over the Lens and the lifted camera, lingering there a beat longer, level and unhurried, until the Lens, to Naila's quiet astonishment, lowered it. Over Naila last. "And you've come to find out how it was done."

"We've come to *try*," Naila said. "Honestly. Full scale. The cramps. With what they had." She heard how it sounded even as she said it and made herself say the rest. "We'll probably make a mess of it."

Quispe looked at her for a long second.

“Yes,” she said, and it was not unkind, and it was not encouragement, and it landed somewhere clean and cold and true. “You probably will.” She let that sit. “People come and they think the stone is a lock and they have brought the key. They have very good keys. Lasers, drones, theories.” The dryness in her voice was a banked fire. “The stone is not a lock. The stone is at rest. It has nothing it is keeping from you.” A small movement of one shoulder, the plateau and the snow and the lake and the whole patient ruin of it gathered into the gesture. “But you may ask. You may *make*. We will see what answers you.”

She did not say *welcome*. She did not need to. She had not handed them a single thing, and Naila understood, standing in the thin loud air with her pulse still complaining, that whatever was going to be given here would be given by this woman, in this woman’s time, or not given at all — and that this was correct, and that she had better get used to it.

“Thank you,” Naila said.

Quispe almost smiled. Almost. “Don’t thank me yet. Thank the air.” She turned to go, then turned back, with the air of someone who had been holding a small unpleasant fact and decided to set it down where it could be seen. “You’re not the only ones who came to make a point about my stones this season.”

She nodded once, past them, toward the site gate.

Naila turned.

She hadn’t looked at the gate when they came in — she’d looked at the stones, she always looked at the stones — but she looked now. Strung along the chain-link, snapping smart and rectangular in the wind that bent everything else, hung a run of banners. Clean type. Good design. A handsome logo, a stylised sun, the kind of branding that cost real money and meant to be trusted.

*FUNDACIÓN VIRACocha*, the banners said. And under it, in a confident serif, the words that had brought the crew across an ocean and up onto the roof of the world:

*WHAT THEY DON’T WANT YOU TO KNOW.*

“Ah,” said Specs.

“They got here first,” Rani said.

The wind took the corner of one banner and cracked it like a sail, and from somewhere beyond the fence — past the broken rows, past the long clean grooves cut for a metal nobody had remelted at altitude since — came the small bright sound of a generator, already running.

## Chapter 2 — The Official Story, Dead Straight

Rani counted the H-blocks before she let herself look at anything else. There were thirteen in the main row, give or take the broken one whose shoulder had sheared off into the grass, and she counted them twice because the first count had felt like a guess and a guess was no use to anyone.

“You’re doing the thing,” Specs said, beside her, not looking up from the rig case they were dragging through the wet grass. “The counting thing.”

“There are thirteen.”

“There are thirteen,” Specs agreed, in the tone of someone agreeing with a weather report. “Fascinating. Hey, careful, this gate’s got a lip—”

The gate had a lip. The whole site had lips, edges, intentions underfoot. Rani had read every paper she could carry up the mountain in her head and on her tablet, and she had built a Puma Punku in her mind out of Protzen and Nair’s measured drawings, out of Vranich’s printed reconstructions, out of grainy plates from a hundred years of expeditions. The site she had built was clean and labelled and lay flat on a page.

The real one did not lie flat. It pitched and heaved where earthquakes and treasure-hunters and centuries had pulled it apart. The red sandstone was redder than the plates ever made it — a sunset colour, almost organic, almost meat — and the andesite was grey-blue and so dense it seemed to drink the thin light. The wind came off the

altiplano with nothing to break it for a hundred kilometres and went straight through Rani's three layers like they were a rumour.

"Breathe slower," Lucía Quispe said, without turning around. "All of you. You walk like the air is going to run out."

"Is it?" Specs asked.

"For you?" Lucía considered them over her shoulder. "Possibly." But the corner of her mouth had moved, and Rani filed that — *the dry one, the warm one underneath* — the way she filed everything, against later need.

Lucía wore a fieldworker's quilted coat and a lanyard with a laminated permit that the wind kept flipping over. She was unhurried in the way of someone who has given this tour four hundred times and still means it. She stopped at the edge of the platform and let the crew arrange themselves, and Rani saw her clock each of them in turn — Naila already crouched at a joint with her face six inches from the stone; Specs fidgeting with a strap; the Lens standing back, camera down at the hip, not raised, not yet. Lucía's eyes rested on the Lens a half-second longer than the others. Then she began.

"Tiwanaku," she said. "Not Inca. People say Inca because it is the only word they know. The Inca came here a thousand years too late, and even they thought the ruins were the work of giants, or gods, or the first ancestors made by Viracocha out of stone at the lake. So you understand — the *Inca* found this place mysterious." A small, pointed pause. "We have always been good at making things our descendants cannot explain."

Rani wrote that down word for word. It would matter.

"The main building here, the platform you are standing on — five hundred to nine hundred of the common era. That is the megalithic phase. There are organic dates in the fill that run earlier, fifth, sixth century, but the great stones, the cut faces, the joints you came to see — after five hundred." She let that sit. "Not twelve thousand years. I say this now so we do not have to circle it. The stratigraphy says one thing. The radiocarbon says one thing. The pottery in the ground says one thing. They all say the same thing, and it is a thing my great-grandmother's people did, on this ground, with their hands."

"The fringe say the dates are contaminated," Rani said. Not to argue.

To get it on the table, the way she'd been taught — *name the claim, label the claim, set it down.*

"They say a lot of things." Lucía's voice didn't change. "When all the evidence agrees and you don't like the answer, you can always say the evidence is the problem. It is a very flexible method. You can prove anything that way, which is how you know it proves nothing." She walked on. "Come. The blocks."

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The largest sandstone slab at Puma Punku weighs something like a hundred and thirty tonnes. Rani knew the number cold and it meant nothing until she stood beside the thing and the number came alive and sat on her chest. It was longer than a bus. It had been *shaped* — leveled, squared, its top surface dressed so flat that the morning damp lay on it in an unbroken sheet.

"Sandstone is the soft one," Lucía said. "Soft is relative. You could carve it. You could not lift it. The quarries are down by the lake, a few kilometres, and even a few kilometres with a hundred and thirty tonnes is a problem that has eaten careers."

"How'd they move it?" Specs asked, and there was the heckle under it, gentle, the *go on, sell me*.

"We don't know," Lucía said, flatly, and Rani's head came up because that was not the textbook answer, the textbook answer was *sledges and ramps and reed ropes and many people* — and Lucía saw her head come up and almost smiled again. "We have ideas. Sledges. Rollers. Reed mats over wet ground — the same reeds they still cut on the islands. Llama-hide ropes, many people, a long time. Every one of those is plausible and there is no surviving sledge, no rope still in the groove, no foreman's notebook. So I tell you the honest thing, which is: we have good guesses and no receipt. The andesite is worse."

She put her palm flat on a blue-grey block as she passed it, the way you'd touch a horse to let it know where you were.

"Andesite is volcanic. It is *hard* — it will turn a bronze edge, it will wear out stone after stone. And the nearest good source is across the lake. Ninety kilometres, some of it. They brought these here."

She did not raise her voice for it. She didn't need to. "Across water. Up onto the plateau. And then they cut it like butter."

The Lens had raised the camera. Rani heard the shutter, three soft frames, and saw Lucía's eyes flick to it — not annoyed, exactly. Measuring. The Lens lowered it again.

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The H-blocks were where the air got strange.

Rani had counted them from the gate. Up close, counting stopped helping. Each one was a single piece of stone cut into the shape of a capital H lying on its back — two flanges, a connecting bridge, every inner angle crisp, every recess to the same depth, the whole row a set of components that looked, to the modern eye, machined. *Looked*. Rani held the word carefully because the word did the work the fringe wanted it to do for free.

Naila had her steel rule out and a feeler gauge and her tongue between her teeth.

"Don't tell me," Specs said. "You can't get the blade in."

"I can't get the blade in," Naila said, with something close to grief. "Razor's nought-point-one-oh. It won't go. There's a seam I can see, light doesn't even come through it, and the thinnest steel I own won't enter." She sat back on her heels. "That's a tenth of a millimetre over — what, sixty centimetres of face? In andesite? With abrasives and patience?"

"With abrasives and patience," Lucía said.

"I believe you," Naila said, which from Naila was a whole speech. "I believe it's possible. I just want you to know that I, personally, a metallurgist, with a workshop and a century of materials science, would weep."

Lucía looked at her for a moment, and something in her face shifted from *tour* to *test*.

"Good," she said. "That is the correct response. The fringe people stand here and feel that same thing — *I could not do this* — and then they make a mistake. They say: therefore no human could. They put the weeping where the thinking should go." She nodded at the seam.



“The flat faces are not magic. Peck the surface with a harder stone, grind it with wet sand, polish it, check it against a flat template, do it again, ten thousand times, with people who have done nothing else since they were children. The precision is real. The method is known. What it costs — the lifetimes, the organisation, the obsession — that is the thing nobody wants to picture, because it is more frightening than a laser. A laser is over in a second. *This* took a society that decided this mattered more than nearly anything.”

The wind dropped, briefly, the way it does on the altiplano, and in the gap Rani heard a child laughing somewhere past the fence, and a market vendor’s call, and a dog. People lived here. People had always lived here. The site was not a tomb. It had a school down the road and a saint’s day coming and somebody had left coca leaves and a little dish of something at the base of the broken H-block, fresh, since the dew.

“And this,” Lucía said, “is the real marvel. The thing your foundation friends will not photograph, because it does not look like science fiction. It looks like a scratch.”

She crouched at a join between two blocks and drew them all down with a turn of her head.

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In the stone there was a channel. Two channels, really, mirrored across the seam — each a clean recess cut into the rock, joined into one shape that spanned the gap. An I, or a double-T, or what the Spanish workers a hundred years ago had called a *grapa*. A cramp.

“They cut this groove into both stones,” Lucía said. “Then they poured molten bronze into it. Across the join. So when it cooled and shrank and hardened, it locked the two blocks together — a metal staple, set into the rock itself. Arsenical bronze. The studies are clear on the alloy.” She traced the air above the empty channel without touching it. “Most of the cramps are gone — pulled out for the metal, centuries ago, melted down. The grooves stayed. A few cramps have been found still in place. The casting channels are everywhere if you know to look. Once you see them you cannot stop.”

“They poured molten metal,” Specs said slowly, “into a stone mould. Up here.” They gestured at the sky, the thin cold dazzling sky. “At —

what's the altitude, Rani—"

"Three thousand eight hundred metres," Rani said.

"At three thousand eight hundred metres. Where it's hard to keep a *match* lit. They smelted bronze, got it liquid, carried it, and poured it into a cold channel before it set, and got it to fill, and got it to fit, and did it again and again all over this platform until the building was *stapled together*." Specs had stopped heckling. "Okay. Okay, that's — I want to do that. Naila. Tell me we get to try that."

"We get to try that," Naila said, and there was a light in her now, the bad light, the one that had cost them money before.

Lucía watched the two of them and did not stop them, and Rani saw — filed it — that she let them want it. That the wanting was part of what she was giving.

"People ask me," Lucía said, standing, brushing her knees, "why it matters whether it was my ancestors or a lost super-race. *Either way it is amazing*, they say. *Why are you upset*." She let the wind take a strand of hair across her face. "I am not upset. I am precise. It matters because they did it. Because a real woman stood where you are standing and knew exactly how hot the metal had to be, and that knowledge died, or scattered, or went underground, and people now would rather hand it to spacemen than admit a brown woman on a cold mountain knew something they have not finished re-learning. That is what is at stake for me. Not a mystery. A name. The right name on the work."

Nobody said anything. The Lens, very quietly, had raised the camera, and this time Lucía let it. Held still for it. Looked down the lens like she was looking at whoever came after.

Rani wrote: *the right name on the work*. Underlined it once. That was the whole book, maybe, in five words, and Lucía had handed it to them in the first hour and Rani had the sense, cold and clear, that they had not earned it yet and might not.

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The handshake arrived before its owner did.

That was how Rani would remember it. A voice came across the platform — warm, carrying, American, pitched to fill a space —

*“Dr. Quispe! There she is, the keeper of the flame!”* — and then the hand, already extended, crossing forty metres of ruin with total confidence, attached to a tall clean-shaven man in a coat that had never been cheap and never been muddy, with two assistants behind him and a third walking backwards in front of him holding a cinema camera, framing the approach.

“Rourke,” Lucía said. Not a greeting. A label, the way she’d labelled the claims.

“Lucía. Lucía, *please*. Ethan.” He arrived and took her hand in both of his, sincere as sunrise, and the camera caught it, and Rani watched the man perform a friendship that the set of Lucía’s shoulders said did not exist. “And these must be the famous skeptics.” The eyes went over the crew, quick, professional, costing them out. They lingered on the rig case at Specs’s feet, and on Naila’s feeler gauge, and on the Lens’s lens, which had not come down. “Metallurgist, yes? Wonderful. *Wonderful*. You’re going to love what we’re about to show the world. We’re past skepticism now. We’re into discovery.” The smile widened, and it reached his eyes, which was the worst part, because he meant it. “You came to watch the experts explain it away. Stay for the part where they can’t.”

Behind him, his backwards-walking cameraman found the row of H-blocks, and the red light came on.



# Chapter 3 — The Impossible Stone

The H-block sat in the grass like a letter the planet had been trying to mail for fourteen hundred years.

Specs got down on hands and knees in front of it before anyone said go, because that was how Specs met things — by getting underneath. Red sandstone, cold enough through their gloves to ache. The face of it dropped away clean: two arms of stone reaching out, the deep recess between them, the inner angles meeting at corners so crisp you wanted to call them machined and then bite your own tongue for it. The shape repeated down the row. Same block, same notches, same depth, again and again, like a font.

“Okay,” Specs said. “Okay, okay. Tell me that’s not a template.”

“It’s a template,” Rani said, not looking at the block, looking at the middle distance where Rani always looked. “Protzen and Nair mapped the geometry. The H-blocks repeat to within a couple of millimetres across the whole array. Modular. They cut a master profile and they held to it.” A beat. “On purpose. Over and over. With patience nobody respects anymore.”

“That’s the part that gets me,” Specs said. “Not that it’s hard. That it’s *repeated*. One miracle, fine. A row of them?”

Naila said nothing. Naila had her cheek nearly against the stone, sighting down the long flat of the inner face the way you’d sight down a rifle or a poured floor, one eye shut, reading the plane for waves. Whatever she saw there had her very still.

The wind came off the altiplano with nothing in its way. Twelve and

a half thousand feet of nothing, the Cordillera white and indifferent on the horizon, the blue lip of Titicaca somewhere behind the haze. Specs had stopped twice on the walk in to breathe like a landed fish. The air was thin enough to make even kneeling feel like work, and that was *before* you tried to imagine pecking a stone flat in it with another stone, all day, for years.

“Razor blade,” Naila said.

Specs handed it over from the kit without asking which one. Naila set the blade against the joint where two finished faces would once have met — here just the single block’s edge and the grass — and tried, gently, to feed the steel into the surface seam of the recess. Not a real joint, just the cut between two planes the Tiwanaku had ground into one stone.

The blade skidded. Wouldn’t bite. The two surfaces met at an angle so close to perfect that a tool engineered to be thin had nowhere to go.

“That’s not fair,” Specs said, delighted, which was their tell. The more impossible a thing got, the wider they grinned. “That is genuinely not fair. Rani. What did they have. Floor it for me.”

“Stone hammerstones for the rough work. Harder andesite to peck the softer sandstone. Bronze tools — arsenical bronze, they’ve found the chisels’ chemistry — but bronze is *soft*, it can’t really cut sandstone, it scores and guides.” Rani’s hands moved, sketching the workflow in the cold air for an audience of one. “So the cutting is mostly abrasion. Pecking to the rough shape. Then grinding — sand, water, a flat reference stone, your own arm, ten thousand passes. Then finer sand. Then finer. Then a polish that fills the light back in. No iron. No steel. No power. Nobody disputes the tools.”

“Nobody *disputes* the tools,” Specs repeated. “And nobody’s done it.”

“Somebody did it,” said a voice behind them. “About thirteen hundred years ago. You’re standing in their workshop.”

Dr. Lucía Quispe had come up the slope without any of them clocking it, which, Specs thought, was a clean little professional humiliation for a crew that prided itself on watching the room. She had her hands in her coat pockets and an expression Specs was already learning to

read: not hostile, not warm, *amused*, the amusement of someone who had watched a great many clever people arrive on her ground with rulers.

"We weren't saying they couldn't," Specs said quickly.

"No. You were saying *you* couldn't." Her mouth did something dry. "That's the more honest version. Go on. You came to make a cramp, I hear. The foundation people made a speech instead. I'd rather watch you fail at a thing than succeed at a press release." She tipped her chin at the block. "Measure it. I'll wait."

So they measured it.

Specs got the rig out — Specs *lived* for the rig — a straightedge of milled aluminium with a dial indicator riding a slide, the whole assembly built in a Cochabamba garage two nights before and zeroed against a granite reference Naila trusted with her life. The idea was simple. Lay the straightedge across the inner face of the recess. Run the indicator. Read the hills and valleys in thousandths.

The needle barely moved.

"Flat to —" Specs leaned in, blew on the dial, read again. "I'm not gonna say it out loud, it'll sound like I'm lying."

"Say it," Lucía said.

"Across this span, on a *sandstone* surface that's been weathering in a sandblaster for a millennium and a half, I'm reading deviations I'd be embarrassed to find on a milled plate." Specs sat back on their heels. "Whatever it was when they finished it, it was *better* than this. We're measuring the ruin and the ruin's humbling me."

"Good," said Lucía. "Now show me yours."

That was the cruel part. That was the part Specs had been hoping she'd skip.

They'd brought a practice block. A slab of the same red sandstone, hauled up from a quarry the long way round with the right permits and the wrong amount of swearing, and on it, over four days, the crew had taken turns trying to peck and grind one honest flat. Period method. Stone and bronze and sand and water and time. Rani had the technique cold from the literature. Naila had the patience of

a person who reads the world as surfaces. Specs had built a little wetted grinding sled because of course Specs had.

The result sat under a tarp by the fence like something they were ashamed to have birthed.

Specs pulled the tarp.

It was a potato. There was no other word and Specs refused to reach for a kinder one. The face *looked* flat from across the site, the way a frozen lake looks flat. Up close it was a country. Low plains where the grinding had bitten deep, stubborn ridges where the grain had fought back, a whole disrespectful mountain range running diagonal because someone — Specs, hour nine, dizzy on the thin air — had leaned harder on one arm than the other and never recovered. They ran the indicator across it and the needle swung like a wiper.

“Forty hours,” Naila said. It was the first full sentence she’d offered the custodian. “Three of us. Modern abrasives we didn’t even admit we were tempted by. And it waves like the sea.”

Lucía crouched by the practice block. She didn’t gloat. That was almost worse; Specs could’ve handled gloating. She looked at the lumpy face the way a doctor looks at an X-ray you’ve been ignoring, and she put her palm flat to it, and she felt the topography with her hand, and she nodded slowly as if it had confirmed something she’d long believed about strangers.

“You went fast,” she said.

“We went *slow*,” Specs said. “We went so slow I aged.”

“You went fast in your *head*. You wanted the flat. You leaned toward it. You were trying to get *done*.” She stood, brushed grass from her knees. “The people who made that —” the H-block, the array, the whole impossible row — “weren’t trying to get done. Getting done is a thing you want when the stone is a problem. For them the stone wasn’t the problem.”

“What was it,” Specs asked, because Specs couldn’t not.

Lucía looked at them a moment, weighing, and Specs felt the door she was standing in front of and felt it stay shut, politely, firmly, *for now*.



“Keep failing,” she said. “You’re closer than the people with the cameras.” And she walked back down toward the town, unhurried, a woman with somewhere to be that was none of their business.

—

The fish, in Specs’s notebook, where Specs heckled himself in the margins because somebody had to, drew a little doodle of a needle swinging on a dial and wrote beside it: *cool cool so the OFFICIAL story is “any patient person could do this” and three trained adults with cheat materials produced GEOLOGY. love that for us.*

And under it, smaller, because Specs was honest even with the fish: *but the tools in the ground are exactly the tools the story needs. no lasers. no widgets. just bronze too soft to cut and sand and somebody who wouldn’t quit. the marvel isn’t the tool. the marvel is the not-quitting. how do you measure that.*

Here is the part nobody disputes, and it deserves to be said with a straight face and a bowed head:

Stone and abrasive can do this. Protzen ground andesite with andesite and dish soap and showed the world the surface coming up sweet and true, and the man was not lying, and the surfaces *can* be had. There is no hole in the *physics*. There is no widget missing from the workshop. Everything in the ground at Puma Punku is exactly what you’d expect a Tiwanaku master to be holding in the year 700 — soft bronze, hard hammerstone, sand, water, a reference plane, and the kind of week that has no end in it. The official story does not need a miracle. It needs *consistency*, repeated across a whole array, to a tolerance that humbled a dial indicator a thousand years downstream, by hands that had, materially, less than Specs had in a garage in Cochabamba.

That gap — between *can be done* and *was done, this well, this often, up here, where you can’t catch your breath* — is not a hole the fringe can fall into. Aliens don’t fix it; aliens are a way of refusing to look at it. The gap is just *the size of the achievement, standing up*. And Specs, kneeling by a potato of their own making at twelve and a half thousand feet, with a needle that wouldn’t lie, felt that size for the first time in their body instead of their head.

“We could do it,” Naila said, very quietly, packing the indicator into

its foam. “Given a year. Given a teacher. Given —” she stopped. “We could *probably* do it.”

“Then why can’t we?” Specs said. Not arguing. Asking.

Naila didn’t answer. Naila had stopped forcing answers somewhere on the walk up the slope, and Specs noticed, and said nothing about it, which between them was a whole conversation.

—

Rani heard the crowd before they saw it. Rani always heard things first; it was the looking-away, all that spare attention pooling in the ears.

“That’s not site noise,” Rani said.

It came from beyond the fence, near the visitor approach where the foundation had its trucks and its lighting and its lanyards — applause, the particular applause of people who’ve been told to clap. A drone went up over the ruins, illegal probably, and hung there like a fat insect taking the angle.

By the time the crew reached the rope, Dr. Ethan Rourke was mid-sentence, and he was *good*. Specs would give him that all day. Clean jaw, easy hands, a voice built for the back row, no notes. He stood beside a fresh-cut block of andesite mounted on a steel cradle, and in his fist he held something that gleamed, and behind him a screen looped slow-motion footage of the same gleaming thing eating into stone like the stone was butter.

“— diamond-tipped,” Rourke was saying, warm, generous, *reasonable*. “State of the art. The best the twenty-first century can put in a human hand. And friends — watch — even *this* leaves a tool signature. A burr. A rhythm. Marks that say *a machine was here*.” The screen showed micro-grooves, ugly and regular. “Now look at *their* faces.” Cut to the H-block, *their* H-block, the array Specs had measured an hour ago. Clean. Silent. No rhythm. No signature. No marks at all. “No burr. No rhythm. *Nothing*. I’m not here to tell you what did this.” His smile said otherwise, gently, with regret, with love. “I’m here to tell you what *didn’t*. A man with a rock didn’t do this. We just tried a man with a *diamond*, and we left fingerprints. They left none.” He let it land. “The honest scientist says one word, friends. *How*.”

The crowd loved it. The drone loved it. The screen loved it.

And Specs stood at the rope with a dial indicator going cold in their pack and a potato under a tarp behind them, and felt the trap close — because Rourke wasn't lying about the diamond, and Rourke wasn't lying about the burr, and the only thing wrong with what Rourke had just said was *everything*, and it would take the crew forty more cold hours and a teacher they hadn't earned yet to find out what.

Lucía was at the far end of the rope. She wasn't watching Rourke. She was watching the andesite block in his steel cradle — fresh-quarried, wrong, *taken* — and her face had gone very still and very tired, the stillness of someone doing arithmetic about what these people would try to carry away next, and from where.

She caught Specs's eye across the crowd.

She did not look impressed.

She looked like a woman deciding how much longer she could afford to merely watch.



# Chapter 4 — Chasing the Groove

The groove was the width of two fingers and it had been waiting for me for fourteen hundred years.

I lay flat on the cold stone, cheek to the andesite, sighting down a channel cut into the block's top face — a clean, T-shaped trough, the bar of the T running into the joint and the stem swallowed by the next block over. Where the two stones met, the channel continued without a stutter. Whoever carved it had cut both halves to meet across a seam you couldn't slip a razor into. The morning sun came in low across the plateau and pooled in the bottom of the cut, and down there, in the shadow of the lip, was a smear of green-grey that wasn't stone.

"That's it," Rani said, not looking at me, looking at the cut. "That's a cramp scar. The metal's mostly gone but the corrosion stained the channel. Arsenical bronze. Protzen and Nair logged dozens of these."

"It poured here," I said. "In the stone. They didn't drop a cast bar in. Look at the way it wicks into the corner."

"That's the consensus." Rani's hands fluttered near the rock without touching it. "Crucible up here, pour molten into the channel, it floods both blocks, cools, and now you've got a bronze staple locking two hundred tons of sandstone together so the whole thing rides out an earthquake instead of shrugging apart. They found cramps still in place at Tiwanaku. Found the channels everywhere. The mold is the building."

I put my finger near the green smear without touching it either. We don't touch. That's the one rule everybody on the crew keeps without being told.

*The mold is the building.* I liked that so much it hurt a little. You couldn't carve a bronze bar to fit a gap this irregular — every joint here a different millimetre. So you didn't carve the metal. You carved the *negative* into the stone, two halves at once across a seam, and let the liquid take the shape the gap demanded. The fit was guaranteed because the fit was the cause. That's not a workaround. That's somebody who understood casting in their hands a thousand years before anybody wrote it down.

"Naila." Specs was crouched twenty metres off at a fallen H-block, the iconic ones, the ones that show up on every documentary thumbnail with a laser drawn over them in red. Specs had their phone out, measuring an inside corner. "Come look at this and tell me it's not a router."

"It's not a router."

"I know it's not a router. I want you to *tell* me." They thumbed the screen. "Inside corner. Sharp. No radius. A spinning bit can't make a square inside corner, it leaves an arc, that's just geometry, so the alien-tool people love this one — *look, no radius, lost technology.*" They grinned up at me, the wind whipping their hair flat. "Except."

"Except a chisel doesn't spin," I said. "You can peck a square corner all day. The sharp corner is the *cheap* version. It's the rounded one that's hard."

"It's the rounded one that's hard," Specs agreed, delighted, and stood, knees popping. "Their best evidence is the easy bit. I love this place."

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Lucía found us like that — three people lying on or crouched over or arguing with the stones, the Lens fifty metres back with the camera up, framing nothing, just watching. She came across the wind-cut grass in a wax jacket with a clipboard tucked under one arm, and she stood over me until I rolled and squinted up at her against the sky.

"You found the green," she said. Not a question.

"The cramp scar." I got up, dusting my front. "In the T-channel. It poured in situ. We were—"

"You were excited." A flat statement, dryly amused, the corner of her mouth doing almost nothing. "I can always tell the ones who come for the metal. The aliens people, they want the corners." She nodded at Specs and the H-block. "The corners are a magic trick. The metal is the work." She let that sit. "Your colleague is right, by the way. About the router."

"He'll be unbearable now."

"They," said Specs, not turning around.

"They'll be unbearable now."

Lucía looked at the channel I'd been worshipping. Something moved behind her face and then chose not to come out. "What do you want to do here?" she asked. "Specifically. You did not fly to four thousand metres to lie on the rocks."

I told her. I told her straight, because she'd see through anything else in a second and I'm bad at the other thing anyway. We wanted to pour one. A cramp. Full scale. Up here, in the thin air, with bronze we smelted ourselves from ore that came out of ground close to this ground, arsenical like the real ones, into a channel we'd cut into stone with the tools they had. Not a demonstration. Not for a camera. Just to see if we could make the thing the official story says they made, the way the story says they made it.

Lucía was quiet long enough that I heard a market seller's call carry up from the town, thin on the wind. Somewhere down the line of blocks a child laughed; two of them were playing a chase game between the megaliths, ducking the fallen H-blocks like they were furniture, because to them they were.

"The Foundation," she said finally, "wanted to do that too. Rourke. Charming man. He brought a forge." She said *forge* the way you'd say *cologne*. "His version had a propane burner and a graphite crucible and a thermocouple and he poured a very nice cramp into a very nice channel he'd had cut with a diamond saw, and he filmed it, and in the film it works beautifully and then a man in a linen shirt says *imagine doing this without modern tools* and the music swells." She tapped her clipboard. "He proved that *he* could not do it without

modern tools. That is the only thing he proved. He filmed his own helplessness and called it the ancients’.

“That’s—” Rani started.

“That’s confirmation bias with a budget,” Lucía said. “It’s not malice. He believes it. That’s worse, in a way. A liar you can correct.” She looked at me. “Why should I let you do better? You’ll either fail, and the fail goes in someone’s video as proof the Aymara needed help, or you’ll succeed, and the success goes in someone’s video as proof anyone with a YouTube channel could do what my ancestors did. Both films already exist in someone’s edit. You’re just here to shoot the footage.”

The Lens lowered the camera. I hadn’t heard them come up. They didn’t say anything. They just set it down by their boot, in the grass, where she could see it was down.

She noticed. Of course she noticed. She noticed everything; she out-saw all four of us before breakfast.

“We fail in private, then,” I said. “If we fail. You hold the footage. You decide what leaves this site. Including whether we leave with anything at all.”

“That’s a generous offer to make with someone else’s hand,” she said, but the edge had gone off it a degree. She looked at the green smear in the channel, the old pour, the metal a culture had walked up here and turned into glue between mountains. “I’ll tell you the part the video people never want, because it isn’t a tool you can buy. The cramps that held — the ones still in the ground, the ones that rode out the centuries — they aren’t only where the engineering needed them. They’re on the lines. The ceremonial lines. The joins people came back to. Offerings still get left at some of them.” She met my eyes for exactly as long as she chose to and not a second more. “You can pour the metal. We’ll see if you can. But you’ll be making half the thing. You don’t have the other half. The other half isn’t a technique.”

Then she handed me a slip of paper with a name and a road and a fired-clay quarry on it, and walked off, and I understood I’d been given something and tested at the same time and that the test wasn’t over.

---



It took us three days to be ready to fail.

Specs built the furnace from a clay-lined pit and a goatskin bellows and a length of copper tube, swearing the whole time that period accuracy was a tax on his soul, then refusing to let any of us help because the rig was *his*. Rani spent two of those days with a craftsman named Don Hilarión whose family had worked metal down the valley for longer than anyone bothered to count, coming back each evening with a notebook full of things she wouldn't summarise because summaries lose the source. The Lens shot the work and not us. Jakobus came by once, at dusk, said almost nothing, crouched by a different block half a kilometre off and ran his thumb along a channel there and looked at the line it pointed down — toward the lake, I thought, but I wasn't sure and he didn't tell me. He nodded at the channel like it had answered a question, and went away again. He never poured anything. He just read the made thing and left.

The ore was the easy part and the hard part. Copper we had. The arsenic — that's what makes it *arsenical* bronze, harder than copper, flows lower, the whole reason it works — that came roasted out of an ore Don Hilarión pointed us to with a look that said *and you'll be careful with the smoke*, because the smoke from that work is exactly as friendly as it sounds. Up here you breathe shallow anyway. We worked the bellows in shifts because two minutes pumping at four thousand metres puts grey at the edge of your sight.

The stone block we cut over the same two days, pecking the T-channel into a slab of red sandstone with hammerstones, abrasive sand, and patience we didn't have. It was rough. It was ugly. It was nothing like the trough I'd worshipped, the one whose lip a craftsman had finished so the metal would wet clean into the corner. But it was a channel and the metal would tell us the rest.

The afternoon we poured, the wind had dropped, which felt like a gift and was probably a trap. Specs read the colour of the melt over the crucible's lip and called it. Rani held the notebook. The Lens stood back. Lucía came, said nothing, stood with her arms folded where she could see everything, the custodian of footage she had been promised the right to bury.

We carried the crucible to the channel on a yoke between two of us and tipped it.

The bronze went in beautiful for half a heartbeat — a bright snake of liquid sun running the channel exactly the way the books promise — and then the stone took its heat. The sandstone was cold, the air was cold, the thin cold air pulled the warmth out of that metal faster than any of my models said it should, and the snake *stopped*. It didn't reach the corner. It froze with its head three centimetres short of the turn, a lump where there should have been a fillet, and behind it the still-liquid metal backed up and found the low place in our rough channel — a hairline I hadn't seen, a flaw in the peck-work — and ran out under the lip in a thin bright tongue that dripped off the block edge and hissed into the grass.

The leak. And as it cooled it pulled, the way bronze pulls, shrinking against stone that wouldn't shrink with it, and the sandstone answered with a sound I felt in my teeth — a small dry *tick* — and a crack opened from the channel wall out toward the edge, fine as a hair, final as a verdict.

For a moment nobody breathed, which up here is its own statement.

I crouched by it. The cramp — *my* cramp — sat half-formed in the trough, blunt and short and leaking, a grey clot where the marvel was supposed to be. A potato. You spend your whole life learning how metal moves and you produce a potato and the truth of it is in your hands before it's in your head: I had not made what they made. I'd made a thing shaped like failing to.

"The stone stole the heat," I said. My voice came out level. "It chilled before it filled. And the channel wasn't tight enough to hold the flow." I looked at the hairline crack running off toward the edge. "And the shrink found the flaw."

"Yes," Lucía said.

That was all. Yes. No comfort in it and no cruelty either. She'd watched a foundation pour a perfect cramp with a propane burner and a diamond-cut groove and call it proof. She'd just watched us pour an honest one, the period way, in the real air, and split the rock and spill the metal into the grass.

I looked up at her. The question was right there, the one the whole official story rests on, the one nobody's published a clean answer to — how, then; how did they get it to fill and hold and *not do this*, a

thousand of these, fit just so, up here in this exact thin breath of air — and she could see me holding it.

She didn't take it from me. She didn't answer it. She tipped her head a fraction toward the cooling lump in the groove, the leak, the crack, all of it, and what she said was:

"Now you know what they knew when it *didn't* crack."

And she nodded at the Lens, who had not raised the camera at all, and turned for the road, leaving us alone with the question and the hardening proof that the easy story was not easy, and the cold coming down off the Cordillera like an answer that hadn't decided yet whether to arrive.



## Chapter 5 — The Potato, Cast in Bronze

The thing sat on the tarp like a dead fish nobody wanted to claim.

Specs crouched over it with both hands on their knees, head tilted, the way you look at roadkill you're pretty sure you ran over yourself. The cramp — their cramp, the third pour of the day, the one that was supposed to *work* — had come out of the mould a sort of lumpen H, one arm fat and the other starved, a cold seam running diagonal across the waist where the metal had stopped caring before it reached the corner. Bubbles pocked the surface. One end had a tail of flash that had crept up the side of the stone channel and frozen there mid-escape, like the bronze had tried to run for it.

"Okay," Specs said. "Okay. So that's not — that's not a cramp. That's a cramp's *understudy*. That's the cramp they bring on when the real cramp has food poisoning."

Nobody laughed. The wind did the laughing, coming off the plateau in long flat shoves that found every gap in every jacket and went straight for the bone. Behind them the H-blocks stood in their rows, andesite gone grey-gold in the late light, and the cold was the cold of three thousand eight hundred metres, the kind that doesn't bite, just leans on you, patient, until your lungs start filing complaints.

"Pour temperature?" Naila didn't look up. She was on her knees beside the stone block they'd been working — a sandstone offcut, not a real piece, Lucía had been very clear about that, a *practice* piece — running her thumbnail along the channel where their bronze had failed to take. There were two channels cut into the test stone. One they'd made. One was the comparison.

“Hot as I could get it,” Specs said. “Hotter than yesterday. The crucible’s fine, the bellows are fine, the charcoal’s—”

“The charcoal’s burning at altitude.” Rani, three metres off, hadn’t looked up from the tablet propped against a rock, knees drawn to chest, voice flat and quick. “Less oxygen, lower flame temperature, slower combustion. You’re fighting the air just to keep the metal liquid. Protzen and Nair don’t model the thermal loss but Lechtman’s metallurgy work on Andean arsenical bronze — she’s good on the alloy, the arsenic content, the cold-working — she doesn’t do the *pour*, nobody does the pour at scale, the pour at scale is—” Rani stopped. Started again. “It’s a gap.”

“It’s a *hole*,” Specs said, and managed half a grin, because it was their job to manage half a grin. “It’s a hole the size of, you know. A cramp.”

Naila put her thumb in the other channel. The original.

Specs watched her do it. They’d all done it, the first day — everyone who comes to Puma Punku does it, the razor-blade thing, the credit-card thing, you slide something into a joint between two stones that were lifted out of the ground thirteen hundred years ago and you find you can’t, the gap isn’t there, the faces meet like the stone was poured as one piece and somebody only *pretended* to cut it in two. But the cramp channels were a different marvel. The Tiwanaku didn’t just cut the stone flat. They cut a clean recess — an H, a double-T, a butterfly — into two adjacent blocks at once, so the channel ran across the seam, and then somebody walked up that hill with molten metal and *filled* it. Poured arsenical bronze into a stone mould the stone never let go of. When it cooled it locked the blocks together. A staple the size of your hand, set into rock so the earthquakes that flatten everything else on this plateau would shrug and roll and find the building still standing.

That was the official story. That was the whole official story, and it was beautiful, and Specs believed every word of it, and it was killing them by inches.

Because the originals fit.

Naila’s thumb moved along the ancient channel and didn’t catch. Not on a bubble, not on a cold seam, not on a tail of flash. Whoever had

poured that one had got the metal in clean and got it in *full*, and the bronze had taken the shape of the carved stone so exactly that there was no daylight between metal and rock, no shrinkage gap, nothing. They'd done it with a crucible smaller than the one Specs was using. They'd done it without a pyrometer or a thermocouple or a single line of Rani's downloaded metallurgy. They'd done it up here, in this air, where Specs' best fire wouldn't hold a temperature long enough to fill a hand-sized mould without a cold shut, and they'd done it again, and again, and again, all down the rows, hundreds of times, and the ones still in the ground still fit.

"How many tries do you think they got," Specs said. Lighter than they felt. "Like. Was there a Tiwanaku version of this exact afternoon? Some guy with a warped cramp going *no, no, the air, the air's the problem, I swear I'm good at this normally—*"

"Stop." Naila said it quietly, which was worse than loud.

Specs stopped.

She sat back on her heels. Her hands were grey to the wrist with charcoal and her face had the locked-down stillness it got when the process wasn't behaving and she was rerunning the whole chain in her head, smelt to pour to cool, looking for the fault, finding only faults, finding nothing but faults and no single one big enough to be *the* fault. Specs knew that look. Specs loved that look on a good day, when she chased a problem so far down she came back up holding something nobody else could've reached. This was not a good day. This was the look turned inward and starting to bite.

"It's not the air," she said. "It is the air, but it's not only the air. The mould's preheated wrong. The channel's cold, it's drawing heat out of the metal before the metal fills it, so the leading edge freezes and the rest piles up behind it." She turned the failed cramp over. The cold seam grinned back at her. "I know that. I *know* that's what's happening. Knowing it doesn't fix it. I'd need to preheat the stone — every stone, all of it, evenly, hot enough that the bronze keeps flowing — and I don't know how they did that without cracking the sandstone, and I don't know how *hot*, and there's no number anywhere because nobody's published the number because nobody's actually—"

"Done it," Rani said. "At scale. At altitude. With period materials. There's no paper. I keep looking. There's no paper."

The wind put its shoulder to all of them at once.

“There’s our hole,” Specs said, very softly, and this time they weren’t joking, and they hated that they weren’t.

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Lucía Quispe had been standing at the edge of the tarp for a while.

Specs didn’t know how long. That was a thing about her — she arrived the way the cold did, without announcement, and you noticed her the way you noticed you’d stopped being able to feel your feet. She had a thermos and two enamel mugs and she was wearing about four more correct layers than any of them, and she was looking at the dead-fish cramp with an expression Specs couldn’t read at all, which was its own information, because Lucía’s face was usually very easy to read and the reading was usually *dry amusement at the gringos*.

“That’s not a bad pour,” she said.

Naila’s head came up.

“For a first week,” Lucía went on, pouring tea, unhurried. “The Fundación’s man — Rourke — he did better than that on day one. You know how?”

“Modern crucible furnace,” Rani said. “Propane. Pyrometer. He had a—”

“He had a video crew,” Lucía said. “And a furnace you could read the temperature off, and a stone channel he cut with a diamond saw the night before, and he poured a perfect cramp on the second take, and the drone got it from above, and it’s very good footage.” She handed Naila a mug. Naila took it without seeming to notice. “He stood right where you’re kneeling and he poured a flawless cramp, and then he turned to the camera and he said *and the experts tell us a Stone Age people did this hundreds of times, with none of my equipment, and you decide whether you believe that.*”

The silence had a different shape now.

“He didn’t lie,” Lucía said. “That’s the thing about him you’ll want to understand before you spend a week hating him. He didn’t say one false word. He cut a channel, he poured a cramp, the metal was good. Everything he did, you can check.” She blew across her tea.



“He just did the easy part with the easy tools and called the hard part impossible. And he’s not even wrong that it’s hard. You’ve spent three days proving him right about how hard it is.”

Specs felt that land somewhere under the sternum. “That’s—” they started.

“That’s the trap, yes.” She looked at Specs directly then, which she didn’t often do, and there was no unkindness in it. “You can do it badly and prove it’s possible but stupid. He can do it well with the wrong tools and prove it’s beautiful but impossible. And in the middle is the actual thing, which is that somebody *here* did it well with the right tools, and we don’t have the recipe, and that’s not a mystery you solve by being clever on a tarp.” She nodded at the original channel. “You’re trying to *do* it. They didn’t do it once for a camera. They *knew* it. Generations of knowing. The number you want — the preheat temperature, the pour rate — that wasn’t a number to them. It was a feeling in the hands. It was *that’s hot enough, pour now*, said by a man whose grandfather said it to him while they poured a cramp into the temple of the people they both came from.”

She drank her tea.

“You can copy a thing without knowing it,” she said. “That’s what he’s doing. That’s what you’re trying not to do. The difference is the whole reason I let you onto this ground.”

Naila was looking at the failed cramp again, but her face had changed. The bite had gone out of the inward look. Something else was working now.

“You said the cramps fail more in some places than others,” she said slowly. “Yesterday. You said it like it meant something.”

Lucía’s mouth did something small. “Did I.”

“You said the ones on the east row are mostly intact and the ones along the south are mostly robbed or broken and I asked if that was looting and you said—” Naila stopped. “You said *not all of it*.”

“I say a lot of things.” Lucía screwed the cap back on her thermos. “Drink your tea before it’s ice. The cold here takes it faster than you’d think.” She stood, knees, then the long way up, the way you stand when you’ve stood up on this plateau your whole life and your body

has made its peace with the altitude in a way the crew's never would. "Tomorrow you preheat the stone. Try cracking it. Crack it three or four times and you'll start to feel the edge of how hot you can go, and that edge is most of your answer, and it's an answer you have to burn your own stone to learn. I'm not going to give you the number." A beat. "I don't have the number. We have something better and worse than a number. Goodnight."

She went. The cold closed behind her like water.

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Jakobus had not said anything all afternoon.

Specs registered this the way you register a draught stopping — by its absence. He'd been off among the rows since midday, the weathered fixer doing the weathered fixer thing, walking slow with his hands behind his back, crouching at a block, standing, walking on, a man reading a made thing the way other men read a face. He came back as the light went long and bronze across the platform, the snow on the Cordillera gone pink, and he didn't pick up the failed cramp and he didn't say anything about the pour. He stood at the edge of the tarp with his notebook open and he waited until Naila looked at him, and then he turned the notebook around.

It was a sketch of the platform. The rows of H-blocks, quick and sure, and along them a scatter of little marks — circles where the cramps survived, crosses where the channels stood empty.

"Lucía's east row," he said. His voice was low and it didn't push. "She's right. The good ones cluster." He laid a finger on the map, drew it down a line that ran through the densest knot of circles — through the cramps that fit, the cramps that hadn't failed, the cramps that thirteen hundred years and the earthquakes hadn't beaten — and the line he drew didn't follow the architecture. It cut across it. It ran from a doorway out toward the lake's far blue edge, dead straight, through the place where the offerings were, where Specs had seen the coca leaves that morning and the little bundle of wool nobody from the crew had left.

"The best work's on this line," Jakobus said. "Not the strongest part of the wall. Not where the engineering needs it most." He looked up, and he didn't look at Specs and he didn't look at Naila; he looked

at the empty space at the edge of the tarp where Lucía had been standing, as though she might still be there, as though the question were hers to take whenever she chose. He closed the notebook.

“So what’s the line for,” he said, to nobody, to all of them, to the wind. “What were they pouring *toward*.”

And he stepped back, the way he always did, and the question stood there on the cold tarp where the dead fish lay, and none of them touched it.



## Chapter 6 — The Adversary's Proof

Rani counted the cables first. That was the thing — before the faces, before the lanyards and the bottled water sweating in the cold, she counted cables, because cables told you who'd planned for what. Four coax runs snaking from the camera trucks. Two for sound. A satellite uplink that meant somebody, somewhere, was going to watch this live, and that somebody had paid for the privilege.

The Fundación Viracocha did not improvise.

They'd put up a marquee at the edge of the fenced zone, white canvas snapping in the plateau wind, the open side framing the ruined platform like a stage flat. Smart. Behind the podium, through the marquee's mouth, you could see the scatter of red sandstone, the H-blocks lying in their broken rows, and beyond them the Cordillera with its snow that never quite left. They'd composed the shot. Of course they had. Rourke's people knew that a stone, properly lit, did half your arguing for you.

"They angled the lectern thirteen degrees off the sightline," Rani said, not to anyone, the way she said most things. "So the camera gets the ruins over his left shoulder and the mountains over his right."

"That's diabolical," said Specs, delighted, chewing a thumbnail. "I respect it and I hate it."

Naila said nothing. She was watching the blocks, not the marquee — Rani could feel it beside her, that particular stillness, the metallurgist gone quiet over a thing that wouldn't fit. The Lens stood a half-step back with the camera lowered against their chest, lens cap on, which

was its own kind of statement at a press event.

Dr. Lucía Quispe stood with her arms folded and her chin level, in the second row of folding chairs, where the foundation had seated the local archaeology service. Not the front. Rani had noticed that too.

Ethan Rourke came out at altitude-pace — the deliberate, unhurried walk of a man who'd learned the hard way what happens when you bound up to a podium at thirteen thousand feet and then can't deliver your first sentence. Clean jaw. Good coat. He set a single sheet of paper on the lectern and didn't look at it.

"Thank you all for coming so far," he said, "to a place that has already come further than any of us."

Rani wrote that down. It was a good line. She hated that it was a good line.

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What Rourke did, over the next twenty minutes, was play it almost completely straight. That was the part Rani hadn't braced for.

He gave the official story its full dress. He named the Tiwanaku culture; he named the period, five hundred to nine hundred fifty of the common era, and he said *the common era* and not *ago* because he wanted the dates to feel real and dated and verifiable. He put a slide up: the H-blocks, photographed raking-low so the cut faces threw their shadows hard. He talked about andesite and red sandstone, about the hundred-and-thirty-tonne monsters, about the quarry ninety kilometres off. He even credited the abrasives — the patient pecking with harder hammerstones, the grinding with sand and water, the templates that let a carver repeat an angle across a dozen blocks.

"This," Rourke said, "is the consensus. And I want to say, clearly, on camera, that the consensus is *magnificent*. These were master engineers. I do not, will not, ever say a word against the Tiwanaku."

He let that sit. Rani saw Lucía's jaw move, a small grinding motion, like she was working a seed loose from a tooth.

"But," Rourke said, and the marquee leaned in.

He brought up the next slide. A flat face, lit so it gleamed. “You can’t slip a razor blade between these. We tried. We have it on tape.” A laugh from the press, a friendly one. “The angles are within fractions of a degree across blocks that weigh more than a bus. And the question I keep asking — the question I think we owe these builders — is *how*. Not whether they were brilliant. Of course they were. But brilliance with sand and stone, at this scale, this precision, this *consistency* —” he opened his hands “— is itself the mystery. And a mystery deserves a hypothesis.”

Rani’s pen stopped.

Here it came. She’d read his foundation’s white papers; she knew the shape of it. And to his credit — she would write this down too, hating it — he labelled it. He said the words.

“Now. There is a *fringe* claim.” He said *fringe* the way a surgeon says *benign* — calmly, to take the fear out of it. “Some people say machine tools. Lasers. Ancient power saws. Some say the stones were poured, like concrete. Some say twelve thousand years, older than every civilization we know.” He shook his head, almost fond. “I want to be honest with you. There is *no material evidence* for any of that. No geopolymer at Puma Punku. No steel, no iron. The stratigraphy doesn’t allow twelve thousand years. I’m not here to sell you aliens.” Another laugh, relieved this time.

“What I’m here to propose,” he said, “is something humbler and harder. *Lost technique*. Not lost magic — lost *method*. A way of working stone and metal so refined that we, with all our advantages, have not been able to put it back together. And I’d like to show you why I think we haven’t even tried hard enough.”

He stepped aside. Behind him, a screen lit with a video.

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Rani knew the demo before it played, because she’d been at the receiving end of it for ten days.

It was the cramps. Of course it was the cramps.

The screen showed the foundation’s people, three days earlier, in a tented workshop with a generator humming off-frame. They’d carved a channel into a sandstone offcut — the U-shape, the same profile

as the real grooves out on the platform, the ones where the Tiwanaku had once locked block to block with poured bronze. They had a crucible. They had, Rani noted, a propane burner the size of a small car, and an induction unit, and a forced-air setup that no Tiwanaku metallurgist had ever dreamed of.

And it *worked*, sort of. The bronze went in glowing. It seethed and steamed in the channel. The narration — Rourke's voice, warm — said: *Even with modern heat, modern alloy control, modern everything — watch the cooling.*

The cast came out. The camera came in close.

It was *ugly*.

Rani felt the crew register it together, a small collective intake. The cramp had pulled away from one wall as it cooled. There was a lip where the metal had skinned over too fast in the thin cold air and trapped a sag underneath. A hairline ran along the stone where the heat had argued with the rock and the rock had cracked.

"This is *our* result," Rourke's recorded voice said, gentle, devastating. "With every advantage. Now ask yourself what the Tiwanaku cramps look like."

The next slide answered. An in situ cramp, photographed in the groove where it had sat for over a thousand years. Arsenical bronze, the alloy confirmed by the metallurgists, poured into stone and *fitted* — filling the channel wall to wall, the metal married to the rock as if it had grown there.

"That," Rourke said, live now, soft, "is not a thing I know how to do. And I have asked. I have read every paper. There is no published, full-scale, high-altitude reconstruction of that workflow — the smelting, the transport of molten metal, the pour, the cooling, the *fit* — that produces what's lying in the ground out there." He turned and gestured at the platform, at the real stones beyond the canvas. "So I'm not telling you the answer. I'm telling you there's a hole. And I think it's the most honest thing in this whole debate."

*Damn him*, Rani thought, and wrote: *he found the real one.*

Because he had. That was the thing nobody in the marquee seemed to feel except, maybe, Lucía and the crew. Rourke had aired the non-



sense and labelled the nonsense — and then, underneath the showmanship, he'd put his finger on the one genuine, citable, unclosed hole in the whole edifice. The cramps. The thing Naila's bronze had skinned and sagged on three nights running. The thing Specs' rig had leaked through. The thing nobody had ever published a clean full-scale answer to.

He was right about the question.

It was what he wanted to do with it that sat wrong.

---

"He's going to walk it back to 'lost civilization,'" Rani murmured.

"How d'you know," Specs whispered.

"Because he said 'lost technique' four times and 'lost' is the load-bearing word, not 'technique.'"

And Rourke did. Smoothly, like a tide coming in over your feet so warm you don't feel the cold till you're standing in it. *Lost technique* became a *sophistication we underestimate* became a *civilization whose full capacities we have not credited* became, by the end, the title slide of the foundation's coming campaign, white letters on the gleaming flat of a stone:

**LOST TECH OR LOST CIVILIZATION? THE PROOF IN THE STONE.**

There was applause. Real applause, from people who'd flown a long way and wanted, very humanly, for the place to be more than it already, impossibly, was.

Rani looked at the slide and felt the specific ache of her job. The hole was real. The leap off it was not. And the distance between those two things — that narrow, vertiginous gap — was the whole book, the whole reason any of them were standing in this wind. *Here is a true thing we cannot explain. Therefore here is an untrue thing we have decided.* The *therefore* was where it all went wrong, and the *therefore* always sounded like generosity. Always sounded like respect for the builders.

She glanced at Lucía.

Lucía had not applauded. Lucía was looking at the platform.

The reception part — there was always a reception part — happened on the apron just outside the fence, with little cups of coca tea steaming themselves invisible. Rourke worked the room the way he'd worked the podium, and to Rani's surprise he came over to the crew, hand out, unguarded.

"You're the makers," he said to Naila. "I heard you've been trying the cramps yourselves. Out here." A note in it that was almost wistful. "How's it going?"

"Badly," said Naila, who did not lie.

Rourke laughed — a real one, the kind that costs something. "Yeah," he said. "Me too." For a second he looked younger and tired than his coat. "Doesn't it just — get under your skin? That we *can't*?"

"It gets under my skin," Naila said, "that you've decided what 'can't' means."

He took it well. That was the trouble with him; he took everything well. "Fair," he said. "Fair. But you'll concede the gap is there."

"The gap is there," Rani said, before Naila could give him more. "It's in Protzen and Nair. It's the unpublished workflow." She didn't look at him; she rarely looked at anyone. "It's a real hole. You're allowed to stand in it. You're not allowed to build a hotel."

For one beat Rourke's face did something complicated. Then he smiled, conceding nothing, and moved on to people who'd clap.

Specs exhaled. "Did you just—"

"He's not wrong about the cramps," Rani said. "That's what makes him dangerous. A liar's easy. He believes it."

---

It was almost over. The trucks were folding their cables — Rani counted them going down, the way she'd counted them up — when one of the foundation's field techs walked toward the fence gate with a clipboard and two carrying straps and a man in coveralls behind him, and that was when Lucía Quispe stopped being in the second row.

She was at the gate before the tech reached it. Rani didn't see her cross the distance; she was simply *there*, small and level and entirely unhurried, standing in the gap with the wind taking her hair.

"You're going where?" Lucía said.

The tech checked his clipboard. He had the foundation's authorization, or believed he did — a letter, a logo, a number. "Block fourteen-B," he said. "We're approved to remove the cramp segment for compositional analysis. It's—" he turned the clipboard, helpful "—non-destructive, we bring it back, it's all—"

"Block fourteen-B has a cramp still seated in the groove," Lucía said. Not a question. "One of three on the whole platform still seated where they poured it." She didn't raise her voice. She didn't fold her arms this time; she let them hang, which was somehow worse. "You want to lift it out for your cameras."

"For *analysis*," the tech said.

"You have the alloy already. It's published. Arsenical bronze." She said it flatly, the way you correct a child who's told you a fact you taught them. "You don't need the cramp. You need the *picture* of taking the cramp."

Behind her the platform lay in its long light, the H-blocks throwing shadow, the channels empty mouths in the stone except for the three that weren't. Rani found she'd stopped writing. The Lens, beside her, had — she heard it, the small mechanical assent — raised the camera, and then, after a breath, lowered it again. Not this. Not for this.

Rourke arrived at a walk, reading the situation, hands already spreading into reasonableness. "Doctor Quispe. I think there's been a—"

"There hasn't," Lucía said. She still hadn't looked away from the tech. "You can analyze anything that's loose. The loose ones are catalogued; I'll have them brought to you. The seated cramps stay seated." Now she did turn, to Rourke, and her voice didn't change at all, which was the thing Rani would remember. "You said you wouldn't say a word against the Tiwanaku."

"I won't. I haven't."

"Then leave the word they wrote where they wrote it," Lucía said. "It's still holding two stones together. That's what it's *for*."

She didn't move. The gate was a metre wide and she filled it the way the cramps filled their grooves — wall to wall, married to the place, not going anywhere.

Rourke looked at her for a long moment. Then he nodded, once, to the tech: *down*. The carrying straps came off the man's shoulder.

"Loose ones," Rourke said. "Tomorrow. Thank you, Doctor."

He walked away with his coat and his composure intact, and Rani understood that he would find another angle, because men like him always had another angle, and that this — this hour, this gate, this small unmoving woman — had cost him nothing he couldn't recoup.

Except.

Lucía stayed in the gate after they'd gone. The crew didn't approach. You didn't, yet. But she spoke, not turning, to the stones or to them or to the cold blue edge of the lake on the horizon.

"He thinks the hole means they were gods," she said. "Or that they had help." A short breath that might have been a laugh, somewhere very tired inside it. "It doesn't mean either."

Rani waited. Held the *maybe* like a coal.

"Come back at the early light," Lucía said, "and I'll show you what it means. If you can keep your cameras down for it."

She let go of the gate then, and the gate swung shut on its own weight, and the wind took the last of the coca steam off the apron and out across the platform, where three poured cramps lay in their grooves, holding.

# Chapter 7 — Night on the Plateau

The cold on the Tiwanaku plain doesn't arrive. It's just suddenly there, the moment the sun goes behind the Cordillera, like a door swinging shut on a warm room. The Lens felt it move through the camp the way he used to feel a change in a crowd before a thing went wrong — a shift in the body of people, a tightening.

He didn't lift the camera. He noted that he didn't, the way he noted everything now: a small ledger he kept against himself.

The fire was dung and scavenged eucalyptus, low and stingy with heat at this altitude — flames going pale and lazy because there wasn't enough oxygen to feed them properly. Even the fire was working harder up here. The Lens found that funny in a way he couldn't have explained to anyone who hadn't carried a stove above four thousand meters and watched water refuse to boil hot.

Across the fire, Jakobus had the lantern.

He'd built himself a little nest out of a folded blanket and a flat stone, and he was sketching. He always sketched at night, off in the margins, never in the photographs. The Lens had tried, once, to make a frame of him at it — the weathered hands, the bowed head, the lovely cliché of the old man and his notebook — and Jakobus had looked up, not unkindly, and waited until the Lens lowered the camera, and then gone back to work. That was all. He didn't perform for the lens, which was perhaps why the Lens kept wanting to photograph him.

What he was drawing tonight was the cramps.

Not the stones. The joins. He had a page of them — those flat little

I-shapes and U-shapes, the casting channels chiseled into the andesite where molten bronze had once been poured to lock block to block. The famous ones. The marvel, when you got down to it, when you stripped away all the noise about the H-blocks and the razor-blade joints. The bronze.

“You’re drawing them like a grammar,” the Lens said.

Jakobus didn’t look up. “They are one.”

“Meaning?”

“Meaning a cramp is a sentence.” He turned the notebook so the lantern caught it. Rows of the little metal staples, sketched in their grooves. “Two stones. The cramp says: *these go together, and they stay together*. Same shape, over and over. Like a word you keep. You see where they used it and where they didn’t, and the where tells you something.”

He set the pencil down and rubbed his hands, and that was him done talking, the Lens knew. Jakobus dealt out a thought and then stepped back from it, every time, as if the thought belonged to whoever picked it up.

—

Naila hadn’t moved in forty minutes.

She was on the far side of the fire with her headlamp on and her metallurgy notes spread across her knees, three printouts and a notebook of her own, and she had the specific stillness she got when the world had narrowed to a single problem and everything else — cold, food, the conversation, the Lens watching her — had simply stopped existing. The Lens had photographed a sniper once who held that stillness for two hours. It had unsettled him then. It didn’t, now. He’d come to understand it wasn’t absence. It was the opposite of absence. It was a person being entirely in one place.

“Arsenical bronze,” she said, to no one, to the page. “They knew. They chose it.”

“Chose what?” said Specs, who was lying flat on their back staring up, ostensibly resting, actually vibrating.

“The alloy. The cramps are arsenical bronze. Copper and arsenic,

not copper and tin — and arsenic lowers the melting point, gives you better flow.” Naila’s finger moved under a line of text. “So when you pour into a cold stone channel at thirty-eight hundred meters, where everything wants to cool too fast, the metal stays liquid a half-second longer. Fills the groove. Tight.”

“Okay.” Specs sat up. “So they were clever. We knew they were clever.”

“We poured tin bronze yesterday.”

“...Ah.”

“And it set up like a brick before it reached the corner.” Naila finally looked up, and the headlamp swung across all of them, blinding, until Specs swore and she clicked it off. In the firelight her face was tired and lit and furious in the way she only ever was at problems, never at people. “Our cramp’s a potato. It pulled away from the wall as it shrank. There’s a gap you could lose a coin in. *Theirs* don’t have the gap.”

“Maybe ours shrank and theirs didn’t because—”

“Because of the arsenic. Maybe. Or because they preheated the stone. Or because the mold geometry’s different than I think it is. Or because they poured hotter than I can get the crucible up here, where the fire won’t—” She gestured at the pale, struggling flame and didn’t finish.

This was the thing the Lens had stopped trying to photograph, because it didn’t photograph. The official story was *clean*. The official story was beautiful and true and not alien at all: the Tiwanaku, ancestors of the Aymara, on their own high ground, smelted bronze and poured it into stone they had cut and ground and fit, and the precision was human precision, hand and template and patience and a thousand years of accumulated genius. No lasers. No lost race. The marvel was the *people*. The Lens believed it. The crew believed it. It was the truest story.

And every single time they tried to *do* it, it came out a potato.

That gap was the thing. Not the gap in the official story — the gap in their cramp. The little measurable space between what the books said the Tiwanaku had done and what four educated, well-equipped,

twenty-first-century people with a generator and good intentions could actually produce on the same dirt in the same thin air. Nobody had ever published the clean version. Nobody had poured the cramps full-scale, at altitude, with period method, and gotten the fit. It was right there in the literature, the absence of it, a shape cut out of the record. Naila had found it and couldn't stop touching the edges.

"It's not that it can't be done," she said, quieter. "It obviously can. It's *there*. I just can't make it stop being magic by doing it."

"That," said the Lens, "is the nicest definition of magic I've heard."

—

Lucía came up out of the dark without a torch.

She did that. She knew the ground. The Lens heard her before he saw her — the particular crunch of the wind-cut grass, unhurried — and then she was just inside the firelight, a heavy shawl and a thermos and that expression she always wore around them, which was warm and amused and watching, the expression of a person who had seen a great many earnest visitors and outlived all of them.

"You're sulking," she told Naila, without heat. "I can see it from the road."

"I'm working."

"Same face, here." Lucía settled onto the flat stone next to Jakobus, who shifted to make room without being asked, the two of them arranging themselves with the easy economy of people who'd spent the day not getting in each other's way at the same site. She put a small cloth bundle on her knee and unfolded it. Coca leaves, dark green, dried. She picked through them, choosing, discarding a stem.

"You want?" She held the bundle out. "For the cold. For the breath. It's not a sacrament, it's just leaves. People treat it like I'm offering them opium."

Specs took some immediately, because Specs took everything immediately. Rani had appeared from the tent at the word *leaves* and was crouching at the edge of the light, not looking at anyone, asking in a fast low voice about the chemistry and the customs and whether



the practice was attested in the Tiwanaku-period record or only colonial accounts, and Lucía answered every question in order, precisely, the historian and the historian recognizing each other.

The Lens took a few leaves and tucked them in his cheek the way Lucía showed him and tasted green and bitter and a faint numbing that crept across his gums. The headache that lived behind his eyes up here loosened by a degree.

“Your friend,” Lucía said to him, nodding at Naila, “thinks she’s failing.”

“She is, a bit. The cramp leaked.”

“Mm.” Lucía chewed, looking into the fire. “And this is a problem because?”

“Because the people who built this didn’t leak.”

“No.” A small smile. “They didn’t.”

She let that sit. The Lens had learned that her silences were the most informative thing about her. She paid out information at her own rate, and the rate was a message: *this is mine, and I will decide what you get*. The Fundación people — Rourke’s lot, clean-shaven and well-lit, with their drone footage and their pre-written press release about a lost super-civilization — they hadn’t learned to read her silences. They talked into them. They thought a silence was an opening. The Lens had watched Rourke do it, that afternoon, by the H-blocks: pour his theory into Lucía’s quiet like she was a vessel waiting to be filled, charming and sincere and so certain, so *kindly* certain, that the stones were beyond her ancestors.

She’d let him finish. Then she’d said, *Doctor, you’ve come a long way to tell me my grandmother’s grandmothers couldn’t lift a rock*, and walked off, and Rourke had stood there genuinely puzzled, because in his story he was the one giving the gift.

—

“Tell me what you see,” Lucía said. To Jakobus, now. She’d reached over and turned his notebook toward her with one finger.

Jakobus considered it like it was the first time he’d looked at his own page. “The cramps aren’t even,” he said. “Across the site. Some

joins are clamped hard, metal in every channel. Some, the channels are cut and never filled. And the *filled* ones aren't where I'd put them, if I were just an engineer holding stones down. They cluster."

"Where do they cluster?"

He turned the page over. He'd made a rough plan — the platform, the rows, dots where the bronze sat. "Here. And here. Lines." He looked up at her, and there was a question in it he was too careful to ask outright.

Lucía took the notebook. Held it to the lantern. The Lens saw something move in her face — not surprise. Recognition, maybe. The expression of someone watching a stranger arrive, by their own road, at the edge of a country she'd been born in.

"You drew the offering lines," she said. "You didn't know it, but you did. Where the bronze is heaviest is where people still come. Where my mother's people leave the coca and the alcohol and the fat, on the days. Those joins—" she touched two of his dots, "—those aren't the joins that hold the most *weight*. If you only cared about earthquakes, you'd cramp the corners hardest. They cramped *these* hardest."

"Why," said Naila. Not a challenge. A genuine, hungry, metallurgist's *why*.

Lucía handed the notebook back to Jakobus. She didn't answer right away, and the Lens, watching, understood she was deciding — not whether she *knew*, but whether they'd earned it. The leaking cramp, maybe, had earned it. The fact that they'd tried and gotten a potato and not blamed the stone.

"You think the bronze holds the rocks together," she said at last. "That's the engineering. That's true, and it's good, and your Tiwanaku were better metallurgists than your foundation friend will ever admit, because he needs them to be too stupid to be human." A dry breath that wasn't quite a laugh. "But the bronze does another thing. You pour metal into two stones at once. The same metal. One pour, two stones — now they're one stone, joined where you can't unjoin them without breaking both. We have a word for what that is. It isn't an engineering word."

She didn't give them the word. The Lens noticed that, and noticed that she'd noticed him noticing, and that she'd done it on purpose.

The word was hers.

“The cramp is a promise,” she said instead, which was as far in as she’d take them. “You make it once, hot, and it’s binding, and you’d have to destroy something to take it back. They poured those where it mattered most that the joining hold. Not the rock. The—” she gestured, the whole plateau, the dark, the people who weren’t here and the people who were. “The rest of it.”

—

The Lens lifted the camera then. Slow. Lucía’s profile against the failing fire, the notebook, Jakobus’s hands. It was a beautiful frame and he wanted it and he held the shutter a half-second from down.

He lowered it.

Lucía had seen. She always saw. “You can have that one,” she said. “That one’s mine to give.” And he took it, and it was a different photograph than it would have been a second earlier — a gift, not a theft, which is a thing that does not show in the negative and shows in everything else.

—

The elder came at the worst hour of the cold, the dead hour before the false dawn, when even the wind had given up.

The Lens was awake because he was always awake then. He saw the small light first — not a torch, a candle stub cupped in a hand against the still air — and then the shape of an old man in a poncho, moving up the path toward the platform with the slow, certain gait of someone who had walked it ten thousand times and didn’t need the light, was carrying it for the stones, not for himself.

Lucía was already up. Of course she was. She met the old man at the edge of the camp and they spoke low in Aymara, and she didn’t introduce him, didn’t perform him for the crew, just touched the Lens’s shoulder once — *stay, watch, don’t follow* — and went with the old man toward the H-blocks.

The Lens watched the candle move into the field of stone. Saw it stop at one block. One particular join. He couldn’t have said which from here, but he knew, the way you know a thing in your chest before your head agrees, that it would be one of Jakobus’s dots. One of

the heavy-cramped ones. One of the joins where the bronze had been poured deepest, where the promise had been made hottest, a thousand years and more ago.

The old man knelt. The Lens saw him set something down against the stone — leaves, a small pour of something that caught the candlelight, dark. An offering. Left where the metal bound stone to stone, where it still bound, where it had outlived empires and earthquakes and was being fed, in the cold dark, by a man who belonged to it.

Beside the Lens, Naila had woken and risen without a sound and was standing with her hands open at her sides, her notes forgotten in the tent, watching the candle and the kneeling man and the stone that she could not make stop being magic.

She didn't reach for the camera. The Lens didn't either.

The candle stayed lit a long time, down among the joins, in the place the books had a hole in, and nobody on the cold plain said a word.

## Chapter 8 — In the Footsteps of Craftsmen

The forge in Tiwanaku town was a lean-to of corrugated tin behind a hardware shop, and it was hotter than anything the plateau had a right to produce. Don Florencio worked a foot-bellows older than the republic, and the charcoal in his clay hearth glowed a sullen orange that Naila distrusted on sight.

“Too cold,” she said, before she could stop herself.

The old man didn’t look up. He was Aymara, broad-handed, with a leather apron worn to the color of the andesite outside, and he pumped the bellows with the unhurried rhythm of a man who had been told he was wrong by foreigners before. “For you, too cold,” he agreed. “For me, hot enough. I am not casting your bronze. I am fixing a tractor part.”

Rani had folded herself into the one plastic chair, notebook on her knees, eyes on the middle distance the way they always were — taking in the room by not looking at it. “The melting point of arsenical bronze runs around nine hundred and fifty degrees Celsius,” she said, to nobody. “Lower than tin bronze. That’s part of why they used it. Maybe.”

“Maybe,” Florencio said, and smiled at the word like it was an old friend.

Naila crouched at the hearth’s mouth and felt the problem in her teeth. They were at thirty-eight hundred meters. The air that fed this fire was thin enough that a match struggled, that her own lungs had spent three days learning to be disappointed. Combustion is

a conversation between fuel and oxygen, and up here the oxygen mumbled. A European bloomery at sea level was one animal. This was another, breathing shallow, asking more of the bellows and giving back less.

She had read, on the flight, that altitude was a footnote in the literature. Standing in front of the fire, it did not feel like a footnote. It felt like the whole sentence.

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Outside, Specs had found a stonemason.

By the time Naila came out blinking into the cold light, they had already disassembled the man's confidence and were rebuilding it as a question. Specs did that. They were squatting beside a block of red sandstone the mason was squaring for a doorway lintel — modern job, modern building — and they had borrowed his hammerstone and were turning it over and over.

"So you don't cut it," Specs said. "You — what's the verb. You *bruise* it."

"You peck it," the mason said. He was younger, maybe forty, a cap pulled low, hands chalked with stone dust. "Small, small, small. You take the high part away. You don't cut. Cutting, the stone tells you no. It cracks where it wants. Pecking, you go where you want, slow."

He demonstrated. Tok. Tok. Tok. Each strike lifted a flake the size of a grain of rice, no more. A pale freckle appeared on the red face. He had been at this block, he said, for two days. It was not even half flat.

Naila watched him work and did the arithmetic she could not help doing, the arithmetic that ran in her like a second pulse. The H-blocks out at Puma Punku had faces measured in square meters. Flat to within fractions of a millimeter — flat enough, the famous claim went, that you couldn't slip a razor blade into the joint. She had slipped a razor blade at one, two days ago, in the wind, with Lucía watching her do it like a teacher watching a child confirm gravity. The blade had gone nowhere.

Tok. Tok. A grain of rice.

"How long," she said, "for one of the big faces. The really flat ones."

The mason laughed and didn't answer, which was an answer.

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Lucía found them at the second forge, the one a kilometer out toward the lake, where a cooperative of metalworkers cast belt buckles and tourist condors and, on commission, replica cramps for a German museum that had paid for the privilege three years running.

"You've been collecting craftsmen," she said. She had a thermos of something hot and three tin cups and she did not offer to share until she'd decided she would. "Like Rourke collects sponsors."

"That's not fair," Specs said, cheerfully. "We collect them and give them back."

"Rourke gives them back too." Lucía poured. "He gave Don Aurelio four hundred bolivianos last week to say, on camera, that no human could carve the H-blocks. Aurelio took the money. Then he told Rourke's man that he could carve the H-blocks, he just didn't have four years to do it, and they cut that part." She drank. "Aurelio thinks this is very funny. He is not wrong."

The fish in Naila's head, the one that heckled, surfaced. *Four hundred bolivianos for the wrong half of the sentence. Cheaper than honesty, which doesn't fit in a thirty-second cut.*

"You came to learn the methods," Lucía said. It wasn't quite a question. Her face did something dry and complicated. "Everyone comes to learn the methods. They want the recipe. They think if they have the recipe, they have the thing." She looked out at the lake, the impossible blue of it, the reed boats small as eyelashes. "The recipe is the smallest part. But fine. Watch Tomás cast. Then tell me you have it."

---

Tomás cast.

He was the youngest in the cooperative and the best, and he worked arsenical bronze the old way because the museum paid for the old way, which meant he had recovered something the chain of hands had nearly dropped. He'd learned it, he said, from a man in Oruro, who'd learned it from his father, who had — here Tomás shrugged,

the shrug that meant *the thread goes back into the dark and I won't pretend to follow it*.

The mold was the stone itself. That was the part the books under-sold and the part that broke Naila's heart a little, watching. You did not cast the cramp and then fit it. You carved the channel into the two blocks — the I-shape, the dovetail throat — and you poured the molten metal *into the stone*, so the bronze took the exact shape of the gap and froze there, gripping. The cramp and the joint were made in the same instant. The stone was the crucible's child and its parent at once.

"You only get one pour," Tomás said. "The metal touches the cold stone, it stops. If your stone is wet, it spits at you. If it's too cold, the bronze freezes before it fills the throat, and you have a cramp with a hole in its heart that you cannot see until the day an earthquake finds it for you." He fed the small furnace. "If it's too hot — the stone, I mean — you crack the channel. So. Not too cold. Not too hot. And you are at four thousand meters and the wind is from the lake."

He poured.

It went wrong, a little, the way real things do. The metal — too cool, Naila saw it, saw the surface skin over a heartbeat early — slumped into the throat and stopped a finger's width shy of the far wall. Tomás clicked his tongue, unbothered. "Today, no. Tomorrow." He chipped the failed pour loose, dropped it back in the crucible. Bronze forgives. Bronze you can melt again and apologize to.

Stone does not forgive. The cracked channel beside him, from some earlier bad day, sat there as a small permanent monument to a moment's wrong heat.

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They tried it themselves the next morning, out at the edge of the fenced zone, with Lucía's permission and Tomás supervising and Jakobus standing at the back of it all, hands in his pockets, watching the way the metal met the rock and saying nothing yet.

Naila had carved the channel herself, in a spare block of the right sandstone, two days of pecking that left her shoulders ruined and her respect for the dead enlarged. Specs had built the furnace — of course Specs had built the furnace — a clever forced-draft thing with



a bellows rig that fed more air against the altitude, which Tomás studied with frank covetous interest and which leaked at a seam Specs swore they'd fix and hadn't.

The bronze went into the crucible. The crucible went into the fire. The fire fought the thin air. They fed it and fed it. Rani read out alloy percentages from a study she'd found, arsenic content, the way the arsenic both lowered the melting point and hardened the casting and, in too high a dose, poisoned the man stirring it — which was perhaps, Rani noted, why so many smiths in so many cultures were depicted as lame, as wounded, as marked. The metal that bound the world took something from the hands that poured it.

"Pour," Tomás said.

Naila poured.

It was a potato.

She knew it before it had finished cooling. The metal hadn't run to the corners of the dovetail. There was a void — she could feel the wrongness of it, a casting that hadn't drunk its own shape. The surface had crawled and pitted where the air found it. When they chipped it free an hour later, the cramp was a sullen, holed, half-formed thing, fat where it should be lean, hollow where it should be full. It would hold nothing. It would have failed the first tremor, and out here the ground tremors like a sleeper.

She held it in her glove and turned it over and the failure was so legible, so honest, that for a second she could only laugh.

"Razor blade," Specs said, deadpan, and produced one, and tried to slip it under nothing, into the gap their cramp would have left, and it went in to the hilt.

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"The transport, even," Rani said that night, because Rani's mind, once it had a thread, walked the whole length of it. They were in the rented room above the market, the kerosene heater making its small inadequate roar, the cold coming through the window like a hand. "The andesite. The hardest of the blocks. The quarry's about ninety kilometers off, around the lake. Nobody's found the road. There are theories — sledges, log rollers, reed mats over wet clay, even ferrying

them across the water on balsa rafts. All plausible.” She turned a page she wasn’t really reading. “None proven. The literature says *plausible* a great deal. It’s a careful word.”

“It’s the word for *we have a story we like and no receipt for it*,” Specs said. They were lying on the floor, staring at the ceiling, fingers drumming. “Which — fine. I do that. I built a furnace tonight on a story I liked. But it leaked.”

“It leaked,” Naila agreed.

She was sitting with the failed cramp in her lap, still, the way you keep worrying a tooth. The official story was not in doubt. That was the thing she kept having to hold onto, the discipline of it. There were no laser tools in the ground. There was no concrete, no Atlantis, no machine that fell from the sky. There was bronze, the right bronze, the boring miraculous arsenical bronze that any honest assay confirmed. There were hammerstones and grinding stones and the marks of patient hands. The recipe was *known*. Every ingredient was exactly what fourteen-hundred-year-old highland Tiwanaku should have, no more, no stranger.

And she, with her education and her alloy charts and Specs’s forced draft and Tomás at her shoulder, had poured a potato.

The fish heckled, quietly, from the back of her skull. *Everything’s in the cupboard. You’ve got the recipe. So make the cake. What’s missing isn’t a thing in the ground. What’s missing is in the hands.*

Nobody had written down, anywhere she or Rani could find, the full workflow done end to end, full scale, at this altitude, in this wind, and shown to come out clean. The cramps in the ground had come out clean. That was the hole. Not a hole where a marvel should be — a hole where a *publication* should be. The dead had done the thing. The living had not yet shown their working.

It was a small hole, measured honestly. It was the size of a razor blade.

It would not close.

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In the morning Don Florencio’s forge was lit again, and Naila brought him the potato.

She didn't know why, exactly. To confess, maybe. She set it on his anvil, the holed and pitted lump of it, beside a tractor part he was straightening, and she stood there in the orange light and the charcoal stink and said, "We tried. To do what they did. The cramps." She gestured vaguely east, toward the fence, the stones, the ancestors. "It came out like this."

Florencio set down his hammer. He picked up the cramp in two thick fingers and turned it the way Naila had turned it, the way the masons turned the hammerstone, the way Jakobus turned everything — *what is it for, where did the hand fail*. He found the void. He felt the cold-shut skin. He understood the whole sad object in about three seconds, which was three seconds faster than she had.

Then he laughed. Not unkindly. A big slow laugh that came up from the apron and filled the tin shed and went out the door into the thin bright morning. He laughed until he had to wipe his eye with the back of a sooted wrist.

"You want to do what the ancestors did," he said, and set the potato down gently, like it was a thing deserving of gentleness, like every honest failure is. "*You*. In your week."

He turned back to the bellows, and pumped, and the fire woke.

"Good luck," he said.



## Chapter 9 — The One Hole: Bronze at Altitude

The laptop fan was losing its argument with the altitude. Specs had it propped on an overturned crate in the cold room behind Lucía's office, three cables daisy-chained to a battery pack, and every forty seconds the fan spun up to a whine like a wasp in a jar and then gave up, defeated, because the air at 3,800 metres is too thin to cool anything properly. Including engineers.

"Same as the bronze," Specs said, to nobody. "Can't shed heat up here. Can't keep it either. The air does whatever it wants and you're just a guest."

Rani didn't look up from the printouts. There were a lot of printouts. Rani had a system — papers fanned in an arc on the floor, sorted by year, then by whether the author had actually poured metal or only thought very hard about pouring metal — and the arc had a gap in it where one category should have been.

"Read me the list," Specs said.

"Which list."

"The good one. The 'somebody actually did it' list."

Rani's hand hovered over the arc, moved left, moved right, settled on nothing. "Protzen and Nair measured the channels. They reconstructed the tools — the hammerstones, the abraders, the templates. They cut sandstone with andesite cobbles and timed it. That's real work, that's the spine of the whole thing." A pause. "They didn't pour cramps at full scale."

“Vranich?”

“3D-printed the blocks. Reassembled the Pumapunku as a digital model, figured out the load paths, where the H-blocks lock.” Rani’s voice warmed the way it only did for a clean citation. “Heritage Science, 2018. Beautiful. It’s a model. Plastic. Nobody melted anything.”

“Lechtman? Somebody did the metallurgy—”

“Arsenical bronze. The alloy’s characterised — copper, arsenic, the cramps are real, the composition’s published, the I-shapes and the U-shapes, the ones still sitting in the grooves.” Rani finally looked at the gap in the arc. “Composition is not casting. Knowing what’s in it isn’t the same as knowing how they got it into a groove cut into a hundred-ton block, on this plateau, with the air this thin, and had it come out tight.”

Specs sat back. The crate creaked.

This was the part of the job Specs hated and loved in equal, useless measure: the moment you go looking for the paper that closes the case, the one that says *and here is exactly how, replicated, at scale, done*, and your hand keeps moving across the shelf and there’s a draught where it should be.

Because the official story is good. That’s the thing people get wrong about it. They think you have to choose between *aliens* and *boring*, and the boring version is supposed to be the one with no marvel in it. But the boring version is insane. The boring version is: a people on the roof of the world, around 600 to 900 of the common era, ancestral to the Aymara who still walk these stones at dawn with offerings — those people cut red sandstone and brutal andesite into faces so flat you cannot post a razor blade into the joint, locked them with interlocking notches and T-grooves and mortise-and-tenon like the stone was thinking about being furniture, and then, *as if that weren’t enough*, they carved channels across the seams and poured molten metal into them to staple the blocks together. On purpose. At a height where Specs’s own blood was thirty percent stubbornness.

No steel. No iron. Pecking stones, abraders, templates, water, sand, patience, and a furnace hot enough to run bronze, kept hot enough, somehow, in air that strips heat off everything.

“So nobody’s done it,” Specs said. Not a question.

“Nobody’s *published* it,” Rani said, which was the more honest and more annoying version. “Full-size cramp, in situ groove, period method, this altitude. Smelt, transport the melt, pour, cool, fit. The whole chain. I can’t find it. I’ve been looking for two days. I keep finding people who did one link and assumed the rest.”

“Assumed it works,” Specs said. “Because it obviously works. Because it’s *right there*. It’s literally in the ground.”

“Yes.”

“That’s the hole.” Specs said it lightly, the way you say a thing you’ve already decided is going to ruin your week. “Not ‘how could anyone possibly.’ We know who. We know roughly when. We know the alloy, the tools, the templates, the whole grammar of it. The hole is that the one move everybody treats as a footnote — *and then they poured the bronze* — nobody’s actually walked back up the mountain and done it the hard way, all the way through, and shown their work.”

Behind them the door let in a wedge of cold and the smell of woodsmoke and Lucía Quispe, who had a way of arriving in the middle of your sentence as though she’d been waiting for the worst possible moment to be polite.

“You found it, then,” she said.

---

She did not gloat. That was almost worse. She pulled the spare stool from the corner and sat with her hands around a tin cup, and she looked at Rani’s arc of papers on the floor with the gap in it, and she nodded once, slowly, like a teacher watching a student finally read the question on the board correctly after twenty minutes of confidently answering a different one.

“The Foundation,” she said, “did a demonstration last month. You heard.”

“Diamond blade,” Specs said. “Angle grinder. Cut a face flatter than anything out there in about an hour and stood next to it for the cameras.”

“And the cameras said: look, *impossible by hand*, it must be lost

machines.” Lucía blew across the cup. “They cut a flat stone. Anyone can cut a flat stone now. They did not cut a flat stone with a rock. They did not pour a cramp. They did not stand in this wind for three weeks keeping a fire at temperature while the air ate the heat. They proved that a modern tool is a modern tool.” A dry tilt of the mouth. “Dr. Rourke is a sincere man. He believes the marvel is too big for the builders. So he makes the demonstration that fits the belief. He is not lying. He is just very good at looking only where the answer he wants is keeping warm.”

“That’s confirmation bias,” Specs said, “with a film crew.”

“It is the most expensive thing in archaeology,” Lucía agreed.

Specs liked her. Specs liked her the way you like a person who will absolutely beat you and won’t pretend afterward that it was close.

“Here’s what I don’t get,” Specs said, and this was Specs’s actual flaw doing its actual job, the impatience, the inability to let a just-so story sit there being smooth. “The grooves. Out on the platform. I measured the cross-sections this morning before the wind came up.” A glance at the open notebook, the sketches, the swearing in the margins. “They’re consistent. Block to block, seam to seam, the channel profile holds. That’s not ‘we figured it out as we went.’ That’s a *spec*. Somebody had a standard cramp and a standard groove to take it. Which means they were casting a lot of these, the same, to a tolerance, and getting the metal to flow the full length and fill the corners and grip both stones — over and over — up here. And we, with thermocouples and propane and a degree, made one yesterday that came out like a—”

“Potato,” Rani supplied, without inflection.

“It came out like a potato,” Specs confirmed.

Because it had. That was the part the foundation never filmed. The crew’s first full attempt at a single cramp: Naila’s crucible bleeding heat into the thin air faster than the charcoal could put it back, the melt going sluggish at the lip; the pour starting bright and arriving at the far end of the groove already pasty, freezing before it kissed the second stone, so that the cramp grabbed one block and merely leaned against the other. They’d lifted the test pieces apart with two fingers and a bad word. The groove they’d cut had spalled at one



corner where the cold metal shocked it. Leaked. Loose. *Ugly*. A bronze cramp shaped like an apology.

And the ones out on the platform, thirteen-hundred years old, sat in their channels like they'd grown there.

"You want to know the alloy thing," Lucía said. It wasn't a guess.

"I want to know if the arsenic's the trick," Specs said. "Lower the melting point, get more working time, fight the cold. I want to know if they pre-heated the groove, ran a fire along the seam first so the stone didn't kill the pour. I want to know if they cast in place at all or carried hot metal in some kind of—" Specs stopped, hearing it, the sound of a person inventing a method out loud to fill a draught. "And the honest answer is I don't *know*, because the paper that tested it doesn't exist, because everybody who could've written it looked at the cramps and went *well obviously they poured them* and moved on to the calendar."

"Obviously," Lucía said, and there was a long, particular silence inside the word.

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She set the cup down.

"I am going to tell you something, and then I am going to take you to see something, and you are going to behave." She looked at Specs when she said *behave*, which was fair, and at Rani when she said *something*, which was also fair, because Rani had already opened a fresh page.

"The mystery is not a problem to me," Lucía said. "Understand that, before you spend three more weeks trying to be the heroes who solved it. People come here and they think the gap in the knowledge is an *embarrassment*. Something to be filled in, claimed, named after them. The Foundation thinks the gap is a doorway to their lost civilisation. You—" a glance, not unkind, at the arc of papers, at the wasp-whine laptop "—you think the gap is a *question*, which is better, I will give you that. It is much better."

She stood, and the cold came off her coat.

"But the gap is also mine. It belongs to this place and to the people who never left it. Every year somebody arrives certain the builders

were too simple to have done this — so it must be aliens, or it must be Atlanteans, or it must be a flood twelve thousand years ago that the carbon and the dirt and the broken pots all somehow forgot to record.” Her voice didn’t rise. It got *quieter*, which was its own kind of loud. “And every year I get to say: no. My ancestors did this. With rock and fire and arithmetic and forty generations of getting it slightly wrong and then less wrong. The fact that *you* cannot do it in a month, with your propane — that is not a wound to me. That is the proof. That is how I know how good they were. The thing you can’t replicate is the size of the thing they knew.”

Specs didn’t say anything. Specs, who always said something, sat on the creaking crate and let the wasp-whine fill the room and felt — for once — that the right move was to be quiet and let it land.

“So when I protect the not-knowing,” Lucía finished, “I am not protecting ignorance. I am protecting *them*. The hole in your literature is, up here, a kind of monument. Do not be in a hurry to pave it.”

Ranì, very softly, to the floor: “I’ll cite you.”

“You will not,” Lucía said. “I am not a source. I am a person. Cite Protzen.” But she almost smiled.

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They might have gone on like that — the warm room, the cold cup, the gap in the arc of papers held open between them like a window nobody wanted to shut — except the door banged a second time and it was the Lens, and the Lens had the look.

Specs knew the look. Everybody on the crew knew the look. The Lens went very still and held the camera *lower* than usual, down at the hip, not up at the eye, which was the tell that the Lens had seen something the camera shouldn’t be the first thing in the room to acknowledge.

“You should come out to the east platform,” the Lens said. Flat. Careful. “Now. Quietly.”

Lucía was already moving.

Out past the lip of the office the wind hit like a flat hand and the plateau opened — wind-cut grass going silver, the Cordillera holding its snow at the far edge of the world, the lake’s blue line, the great scattered red bones of the place. And on the east platform, where one

of the andesite blocks lay with a cramp still bright in its groove — one of the *good* ones, one of the tight ones, one of the ones that fit *just* so and nobody on Earth had published how — there were people in clean jackets. A small generator. A frame of yellow webbing slings being walked around the block by careful, professional hands. A flatbed at the edge of the access track with its tail down.

Rourke was there. Hard hat tucked under one arm, the wind in his good haircut, talking to a man with a clipboard and gesturing at the block the way you gesture at a thing that is already, in your head, on its way somewhere else.

Specs counted the slings. Counted the anchor points. Read the rig the way Naila reads a weld, the way you can't help reading a made thing once you know how things are made — and what the rig was *for* was lifting. The rig was for taking it off the mountain.

"He wants the alloy," Specs said, very low. "He wants the cramp in a lab. He wants to be the one who finally measures how they poured it." A breath that the thin air made too small. "He wants the hole. He wants to fill it with his name."

Lucía had already stopped walking. She had gone still in a different way than the Lens — not the camera-low stillness of a witness, but the planted stillness of someone deciding, in real time, what she was willing to lose.

She handed Specs the tin cup. She still had it. She'd carried it all the way out into the wind.

"Hold this," she said.

And she walked out onto the platform toward the men and the slings and the bright, tight, unsolved bronze, with the snow at her back and the whole long argument of her ancestors behind her like a second wind.



## Chapter 10 — A Race for the Cramp

The most intact cramp groove at Puma Punku is the length of a man's forearm, and on the morning it became a battlefield it held nothing but a finger of rainwater and the reflection of two foundations who both wanted it.

It sits in a red sandstone block on the eastern terrace, half-shadowed by a fallen lintel, and it is — Rani had said this three times the night before, in the voice she used for things she needed everyone to *re-tain* — the cleanest surviving casting channel on the platform. Not the most famous. The most *legible*. The walls of the I-shape are crisp enough that you can read the draft angle the founders left, the slight flare that let molten bronze find the bottom corners. Most of the cramps were prised out for scrap centuries ago, the channels left gaping like sockets after teeth. This one had kept its tooth long enough to leave a perfect mould of its own absence.

Naila got there at first light because she could not stop thinking about that flare.

She wasn't alone. Across the groove, kneeling with the unhurried confidence of a man who has booked the room, was Dr. Ethan Rourke of the Fundación Viracocha, photographing the channel with a phone in a rig worth more than the crew's truck.

"Beautiful, isn't it," he said, not looking up. "You can see the failure of nerve."

"The what."

"The retreat." He angled the phone. "Whoever poured this lost con-

fidence at the corners. You can see where the metal didn't reach. That's a hand-pour. Inconsistent. Human." He smiled at her, warm, sincere, the most dangerous kind. "We're going to scan it at fifty microns. Cross-section the residue if we can lift a flake. Then we'll know what they couldn't do."

"Lift a flake," Naila repeated.

"Non-destructive sampling. Negligible." He said *negligible* the way a man says it when the thing being negligible isn't his.

She looked at the groove and did the thing she does, the thing her brain does whether she asks it to or not — read the flow. Bronze poured into stone doesn't fill a channel the way water fills a glass. It fights. It skins over where it touches cold rock, it slumps, it traps gas in the undercuts. Whoever cut this groove had cut it knowing all of that. The flare wasn't a failure of nerve. It was someone who had poured before, and lost a casting to a sharp corner, and never lost one to a sharp corner again.

She didn't say that. She'd learned, lately, to hold a thing in her mouth before spending it.

"You think they couldn't do it," she said instead.

"I think nobody's *shown* they could," Rourke said, and there — there it was, the one true thing in him, sitting in the middle of all the wrong ones like a good cramp in a bad wall. "Find me the paper. Full scale. Arsenical bronze, poured into cut stone, at this altitude, with what they had. Show me the reproduction and I'll pack up and go home and write a very humble correction." He spread his hands. "I've looked. There isn't one. There's *the precision is within reach of hand tools* — and I love that sentence, I do, it's the most confident sentence in archaeology and it has never once been cashed."

*He's not wrong about the gap*, the fish in the back of Naila's skull observed, the heckler that never preached. *He's just packed his whole cathedral into it.*

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The crew's own attempt was forty metres away under a tarp, and it was a potato.

They'd cut a channel into a salvaged sandstone offcut — Specs grinding it with a hafted andesite cobble and river sand until his shoulders gave out, then handing off, then taking it back, two days for a groove the length of Naila's hand. That part had worked. That part had been almost holy: the grit hissing, the dust turning red, the shape coming up under their hands exactly as flat as patience made it. *The precision is within reach.* True. Slow, but true.

The bronze was the heartbreak.

Arsenical bronze wants to be poured hot and poured *fast*, and at twelve thousand eight hundred feet the air will not let you. Specs' bellows-driven furnace fought the thin atmosphere for every degree. The melt came up sluggish, the colour wrong. When Naila tipped the crucible the metal hit the cold stone channel and *skinned* — a grey hide forming on the surface even as the body still moved underneath — and where it should have flared into the corners it pulled back, shy, sulking, leaving the undercuts hollow.

They'd broken the block open after to look. The cramp inside was a sad thing. Porous. Bridged at the top, hollow at the heart, never touching the floor of the channel along a third of its length. Lift the two stones it was meant to join and they'd come apart like a cracker.

"It would hold nothing," Rani said quietly, looking at the cross-section. "It's a casting of our own failure of nerve."

"Don't," Specs said. "Rourke owns that line now. I refuse to give him royalties."

But they all stood there a moment, looking at the gap in their own bronze, the exact gap Rourke had pointed to in the ancient one and called *human* — except the ancient one had bridged it. The ancient one had reached the corners. The hollow in *their* cramp was the size of everything they didn't know.

This was the trouble with the easy story. You poured it and it came out hollow.

---

By midmorning the eastern terrace had two flags planted on it, in spirit if not in fabric.

Rourke wanted the intact groove cordoned for his fifty-micron scan

and his negligible flake. Specs wanted forty minutes alone with it and a silicone mould-kit to take a perfect negative — non-contact, he swore, paint it, peel it, learn the draft angle the old founders actually used, *then* maybe their next pour wouldn't come out a cracker.

Both of them wanted the same forearm of stone. Both of them, Naila thought, with a cold honesty that did not make her proud, wanted to *take* something from it back to their own story.

The LENS had been photographing none of this. They'd been sitting on the lintel with the camera in their lap, lens cap on, watching the groove the way you watch a thing you've decided not to shoot. It was the stillest Naila had seen them. When she sat down beside them the LENS said, without turning, "Everybody's framing it. Nobody's looking at it."

"You're not shooting."

"No." A long breath. "It's not mine to take either."

Jakobus was at the edge of the terrace, where he liked to be, off to one side with his hands behind his back, reading the wall. Not the groove — the *wall*, the long course of blocks the groove belonged to, his eyes tracking left to right and back like a man reading a sentence in a language he half-knew. He hadn't said a word all morning. He never raced for the front. The road, never the lead. When the moment came he would hand it to someone with more right to it than he had, and fade back, and that was the whole of him.

It was Lucía who came up the terrace and stopped it.

---

She came without hurrying, which was how you knew she'd seen it coming for an hour. Dr. Lucía Quispe, site custodian, walking her own ground in a wind that flattened the grass, two foundations turning toward her like flowers toward a sun they intended to harvest.

"Doctora," Rourke began, smooth, already reaching for the register of one professional to another. "I've filed for non-destructive access to the eastern channel. I have the permits in—"

"You have permits from La Paz," Lucía said. "This is not La Paz." She said it the way you'd note the weather. Then she looked at Specs,



who had the mould-kit half out of his bag and the decency to stop. "And you. You want to peel it."

"A negative," Specs said. "Paint and peel. It doesn't touch—"

"It touches it," she said. "Silicone in the channel. Pigment from the paint sitting in stone that's held offerings since before your country existed." Not unkind. Just level. "You want to learn the draft angle." She'd heard him say it; of course she had. "I know the draft angle."

That landed in the quiet differently than anything Rourke's instruments could.

"My grandfather's brother," she went on, "could tell you why the corners flare without a single micron of scanning. The men who cut that channel poured before. They lost castings to sharp corners. They never lost another." Her eyes went to Naila for half a second, and Naila felt the floor tilt, because that was *her* thought, the one she'd held in her mouth and not spent, and here it was, older than the holding, given back to her by someone who'd carried it for forty years.

The fish in her skull went very quiet.

"You think it's a problem to be solved," Lucía said, to all of them, to the wind. "A workflow you haven't reconstructed. And you're right — nobody has. Your paper doesn't exist, Doctor Rourke, and I would dearly love to read it, and I do not think you'll be the one to write it." She let that sit. "But I will tell you what's wrong with both of you. You both think the groove is *information*. A flake to cross-section. An angle to peel. You're both standing over the most legible cramp channel on this platform and the only thing neither of you has done is *leave it alone*."

Rourke opened his mouth. To his credit — and he had it, the man had real credit, it was what made him such a clean adversary — he didn't fill the silence with funding.

"You're risking your position," he said instead, and it wasn't a threat. It was almost gentle. "La Paz signed my access. If you deny it on-site you'll have a complaint by tonight. You know that."

"I know that," Lucía said.

She did know it. Naila could see it on her — the cost arriving in real time, the calculation of a woman who balanced preservation against

tourism against ancestors against a phone in La Paz that would ring whether she was right or not. She was standing on her own ground putting her own footing at risk, and she was doing it slowly, on purpose, so everyone present would have to watch her choose.

The wind came down off the Cordillera with the lake's cold in it.

"Here is the access I'm granting," Lucía said. "To the eastern channel. Today and until I say otherwise."

She crouched. From the bag over her shoulder she took a small twist of cloth, and from the cloth three coca leaves, and she laid them in the bottom of the groove, in the inch of rainwater, over the absent cramp. Smoothed them with two fingers. Said something low in Aymara that was not for any of them, and was not translated, and would not be.

Then she stood.

"None," she said. "No scan. No mould. No flake. The block is closed." She looked at Rourke, and there was no triumph in it, which was somehow worse for him than triumph. "Write your foundation. Tell them the custodian denied access to the eastern channel for reasons of ancestral significance. It's true. It will hold up, because it is the actual reason, and the actual reason is the strongest permit there is." She turned to Specs, and her face eased, a fraction, the dry warmth coming back into it. "And you. Your cracker of a cramp under the tarp — bring it to me tonight. Not to scan. To talk about. There's a difference, and I think you're starting to find it."

She walked off the way she'd come, unhurried, the leaves left in the water behind her.

For a long moment nobody on the terrace moved. Two foundations, both denied, standing over a forearm of red sandstone that had outlasted empires and would now outlast them both, holding its one tooth, keeping its draft angle, telling neither of them how the corners got filled.

Rourke was the first to laugh — short, real, the laugh of a man who has lost cleanly to someone he's just decided to respect. "She out-read both of us," he said, to no one, to Naila, packing his rig away. "Before we got here. She's known what that groove is for since she was a child." He shook his head. "I still want the paper."

“So do I,” Naila said. It was the truest thing either of them said all day.

The LENS hadn’t moved from the lintel. Lens cap still on.

Jakobus came back from the wall then, easy, his reading done, and he didn’t go to the groove. He stood a little apart, where he always stood, and he looked at the three leaves floating over the empty channel, and he asked it the way he only ever asked it — quietly, of the made thing, of all of them, of nobody:

“What’s it for, then. A groove with nothing in it.”

Nobody answered him. He didn’t want them to yet. He tipped his head a degree, like a man filing a question for later, and stepped back, and let the wind have the terrace, and the empty channel hold its inch of sky.



# Chapter 11 — The Ceremony of the Stones

The cold at four in the morning on the altiplano is not a temperature. It is a position you take up against the world, knees pulled in, breath rationed. The Lens had learned this in other high places, other dark hours before light, and his body remembered before his mind did: stay low, stay still, let the dark resolve itself.

He had the camera in his lap and he was not raising it.

That was the discipline he'd promised himself coming up the hill behind Lucía's torch. She had said one thing at the gate, dry as the wind: "You can come. You don't shoot until I say." And then, almost kind: "It will be hard for you. Your hands will want to."

She was right. His hands wanted to.

Below them the platform of Puma Punku was still only a darker dark, the great fallen blocks lumped against the lake's faint grey. Up here the sky had not yet decided to be sky. The Cordillera held a line of snow that caught nothing yet, and the air was so thin it had a taste, mineral, like sucking a coin. Naila was beside him with her hood up, and even hooded he could see her doing the thing she did, scanning, fitting, her eyes ticking from stone to stone the way other people's eyes tick across a sentence. Rani sat a little apart with a thermos he wasn't drinking, lips moving over something — a date, a citation, a fact held to like a warm stone. Specs, for once, was not building anything. Specs was just shivering, and watching, which for Specs was a kind of prayer.

There were maybe twenty people on the rise. Most of them were not

the crew. That was the point.

---

They had earned this much, the official story, and the Lens turned it over while the dark thinned, because turning over what was *known* steadied him the way breathing did not.

It was a human site. That was the whole of it and the marvel of it. The people the archaeologists called Tiwanaku — ancestors of the Aymara who were sitting on this cold hill right now, in fleece and bowler hats and woven layers — had quarried red sandstone and hauled volcanic andesite, some of it from across the lake, some of it from quarries that the careful papers put as much as ninety kilometres off. They had shaped it. They had cut the H-blocks so their interlocking shoulders matched their neighbours along a row, faces so flat you brought a razor blade and went home with the razor blade. No steel in the ground. No iron. Bronze and stone and abrasive sand and, the consensus said, time and templates and a genius for the repeating unit. Protzen and Nair had measured it block by block and found the precision real and the precision *reachable* — pecking, grinding, polishing, the slow tyranny of patience. Five hundred to nine hundred and fifty of the common era. People. Hands. No saucers.

And the metal. The metal was the thing that made the Lens love them.

They had cut channels into the stone shaped like a stretched H, or a fat I, or a butterfly with its wings folded, and into those channels they had poured molten bronze — arsenical bronze, the studies said, alloyed deliberately — so that when it cooled it locked two five-ton blocks together with a key of metal sunk into both. Cramps. Done in Egypt, done in Greece, done here, on a plateau nearly four kilometres into the sky where water boils lukewarm and a fire fights for every degree.

*That's* the marvel, the books said, and they were right, and the Lens believed them.

He just also had the memory of three days ago, very fresh, of Naila kneeling over their own poured channel with her face going still in a way he'd photographed on people standing in rubble.

---

Their cramp had been a potato.

He had it on a card somewhere, the whole sad sequence, though he hadn't shown her the frames. Specs's rig had finally held a charge of charcoal hot enough — barely — after they'd jacketed it and worked a bellows in shifts because the thin air starved the coals like it starved them. Naila had melted the bronze. She had cut the channel into a sandstone offcut with bronze chisels and a slurry of sand and water and her own knuckles, two days of it, and the channel was *good*, the Lens would give her that, the channel was true.

And the pour had gone wrong in the way she'd told them it might and hoped it wouldn't.

The metal hit the cold stone and the cold thin air and it didn't *run*. It grabbed. It skinned over while the heart of it was still trying to crawl into the corners, and where the corners were was where it didn't go, so the cramp came out with a foot missing, a bubble like a blister, and a hairline where the stone had taken the thermal shock and answered with a crack you could lose a fingernail in.

"It's a model," Naila had said, flat, to no one, holding the ugly thing. "It's not a join. A model of a join."

"We're at altitude," Specs had offered. "They had to deal with that too."

"They had to deal with it," Naila said, "and they *dealt with it*. Look at the in situ ones. Full feet. No blisters. The alloy's consistent block to block — that's the part nobody —" She'd stopped. Started again, smaller. "Nobody's published the whole chain. Smelt to transport to pour to fit, full scale, up here. I went looking. It isn't there."

The fish, somewhere in the Lens's head where the fish lived, surfaced and blew a bubble. *So the easy answer's got a hole in it the size of your missing foot. Cute.*

The fish did not get a vote on a cold hill at dawn. The Lens shut him in the thermos.

---

Light came not as a line but as a decision the lake made. The grey turned the colour of a struck match, and the snow on the Cordillera went the way snow goes, pink before it's white, and the great blocks of

Puma Punku stopped being lumps and became *cut* — every ground edge catching the low light and throwing a thin gold seam, so the whole field of fallen geometry looked, for ninety seconds, like a thing that had just been finished and laid down to set.

Lucía stood. She wore the same field jacket she'd worn all week, but over it now a woven cloth the colour of the snow-pink, and she did not look at the crew at all, which was correct, which was the whole architecture of the morning.

She spoke in Aymara, low, and the elders answered low, and a younger woman unfolded a textile on the flat top of a sandstone block and laid things on it. Coca leaves in threes. A small fire of something resinous that the wind tried to take and didn't. Sweets in bright paper, absurd and exactly right, the colours of a Tiwanaku market against four-thousand-year-old stone. Alcohol tipped to the four directions, a few drops to the ground first — to her, to the earth, before anyone's mouth.

The Lens's hands wanted the camera so badly his fingers ached. He kept them flat on the cold case.

He watched instead the way you watch when you've taken the picture away. He watched Lucía's feet. She moved along a line — not a path, there was no path — from the block with the offering toward a fallen H-block, then to another, then to a long andesite slab with its channels still cut into the upper face, dry now, empty of their bronze for a thousand years. She did not move at random. She moved the way Naila's eyes had moved.

And here is the thing he would not say aloud, because saying it would have made it his, and it was not his:

She walked the cramps.

Where she stopped to tip a few drops, where the woman set a knot of coca, where an old man pressed his palm flat to the stone and held it — those were the places where the joinery ran. The channels. The seams. Not the biggest blocks. Not the prettiest faces. The *joins*. The morning's reverence followed the metal that wasn't there anymore, the lines where stone had been bound to stone, as though the binding were the holy part and the block merely the thing bound.

Beside him Naila had gone very still in her particular way, the way



that meant a pattern had locked.

---

The sun cleared the lake's far edge and it was day, suddenly and without ceremony, and the ceremony folded itself up. The textile was gathered, the small fire let die. The old man stayed a moment with his hand on the stone. Children who had been silent stopped being silent and went, shrieking, to play tag among hundred-ton miracles, which is the most appropriate thing children have ever done.

Lucía came to them then. She accepted the thermos Rani held out, drank, handed it back. She looked tired and not at all sad about being tired.

"Now," she said to the Lens. "Now you can shoot, if you want. The stones don't mind the stones being photographed."

He didn't lift it. He found he didn't want to. He found the morning was already where he kept things, behind his eyes, and the camera would only make it smaller.

She watched him not lift it and something in her face eased, a half-degree.

"You walked the joins," Naila said. Not a question. Naila didn't ask questions she already had the answer to; it was one of the things that got her uninvited from dinner parties. "The offering line. You didn't go to the big blocks. You went where the cramps were."

Lucía considered her. "You count things."

"I can't not."

"Mm." Lucía looked out at the field, at the children, at the gold seams going ordinary in the flat light. "Your foundation friends" — she let the word sit, *friends*, with a tilt that wasn't quite contempt — "they want to know how. They have very expensive ways of asking how. Lasers. A scanner that costs more than this town earns in a year. They will take a block to a laboratory to ask it *how*, and it will tell them, because a block in a laboratory has nothing else to do." She turned back. "You broke your little cramp. The foot didn't pour."

"You heard about that."

“Everyone heard about that. The bronze didn’t run, and you didn’t pretend it ran.” A small, dry warmth. “That is more honest than the laser.”

“We can’t reproduce it,” Naila said, and it cost her, the Lens could hear it costing her, the metallurgist saying the thing aloud. “Not cleanly. Not the alloy that consistent, not the fit, not at this altitude. We’ve got better tools and worse results. I keep thinking we’re missing a step in the *making* —”

“You are,” said Lucía.

The crew went quiet the way a room goes quiet.

“But not the step you mean.” Lucía crouched, easily, by the dry channel on the andesite slab, and ran one finger along its cut shape — the stretched neck, the wide feet, the place the metal would have keyed. She did it the way you touch the seam of an old coat someone you loved had worn. “You think it is a problem of heat. Of flow. Of how to make the molten thing fill the cold thing.” She looked up at Naila. “When did you pour yours?”

Naila blinked. “Afternoon. When the rig finally —”

“Alone?”

“There were four of us. Specs on the bellows, Rani timing, the Lens —”

“Working,” Lucía said. “Hurrying. The light going. Wanting it to come out so you could write it down.” She stood, brushing andesite dust from her knee. “The people who poured this did not pour to write it down. The pour was the smallest part. By the time the bronze went in, the binding was already most done. The agreement to bind. Whose stone, whose channel, whose metal, brought from where, given by whom. Who stood here. What was said. The metal closes a thing that the people had already decided to close.” She gestured, light, at the whole gold-seamed field. “We don’t lay coca on the corner of a block. We lay it on the join. You noticed. Good. The join is where two things that were apart agree to be one thing and stay. The stone holds the stone. We hold each other. The cramp is —” she paused, and the pause was a choice, a gift she was deciding the size of — “a picture of that, that you can stand on. That outlasts the people who

poured it. Twelve, thirteen hundred years. Earthquakes. It is still holding.”

The fish, in the thermos, said nothing. Even the fish knew when to shut up.

“That’s not how-it-was-made,” Naila said slowly. “That’s what-it’s-for.”

“Those are the same question,” said Lucía. “You only think they are two questions because you were taught to ask one at a time.” She gathered her pink cloth closer against the wind. “Your friends will get their scan. They’ll learn the percentage of arsenic. They’ll publish *how*. And they will still not be able to do it, and they will say *therefore, advanced beings* —” the tilt again, almost fond, almost weary — “because to a hammer everything is a nail it cannot reach, and to that man everything is a secret only gods could keep.”

She started down the rise, then stopped, half-turned.

“You can come tomorrow too,” she said. “You don’t shoot until I say.”

---

She left them on the cold hill with the day going hard and bright and ordinary around the children’s shrieking.

Naila had crouched at the dry channel where Lucía had touched it. The Lens watched her — watched her finger find the shape, the neck, the feet, the place the foot of *their* cramp hadn’t poured. And he saw it happen, the thing he’d been waiting a week to maybe photograph and was glad now he hadn’t: the pattern in her unlocking and relocking into a different shape.

She’d been looking at it as a joint. A mechanical solution to the problem of two heavy things wanting to slide apart. Tab and slot. Key and keyway. Solve the flow, fill the feet, done.

Her finger stopped at the narrow waist of the channel, where the two feet drew toward each other and almost met.

“It’s not the gap it closes,” she said, to the stone, to the cold, to no one and all of them. “It’s the two things it’s closing it *between*.”

She didn’t say *I realised*. She didn’t say anything else at all. She just kept her finger in the dry groove a long while, in the place where

the metal had been, where for thirteen hundred years a thing held by being a picture of holding — and the Lens left his camera in his lap, and let the morning be a thing he kept, and let the question stay exactly as open as it was.

Up the hill, the foundation's white truck was coming through the gate.

## Chapter 12 — Blueprints in the Earth

Rani liked the cramps best at the hour when the sun came in low and sideways, because then the grooves filled with shadow and you could see them the way you'd see a sentence — a line of dark notches running across the joint between two blocks, each one cut to the same fat keyhole shape, each one waiting for the metal that had long since been pried out by people who needed bronze more than they needed mystery.

She had a clipboard. Specs called it a relic. Rani called it correct. On it she'd drawn the eastern platform in plan, the H-blocks in their rows, and over each joint where a cramp socket survived she'd inked a small black tally, the way you'd mark a concordance.

"You're counting them," Jakobus said. He'd come up on her left, downwind, which she appreciated, because she didn't have to manage the eye-contact thing and the smell-of-coca-and-diesel thing at the same time.

"There are sixty-one I can confirm," Rani said. "There are at least nine more where the socket's there but spalled. The Protzen and Nair survey put the total higher across the whole complex, but they were working the catalogue, not the standing joints. I'm only counting what I can put my finger in."

Jakobus crouched. He had a way of looking at a worked surface like he was about to be quizzed on it later — patient, a little wary, the look of a man who'd been lied to by objects before. "Put your finger in this one," he said.

She did. The socket was cut like a flattened figure-eight, two lobes joined by a narrow waist, deeper than a thumb, and the walls were *smooth*. Not polished-smooth. Cut-and-then-worried-smooth, the way river stone is smooth, every facet of the tool stroke ground down until your fingertip found nothing to catch on. They'd poured molten arsenical bronze into that — the alloy was published, the studies were real, the metal had cooled into the lobes and locked the two stones into one stone, and then somewhere in the long centuries between then and now a person with a fire and a need had dug it out.

"Sandstone, this side," Rani said. "Andesite, that side. They cramped *across* the materials. Soft to hard."

"Which moves how?"

"Differently. In an earthquake. In a frost." She'd read it three times to be sure she had it. "So you wouldn't expect a rigid metal bond to survive there. And mostly it didn't — that's why they're empty. But the *sockets* survived. The geometry of the join survived even when the metal failed."

Jakobus made a sound that meant *go on* and *not yet* at once.

---

Here is the official story, which Rani held in both hands like the good and serious object it was.

The Tiwanaku built this. Ancestral to the Aymara, on their own high ground, between roughly five hundred and nine hundred fifty of the common era — the megalithic phase late in that window, the radiocarbon honest, the stratigraphy stacked the way stratigraphy stacks when nobody's cheating. They quarried red sandstone and they hauled andesite from a long way off, and they cut both with stone and bronze, no steel, no iron, no power, and they fit the faces so flat that the cliché about the razor blade is a cliché because it keeps being true. And to hold the great courses together they carved these sockets and poured bronze into them, hot metal into cold stone, at thirteen thousand feet where water boils tepid and fire fights you for every degree.

That was the marvel. Not a beam from the sky. *That*.

Rani believed it the way she believed a well-sourced footnote: com-

pletely, and with the door left open behind her.

---

“Look at where they aren’t,” Jakobus said.

She had been looking at where they were. That was the difference between them — she read the text, he read the margins.

He walked her west, along the line of fallen blocks, and pointed with his chin, never his hand, at a joint as tight as any she’d marked. “Two-hundred-ton problem, that pair. Mass wants binding. Earthquake country. By the engineering you’d cramp it twice.”

“It isn’t cramped at all.”

“No socket. No groove. They trusted the fit.” He moved back east, to a joint between two smaller blocks, blocks a strong crew could have shifted with levers and prayer. “Here. Cramped. Three sockets in a span you’d think needed none.”

Rani stopped. Her pencil hovered. This was the wrong shape. If cramps were structure — and they *were* structure, the literature was clear, they were genuine engineering, no one sane disputed it — then the heavy joints should carry the most metal and the light joints the least. That was the blueprint any sober engineer would draw.

The earth had a different blueprint.

She went back through her tallies, fast, the way she went through anything when the pattern tipped, and the room of her attention closed around the clipboard and shut everyone else out. Specs would have called it gone-fishing. Rani called it *reading*. The heavy structural joints: lightly cramped, sometimes not at all. The light joints, the ones near the channels, the ones along the line that ran toward the sunken court — cramped, re cramped, lavished with metal.

“They spent the bronze where it wasn’t needed,” she said.

“They spent it where it *meant*.” Jakobus crouched again, by a T-shaped channel cut clean through a threshold stone, water-grooves, drainage, plumbing for a building that drained nothing now but wind. “Engineer pours metal to hold a load. These people poured it to hold a *line*.”

---

That night the cold did the thing it did, came down off the Cordillera like it had business, and Specs's third test rig leaked from a new and creative seam, and Naila's small crucible of bronze went pasty before it filled the practice socket they'd cut in a sandstone offcut, so that what they pulled out at dawn was the same lumpen ingot-shaped *potato* they'd been pulling out all week, a cramp that no Tiwanaku metallurgist would have signed.

"It's the heat," Naila said, not as an excuse, as a fact she resented. "At sea level this flows. Here the metal sheds heat into the air before it sheds it into the stone, and the stone's a heat sink anyway, and the socket walls chill the leading edge and you get this." She turned the ugly thing in her hand. "A cold shut. A seam where two flows of metal met and refused to marry. You can see it. They didn't get cold shuts. I've looked at the photos of the in situ pieces a hundred times. The pours are *clean*."

"So they preheated the stone," Specs said. "Charcoal in the socket, burn it down, pour into a warm channel."

"Maybe. There's no published full-scale test that closed it. That's the whole" — Naila stopped herself before *hole*, because they'd worn the word smooth — "that's the whole unfinished business. Nobody's poured a bronze cramp the size of these, into stone, at this altitude, with what they had, and shown the workflow that gives you what's in the ground. People assume it. People are *probably right*. But assume isn't the same as did."

The fish, who had ridden up the altiplano in a thermos and lived now in a clip-top jar by the camp light, regarded the potato through the glass.

*You'd think, the fish observed, a species that can land a robot on a comet could pour soup into a hole at altitude. Just an observation. Carry on.*

Nobody answered it. That was the rule.

---

Lucía Quispe found them mid-morning, at the threshold stone, with Rani's plan unrolled and weighted at the corners with cramp-potatoes because they were heavy and good for nothing else.



She didn't look at the crew first. She looked at the drawing. Then she looked at where Jakobus was standing, which was three steps back, hands in his pockets, having handed the spot to Rani the way he handed everything.

"You've drawn it as a load diagram," Lucía said.

"I drew it as a count," Rani said, to the clipboard. "It came out not matching the loads."

"No." Lucía crouched, knees popping, unhurried, a woman on her own ground who had crouched here ten thousand times. Her finger went to the lavished joint, the over-cramped one, the engineer's nonsense. "And you want me to tell you why."

Rani made herself say the true thing. "I want to know if it's mine to ask."

Something moved in Lucía's face. Not warmth, exactly. The lowering of a thing she kept up because outsiders mostly earned it. "That," she said, "is the first sensible question any of you has asked me." She let it sit. Then, because she was also dryly funny and tired of solemnity: "The other foundation never asks. They tell."

She put her hand flat on the threshold, the worn channel running away from her palm toward the sunken court, toward the place where, even now, there were three small things left at the stone's foot that the crew had been careful not to photograph: a twist of wool, a scatter of dark leaves, a splash gone to a stain. Offerings. This week's. Last week's was under it.

"You're looking for where the weight is," Lucía said. "Ask instead where the *water* goes. Where the sun comes in at solstice. Where the dead are carried and the living stand to watch. Where the channel runs and the line runs and the eye runs, all three, and *meet*." Her finger traced it, joint to joint, and Rani's tallies lit up under it like a constellation that had been waiting for someone to name the figure. The over-cramped joints fell on the meeting-line. Every one.

"They bound the stones hardest," Lucía said, "where the meanings crossed. Not where the mass was. A wall that only holds itself up is just a wall. Here —" she pressed the threshold, "— here they joined the water to the light to the path of the procession, and they made the joining *visible*, and then they poured metal into it so it would not

come apart.” She straightened, slow. “The cramp is structure. Yes. Your engineer is right. But your engineer is reading half the sentence and calling it the whole book.”

Rani wrote nothing. For once. The clipboard hung at her side and she let the woman talk, and the wind came across the grass, and a child somewhere past the fence was shouting at a dog in Aymara, and the marvel got *bigger* instead of smaller, which Rani had not known a marvel could do.

“This reading,” Lucía said, and now she looked at Rani directly, which Rani returned for exactly as long as she could, which was not long, and which Lucía accepted with a flick of the eyes away, kind, “this is mine. I’m giving it to you. I can also not give it. You understand the difference now, I think.”

“Yes,” Rani said. “Thank you.”

“Don’t thank me yet. You’ll get it wrong in the book. Everyone gets it a little wrong.” A thin smile. “Get it less wrong than the foundation.”

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The foundation got it wrong that same afternoon, in a way that had nothing to do with the book.

Specs heard it first — Specs, whose attention scattered across everything was a kind of early-warning radar the rest of them lacked. A motor. Not a generator note, not a vehicle on the access road. Higher. Whining under load.

“That’s a core drill,” Specs said, already moving. “That’s a *diamond* core drill. Somebody’s coring.”

They came around the western platform together, Rani with her clipboard clutched stupidly to her chest, and there in the lee of the great fallen andesite block — the cramped one, the threshold’s cousin, the one on the meeting-line — were two of Rourke’s people in clean field gear, and a tripod-mounted rig clamped to the stone, and a wet line of slurry running pale down the red-grey face where the bit had already bitten. Going for a cramp socket. Going for the metal residue still ghosting the walls of it, the alloy signature, the thing you could carry away in a sample bag and put in a press release.

Not Rourke himself. Rourke was twenty metres off with a tablet, and

when he saw them his face did the thing his face did — the open, sincere, *let-me-explain* arrangement of a man who genuinely believed he was the hero of a documentary.

But it was Lucía who came over the rise behind them, and it was Lucía the drill stopped for, because she said one word, flat and carrying, and the technician's hand left the trigger like the thing had gone live.

"That block," Lucía said, walking, not hurrying, her voice the level voice of someone who has been here before and lost, "is on the register. You don't have the permit for invasive sampling. You don't have *any* permit I signed."

"It's a non-destructive—" Rourke began.

"It's a hole," Lucía said. "It's a hole in a thing that's held together for fourteen hundred years, and you're making it to prove a story the stone never told you." She stopped at the rig. Put her hand on the wet slurry-line, brought her fingers away pale. Looked at the colour of it on her skin the way she'd looked at the offering-stain that morning, with the same long custodial patience, the same refusal to be rushed.

Then she turned, and she did not solve anything for anyone, and she did not hand the camera to the crew, and she said to Rourke, in the driest voice Rani had yet heard her use:

"You came all this way to drill into the one thing nobody can yet explain, so you could explain it. Put the bit down. I'm calling the *Ministerio*."

Rani looked at the half-cut socket — at the bright wound in the red stone, at the metal-ghost the foundation had been an inch from taking — and understood, the way she understood a citation gone missing, that the hole they all chased and the hole now being drilled were the same shape, and that one of them belonged to the place, and one of them did not.

She did not write that down. Some sentences you carry uncut.



## Chapter 13 — The Confrontation

Specs heard the generator before they saw it: that flat industrial hum that doesn't belong anywhere near a thing made by hand. It came across the platform on the wind, mixed with the smell of diesel and the cold that lived in the stones even at noon, and Specs stopped halfway through coiling the slack on their leaky pressure rig and just listened, the way you listen to an engine you didn't start.

"That's not ours," they said.

"It is very much not ours," Rani agreed, from where they were crouched over a notebook, copying a T-channel by hand because the photos never got the depth right. Rani didn't look up. Rani never looked up. But the pencil had stopped.

The Fundación Viracocha had set up beside the Plataforma Lítica, where the big red sandstone blocks lay in their broken ranks like a sentence somebody had unsaid. White marquee, white cases, a man in a clean fleece directing two technicians toward one specific block — and Specs knew which one before they got close, because Jakobus had been quiet about that block for two days. The one with the bronze cramp still seated in its channel. A little dovetail of metal, green-skinned, sunk into stone on both sides of a joint, holding two pieces of the world together since before the word *Bolivia* existed.

Dr. Ethan Rourke turned as they came up. He had a clipboard and the easy face of a man who had never once been told to leave.

"You found us," he said warmly. "I was hoping you would. We're documenting before extraction. Strictly reversible — we've got the

cradle, the polymer wrap, the whole protocol. I'd actually love your metallurgist's eyes on this. The arsenical content alone—"

"Extraction," Specs said.

"For analysis. You can't read the casting properly *in situ*. The channel's occluded, the alloy's surface-altered, we need it on a clean bench." He said it the way you'd say *of course you take the patient to the hospital*. Reasonable. Sincere. That was the thing about Rourke that nobody could ever quite get a grip on — he believed every word. He thought he was rescuing the truth from people too timid to dig it out.

Specs's hands had gone still, which for Specs was a kind of alarm bell. They were not a person who went still.

"You've got a permit for that," they said. It wasn't a question, but it was hoping to be.

"We've got a permit to work the zone." Rourke smiled. "There's a—latitude—in how that reads."

There is a particular sound a custodian makes when she crosses a hundred metres of ruin without hurrying. Specs would think about it later. It wasn't fast. It was just *coming*, and you could hear it the way you hear weather change.

Dr. Lucía Quispe arrived without raising her voice, which somehow made everyone else's voices feel too loud already.

"Dr. Rourke." She looked at the cradle, the wrap, the two technicians who had suddenly found other things to study. "You are going to take the cramp."

"We're going to *borrow* a single element for non-destructive—"

"It is not an element." Her tone didn't change. That was the unsettling part. Specs had spent a week watching Lucía be dryly amused by absolutely everything — by their leaking rig, by Naila's hyperfocus, by Rani reading footnotes aloud at altitude until they nearly fainted. The amusement was gone now and what was underneath it was very old and very level.

"It is a joint," she said. "You don't borrow a joint. You break a thing to take a joint."

“With respect, Doctor, the block has already separated along that line. We’d be lifting a free piece of metal from an—”

“It has not separated.” She crouched, two fingers on the channel, not touching the bronze. “The cramp holds. That is the entire point of the cramp. That is *what it is for*.” She looked up at him, and Specs saw her clock the technicians, the cradle, the size of the foundation’s checkbook, the size of her own — and decide anyway. “You pull it, the joint opens, and you will have proved nothing except that a thing built to hold will stop holding if you remove the thing that holds it.”

Rourke had the grace to look pained. He genuinely did. “Lucía. If this site is what I think it is — if the precision here is beyond the toolkit we keep insisting on — then the *world* deserves to know. You of all people—”

“Don’t,” she said, “tell me what I deserve to know about my grandmother’s ground.”

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Here is the official story, and Specs would be the first to tell you it’s a good one.

The Tiwanaku built this. Ancestral Aymara, on their own plateau, somewhere across the four-and-a-half centuries after 500 CE, hauled red sandstone and andesite — andesite from a quarry that may be ninety kilometres off across cold open country — and shaped it with stone hammers and bronze and patience and sand and water until the faces came up flat enough to argue with a razor blade. Then, where two blocks met, they cut a channel across the seam, shaped like a wide-shouldered I or a fat double-T, and they poured molten arsenical bronze into the living stone. It cooled. It gripped. It made two stones into one stone. The whole platform is stitched with these. Some are gone, robbed for the metal centuries back, leaving only the empty sockets like extracted teeth. A few — a very few — are still in.

It is not magic. The metallurgists who’ve published on it are clear, and Specs respected them for it: arsenical bronze, real alloy, real smelting, no signature of anything that didn’t belong to these people on this ground. The fringe wants lasers. The fringe wants a lost super-race. The record keeps handing back exactly what you’d expect from

skilled humans working hard. That's the whole boring, dazzling answer.

Boring, dazzling — and the crew had spent six days failing to do it.

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"Tell him about the rig," Naila said quietly. She'd come up beside Specs without Specs noticing, which meant she was somewhere past angry into the cold flat place she went when the work got serious.

Specs almost laughed. The rig. Their beautiful, leaking, third-attempt rig, sitting fifty metres away in disgrace.

"We tried this," Specs said to Rourke, and they heard their own voice go fast the way it did when the thing was true and they couldn't get it out quick enough. "You want to know if it's hard? We tried to pour one. Full scale. Period method, best we could reconstruct. Up here. In the air you're breathing right now, which by the way has about two-thirds the oxygen your lungs think they ordered."

Rourke's expression flickered — interest, real interest, the thing in him that wasn't an agenda.

"And?"

"And the bronze wanted to cool before it filled the channel. Cold stone sucks the heat right out of the melt — the channel's a heat sink, the whole block is a heat sink, and the melt skins over and then you've got a casting that *looks* like a cramp and fits like a potato." Specs heard Rani exhale a small huff that meant *that is exactly the word*. "We preheated. We poured fast. The fit still leaked. The groove cracked on the second one — thermal shock, the stone didn't like the heat any more than the metal liked the cold. And that's *us*, with a thermocouple and a propane torch and a guy who's read every paper there is. They did it again and again and again across that whole platform and the ones that are left are *tight*. Tighter than ours. Consistent. Every time."

"Which," Rourke said, leaning in, "is exactly my point. If it can't be done with the toolkit—"

"It can be done." Naila's voice cut clean across his. "We just can't do it yet. Those are different sentences. You keep collapsing them."



For a second nobody said anything, and the wind came across the platform and found every empty cramp-socket and made a sound in them, a low even tone, almost a chord.

It is, in the cold honest light, the one hole nobody's closed. Not the precision — pecking and grinding and abrasives will get you machine-flat if you've got the years. Not the dating. Not the transport, exactly, though ninety kilometres of andesite is its own argument. The hole is this: the full bronze-cramp workflow — smelt, carry the melt, pour into carved channel, cool, fit, *repeat with that consistency at this altitude* — has not been published as a clean reconstruction at full scale. Other cultures cramped their stones, Egypt, Greece, sure. But the in-situ casting here, the way the metal went liquid into the rock and bound it, the whole act, start to finish, hasn't been put back together in a paper anyone can hand you.

Which means the foundation could not actually answer the question they were about to break a thing to ask.

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"Here's what I don't understand," Specs said, and they meant it, which was the dangerous part. "You think this proves a lost civilization. We think it proves the *opposite* — that it's hard, that it's human, that human's the marvel. And you and us, we're both standing here in front of the only people who actually know what it's *for*, and only one of us is asking them."

Rourke looked at Lucía. Lucía had not moved from her crouch, two fingers a hair above the bronze.

"What is it for, then," he said, and to his credit there was no sneer in it, just the question, finally, the right one and three weeks late.

Lucía didn't answer him.

She answered the cramp.

"This one," she said, low, and they had to lean to hear, "is on the line. You don't know there's a line because the marks of it are mostly gone now, but my grandmother's grandmother knew, and the offerings still come this way, every year, the families walk it." Her finger traced the air above the channel, along the seam, out past the block toward — Specs followed it — toward nothing the foundation's instruments

would ever register, the open plateau, the snow on the Cordillera, the blue lip of the lake. “The strongest joints are on the lines that matter. Not because the metal is better there. Because the *making* was. You join the stone where you are joining everything else. The cramp holds the blocks. The line holds the cramp. The people hold the line.”

Jakobus had come up at some point — Specs never saw him arrive, you never did — and he was standing a little back, hands in his pockets, looking at the socket and the seam with the expression of a man reading a sentence in a language he respected too much to translate out loud. He caught Specs’s eye and tipped his head a half-inch toward Lucía. *Hers. Not yours.* Specs hadn’t been about to. But it was good to be reminded which way the camera should point.

“You could publish that,” Rourke said, and something in him had gone quieter. “The ceremonial dimension. With the metallurgy. It would be—”

“I could,” Lucía agreed. “I might. When I choose. Not when you uncrate it.” She stood, slowly, the altitude in her knees same as everyone’s. “You will repack the cradle, Dr. Rourke, and you will leave the cramp in the stone, and you will document from outside the wrap, because if you take that joint I will have the *ministerio* and the *comunidad* and every camera these three have brought down on this marquee before your generator cools, and you will spend the rest of your funding cycle explaining to a Bolivian court why you thought the latitude in a permit was a key to my ancestors.”

It was not a threat the way a threat is. It was a survey of facts. She was risking her post saying it — Specs understood that, watching her — because if the foundation had friends in the right offices, and foundations always did, a custodian who said *no* to money could find the ground taken out from under her. She said it anyway. She’d decided before she crossed the platform.

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Rourke stood very still. The technicians watched their boots. The wind went on playing the empty sockets like a row of pipes nobody owned.

Specs did the thing they were proud of, after, even though it cost them — because the truth was they *wanted* that cramp on a bench,

wanted to read the alloy, wanted the answer their rig kept failing to give. Every cell in them wanted it.

"We'll back her," Specs said. "Whatever it costs. And just so we're clear — we're not taking it either. Not to a bench, not to a paper, not for us." They felt Naila's hand close briefly on their shoulder, hard. "It stays in. The question stays open. That's allowed. That's actually the whole—" they stopped, because they'd nearly preached, and you don't, you just don't. "It stays in," they finished.

Rourke looked at the four of them, and at the fixer behind them who hadn't said a word, and at the woman in front of all of them with her two fingers and her hundred generations.

"Pack the cradle," he said to his technicians, not loudly.

He turned back to Lucía. The charm came down over his face again, the professional weather, but underneath it Specs saw the part that had been losing the argument with itself for years and would go on losing it.

"This isn't finished," he said. Not unkind. Just true.

"No," Lucía agreed, and there it was again, the dry thing surfacing through the level cold, almost a smile. "Things that hold rarely are."

The generator coughed and died. In the sudden quiet the plateau got very large, and the cramp sat in its channel exactly where it had sat for a thousand-some years, holding two stones together along a line that ran out past all of them toward the snow, and not one person standing there could have poured it again.

Rani picked the pencil back up. Somewhere below, a market seller called a price, and a child laughed among the stones, and the lake kept its blue, and nobody, for once, took anything that wasn't theirs.



## Chapter 14 — The Second Attempt

The mould was warm this time. That was the lesson, and it had cost them a week to learn it.

Naila crouched over the channel they'd cut into the practice block — a slab of red sandstone Lucía had pointed them to in the rubble field, already split, already worthless, so no one could accuse them of carving into the past. The groove ran the shape of the in-situ cramps: that fat I, two square mouths joined by a waist. She'd traced the real ones with calipers until the custodian's assistant started giving her a look. The dimensions lived in her hands now. The trouble had never been the shape.

The trouble was the metal.

"Charcoal's good," Specs called from the bellows pit. They'd rebuilt the furnace three times, and the third one finally held heat the way the thin air wanted to steal it. At thirty-eight hundred metres the fire fought you. Every flame ran hungrier, leaner, the oxygen spread out and reluctant, and the crucible took longer to come up than the books — written at sea level, by people who'd never gasped walking to their own car — said it should. "She's near. Naila. She's near."

"Don't tell me near."

"I'm telling you near."

The fish, riding in the chest pocket of Naila's coat where it had no business being and no reason to talk, observed that *near* was also what the last cramp had been, right up until it wasn't, and that per-

haps near was the word people used the way other people used *probably fine*.

She ignored it. She'd learned to ignore it the way you ignore your own pulse.

The first attempt — chapter of its own, that disaster — had failed because she'd treated the stone like a foundry would: cold mould, hot pour, let physics do the rest. The bronze had hit the cold sandstone and seized. Skinned over before it filled. They'd chipped out a thing that looked like a melted spoon, porous as bread, and the groove around it had spalled where the thermal shock cracked the rock. A potato. An honest potato. Rani had photographed it from four angles and filed it under *do not repeat*.

The locals had not laughed at them. That was the thing Naila kept turning over. A foundry crew at sea level would have laughed. Here the old man named Eulogio — Lucía's cousin's husband, a smith who still cast bronze for festival ornaments the way his father had — had only looked at the seized lump for a long moment and then mimed warming his hands at a fire. *The stone is cold*, he'd said, through Lucía, who translated with the patience of a woman who did it forty times a day. *You poured into a cold thing. Did your grandmother put bread in a cold oven?*

So now: preheat.

They'd built a fire of dung and brush in the channel itself and let it burn down twice, and Naila had her bare hand flat against the sandstone an inch from the groove, reading the warmth up through her palm the way she read everything, by feel, by the slow honest argument of materials. The stone was hot enough to make her want to pull back. Not hot enough to crack. That window — Eulogio had shown them with his hands how wide it was, and it was not wide.

The official story did not require any of this.

The official story, stated straight and stated with respect, was clean and it was sufficient. The Tiwanaku — ancestors of the Aymara who walked these stones every morning, who left coca leaves in the joints while the crew sweated over their furnace — had cut these channels with stone and bronze tools, no steel anywhere in the ground to argue otherwise, and into them they had poured molten arsenical bronze to

lock block to block. U-shaped cramps. I-shaped cramps. Metallurgists confirmed the alloy. The channels were right there, mouth and waist, in stone you could touch. No lasers. No lost machines. No one from anywhere but here. A supreme feat of hand and patience and a thousand years of accumulated skill, and the marvel of it — the real marvel, the one that didn't need aliens to be astonishing — was the metal. People standing where Naila stood now, in this air, with this fire fighting them, had done it consistently, block after block, fit after fit, for generations.

That was the whole story. It was true. Naila believed it.

She just could not do it.

"Crucible's ready." Specs's voice had gone tight, which meant they believed it too. The graphite crucible came up in the long iron tongs glowing the deep orange of bronze at temperature, and the surface of the melt moved like something alive, like mercury's serious cousin, a skin of dross sliding as Specs tilted it for the skim. "Rani, leaves."

"Coca and a little tallow." Rani read it off the notes without looking up, the way Rani read everything, eyes on the paper because the paper didn't ask anything back. "Eulogio said the flux is — he didn't say flux. He said it *helps the metal forget itself*. I wrote it down exactly. I don't know what it means."

"It means it wets the channel," Naila said. "Lowers the surface tension so it flows into the corners instead of balling up. He's describing flux. He just doesn't call it that."

Across the rubble, on a fallen block worn smooth by five hundred years of weather and sitting, Dr. Lucía Quispe watched them with her coat collar up and her hands around a tin cup, and said nothing.

She'd been there an hour. She hadn't offered to help. That wasn't coldness — Naila had stopped reading it as coldness around the time of the first potato. It was something closer to courtesy. The kind you extend to people who have asked, finally, the right way: by failing in front of you and not pretending otherwise. Lucía would tell them what the cramps *meant* when she chose to, and not before, and that was hers to choose. She had told them once, flatly, *You keep asking me how. The how is in the museum. Ask the right question and I'll think about answering it*. Naila was still working out what the right

question was. She suspected it wasn't a metallurgy question.

But Lucía had pointed them to Eulogio. That was a gift, and Naila knew the weight of it.

"Now," Naila said.

Specs poured.

There is a moment in a casting that you cannot describe to anyone who hasn't stood over one. The metal leaves the crucible in a rope, bright and shivering, and it goes into the dark mouth of the channel and for a second you can't see anything, you can only listen — to the hiss, to the way the stone takes the heat, to whether it accepts the metal or rejects it. Cold rock screams. This stone sighed. The bronze ran down the waist and filled the second mouth and rose to the lip and Naila watched it crawl into the square corners — actually *crawl in*, fill them, the tallow-and-coca flux doing its work, the metal forgetting itself just as the old man had said — and for the length of one held breath it was right.

It was right.

"Look at it," Specs whispered. "Look at it *go*."

"Don't talk to it," Naila said. "Don't talk."

The fish noted that they were all talking, that the only one not talking was the woman on the rock, and that this seemed like data.

They let it sit. You always want to crack the mould early — every fibre of the impatient animal in you wants to see — and every smith who ever lived has learned the hard way to wait. Specs jittered. Specs's whole arc was the jitter, the hands that needed to be doing the next thing, and Naila put a hand flat on their forearm without looking and held it there until the jitter eased to a thrum. They waited while the bronze went from gold to straw to the dull liver-brown of cooling metal. They waited while the warmth bled out of the stone and the high cold of the plateau came back in around them, that thin blue cold that smelled of dust and far-off snow, and a market woman's call drifted up from the town below, and two children chased each other along the top of a fallen H-block as if it were a garden wall and not the most precisely cut object for a thousand miles.

"Closer," Rani said quietly. Rani had the calipers now, hovering, not



yet touching. “The fill’s complete. No porosity I can see on the surface. The first one had — that one was Swiss cheese. This one’s solid.”

It was the closest they’d come. Naila could see it without measuring — the cramp sat *in* the channel instead of shrinking away from it, the two mouths full and square, the metal kissed up tight against the sandstone walls. A week ago they’d had a melted spoon. Now they had a cramp. A real one. Ugly at the edges where the pour-gate had been, a flaw any of them could fix with a file, but a cramp, an actual lock of bronze sitting in stone the way the ancient ones did, and Naila felt the thing she always felt at the edge of getting it, the pull toward *we did it, it’s solved, close the file* —

— and that was the pull Lucía was watching for. Naila understood that, suddenly, looking up and catching the custodian’s eye across the rubble. Lucía wasn’t watching the cramp. She was watching Naila’s *face*. Watching to see whether the outsider would mistake the closer fit for the answer.

Naila made herself put the calipers down.

“It’s closer,” she said, carefully, to the air, to the fish, to the question she still hadn’t found the shape of. “It’s not the same. Look at the fit at the lip — there’s a hair of a gap on the south corner. The real ones, you can’t get a blade in. I can get a blade in this.”

“You can get a *thought* in this,” Specs said, deflating only a little, because Specs’s recoveries were as fast as Specs’s crashes. “But Naila. The first one didn’t even fill. This one filled. We learned the heat, we learned the flux, we got the ore from — where’d Eulogio say the ore—”

“Local,” Rani said. “Arsenical copper from the same belt the originals tested to. We got that right. We did everything right.” A pause, the honest autistic pause that didn’t soften the next part for anyone’s comfort. “It’s still wrong.”

And there it was, the thing the books didn’t close. Naila had read every paper Rani could find. Protzen and Nair on the architecture, careful and humane and full of love for the builders’ skill. Vranich modelling the blocks in three dimensions. Janusek on the culture, the ancestors, the living thread. Not one of them — not one — published a

full-scale workflow for this. Smelt to transport to pour to cool to fit, at this altitude, with this air, repeated to this consistency. The cramps existed. The alloy was known. The channels were measurable. The *making* of them, the actual sequence of a person's hands and a fire's behaviour and a stone's temperature on this exact ground, had never been cleanly reproduced and written down. Other ancient places had cramps — Egypt, Greece. Nobody had nailed *these*. The crew had the best modern knowledge, period materials, a master smith's guidance, the right ore, the right heat, and they had produced a thing you could slide a blade into.

Closer. Not it.

"Two days of preheat we couldn't have sustained for a whole platform," Naila said, mostly to herself. "How many cramps in this site? Hundreds. Each one a fire, each one a pour, each one this window of heat you can't miss. And they didn't miss. They didn't get *closer*. They got it." She looked at the children on the H-block, at the coca leaves brown in the joints from this morning's offering, at a fit so old and so good it had outlasted the empire that made it and three more after. "We're missing something. Not technology. Something in the *doing*."

Lucía spoke for the first time in an hour.

"Now," she said, dry as the dust, "you're getting warmer than your fire."

Naila turned. The custodian was almost smiling, the tin cup steaming, her eyes on the cramp and then on Naila with that measuring patience.

"You think it's a temperature problem," Lucía said. "Or a flux problem. Or an ore problem. You keep it in your hands because your hands are clever and I respect that — Eulogio respects that, he doesn't take just anyone to the fire." A tilt of the head. "But you're casting it like a thing you make once and walk away from. Ask yourself who else was standing at that fire when the real ones were poured. Ask yourself what they were saying while the metal ran. You think it was *don't talk to it*."

She didn't say more. That was hers to give or hold, and she was holding it, the way she'd held it all week, the door open just enough

to let the cold draught through.

Naila looked back down at the cramp. It had gone fully dark now, the warmth all bled away, the south corner with its blade-width gap, the closest they'd ever come.

And as she looked, the stone spoke.

A sound like a knuckle popping, small and absolute, came up out of the channel — the bronze had cooled faster than the sandstone around it, pulled tight as it shrank, and somewhere in that argument between metal that wanted to be smaller and rock that wasn't ready to let it, something had to give. A hairline split ran out from the south corner across the lip of the groove, threading the stone, and the cramp shifted a sixteenth of an inch and sat loose in its bed.

Nobody moved.

Specs let out a long breath that fogged in the cold. Rani wrote something down. The fish, in Naila's pocket, said nothing at all for once, which was its own kind of comment.

Lucía rose from the worn block, brushing dust from her coat, and walked over to look down at the crack with neither triumph nor pity, only a long custodian's familiarity, as if she'd watched this exact thing happen to clever hands before.

"That," she said, "is the question I'll answer. When you stop asking it the way you're asking it." She set her empty cup down on the stone. "Come tomorrow. Early. Before the tour buses. I want to show you something, and it isn't a furnace."

She walked off toward the town, and the children waved at her from the H-block, and she waved back, and Naila stood over the cracked cooling thing they'd made — closer, so close, *not it* — and understood that the gap she could fit a blade into was not the gap that mattered.



## Chapter 15 — Reading the Failure

The cramp had come out of the mould looking like a dog's hind leg.

Rani held it up to the morning light — that thin, knife-edged plateau light that made everything look both very near and very far — and turned it slowly, the way you'd turn a coin you suspected was fake. Arsenical bronze, the right alloy by their best reading of the spectrometry, the same recipe Naila had chased for three days. But the surface was porous as a cracker. One arm had slumped where the metal cooled before it filled. A bubble the size of a thumbnail sat at the elbow of the U, a frozen breath that should not be there.

In the actual blocks, forty metres away behind the rope, the cramps fit their channels like the channel had been carved *around* them. Rani had spent yesterday afternoon lying on her stomach in the wind-cut grass, nose nearly in the stone, looking at one of the in-situ pours — the dull green-grey of ancient bronze still seated in its T-shaped bed, no gap, no slump, the metal having found every corner of the carved groove and held it for a thousand years through earthquakes that had toppled the rest of the platform into the scatter of giants it was now.

That one fit. This one was a dog's leg.

"Read it back to me," Naila said. She was sitting cross-legged by the cold furnace, sleeves shoved up, soot to the elbow, and she had the flat un-smiling attention she got when a thing had refused her. "All of it. From the start."

Rani liked this part. Reading the failure was just sourcing, really —

finding the original instead of the retelling. The retelling was *we couldn't do it*. The original was a chain of specific small wrongs, each one checkable, each one with a footnote in the world.

"One," she said, not looking at anyone, looking at the cramp. "Air. We're at three thousand eight hundred metres. The boiling point of water is eighty-eight degrees here, not a hundred." She knew this the way she knew most numbers — it had simply stayed once it arrived. "Less oxygen per breath of bellows. The fire ran cooler than the charts assumed because the charts assumed sea level. Specs measured the flue at—"

"Hundred and ten under target," Specs said, from where they lay flat on their back in the grass with their hat over their face. "We were pouring cold. Metal that's barely liquid is metal that wants to stop being liquid. It hit the stone and the stone drank the heat and it set before it travelled. That's your slump arm. That's your bubble. Cold pour, every time."

"Two," Rani went on. "The alloy. Arsenical bronze." She let the words sit a second because they were good words, and because the official story lived inside them and the official story deserved to be said straight, with respect, before anyone touched it. The Tiwanaku metallurgists — the ancestors of the Aymara families who walked past these stones now with their hats and their bundles and their entire ownership of the place — had alloyed copper with arsenic, deliberately, knowingly, getting a metal that flowed better and hardened harder than copper alone. That was real. That was in Lechtman, that was in the published spectrometry, that was sitting green in the grooves out there past the rope. Nobody mainstream argued it. The marvel was not a mystery. The marvel was *craft*. "Arsenical bronze flows beautifully. When it's hot enough. Ours might've been a hair off ratio — Naila said the arsenic burns off, you lose it to the air as you melt—"

"You always lose some," Naila said. "They knew you lose some. They compensated. We're guessing at the compensation."

"Three." Rani crouched and put her finger near, not on, the cracked groove they'd carved into their practice block of red sandstone — the soft stone, the easy stone, and even it had split when the metal went in. "Groove prep. The channel wall flaked when the hot bronze hit

it. Thermal shock. The stone we cut was dry and cold and the metal came in like a slap.”

“Theirs didn’t crack,” Specs said, hat still over their face, voice gone quiet under it.

“No,” Rani agreed. “Theirs didn’t.”

The wind moved through the H-blocks with the small dry sound of someone turning pages. Far off a market woman called a price toward the road, two notes, falling. A child ran along the rope line trailing a stick across the stanchions, *tk tk tk tk*, and a custodian’s voice — not Lucía’s, one of the younger ones — said something Aymara and gentle and the child laughed and ran the other way. The place did not care about their cramp. The place had its own day to get through, the same one it had been getting through since before five hundred CE, and it would get through this one too whether or not these strangers with their furnace got the metal to flow.

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“Okay,” Specs said. They sat up. The hat fell off. There was the look — Rani knew the look, the kinetic kindling behind the eyes, the *I-have-a-thing* — and she felt the small dread she always felt when Specs got the look, because the look was usually right and usually a shortcut and the shortcut was usually the problem. “Okay. So we’re losing heat at the pour. We can’t make the air thicker. Can’t change the altitude. But.” They held up a finger. “Modern flux. A modern crucible coating. I can preheat the groove with a torch right before we pour — propane, two minutes, take the stone up to two hundred and the thermal shock just *goes away*, the bronze flows further before it sets, we get a clean fill on the first—”

“No,” Naila said.

She didn’t say it loud. That was the thing about Naila. She said *no* the way she’d read a flow line in a casting — flatly, because it was simply where the metal would or wouldn’t go.

Specs spread their hands. “It’s a *control*, Nai. We prove the geometry’s achievable. We isolate the variables. We come back to period method with the answer in our pocket—”

“Then we haven’t proven anything.” Naila picked up the dog’s-leg

cramp and set it down again, and Rani saw she did it because her hands needed somewhere to put the *no*. “If I torch the groove I’ve shown that *I* can do it with a torch. I already knew I could do it with a torch. The Fundación already did it with a torch.” Her mouth went thin. Two days ago Rourke’s people had run their demo for the cameras up by the road — clean-cut Ethan Rourke in a good jacket, a diamond saw, a polished andesite face cut in eleven minutes, and the headline already written: *MODERN TOOLS REQUIRED. JUDGE FOR YOURSELF*. The wink in their version was a sneer; they’d cut a stone with a machine and called the absence of the machine in the ground a *mystery*. They’d taken the marvel and bent it toward their lost golden empire and they’d done it, technically, by the rules. That was the maddening thing about Rourke. He played fair. He just couldn’t see the thing he wasn’t looking for.

“If I torch the groove,” Naila said, “I’m them. I’ve proven the thing’s impossible *without* the torch. That’s their whole story. Hand it to them gift-wrapped.”

Specs opened their mouth.

“The question isn’t *can a hot groove be filled*,” Naila said. “We know it can. The question is the one nobody’s published — whether you can get a consistent, tight, period-tooled cramp, full scale, *at this altitude*, with what’s in this ground. And every time we cheat the air or the heat or the stone, we stop asking that question. The potato is the data. The potato is the *whole point*.” She looked at the cramp. “I’d rather have an honest dog’s leg than a lie shaped like a cramp.”

Silence. Specs picked their hat up off the grass and turned it in their hands. ADHD didn’t make you wrong; it made you *fast*, and fast hated the long slow honest path that didn’t finish today. Rani knew that about Specs the way she knew the boiling point of water at altitude. She watched them fight it and lose to it on purpose, which was its own kind of patience.

“Fine,” Specs said. “No torch.” A beat. “I still think you’re wrong about the flux.”

“I might be wrong about the flux,” Naila allowed. “Be wrong about the flux on period materials. Find me a flux they had.”

“Now *that*,” Specs said, brightening, the kindling catching legiti-



mately this time, “is a research problem,” and they both looked at Rani, because finding what they *had* was the thing Rani was for.

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Lucía had been standing at the rope for a while.

Rani didn’t know how long. She wasn’t good at the room — she’d never been good at the room, had learned everything she knew about people the long way, from footnotes and from watching their hands — but she’d learned to clock Lucía, because Lucía was worth clocking. Dr. Lucía Quispe stood with her arms folded inside her jacket against the cold and watched their furnace with an expression Rani couldn’t fully source. Dry. A little tired. Something underneath the dry.

“You’re arguing about whether to cheat,” Lucía said. Not a question.

“We decided not to,” Rani said, because it was true and because Rani said true things.

“I heard.” Lucía came under the rope — her rope, her site, her permission to cross it — and crouched by the cracked practice block. She looked at the split in the groove for a while. She had a way of looking at the stones that was nothing like the way the crew looked at them, or the way Rourke looked at them. The crew looked to *solve*. Rourke looked to *prove*. Lucía looked the way you look at the back of a relative’s hand.

“You want to know why your cramp slumped,” she said.

“We know why it slumped,” Rani said. “Cold pour. The air—”

“You know why the *metal* slumped.” Lucía stood, knees cracking, and she almost smiled. “You don’t know why *yours* slumped and theirs didn’t. You think it’s a variable you haven’t found. A flux. A ratio. A temperature.” She nodded out at the in-situ cramp, green in its groove a thousand years on. “Maybe. Maybe there is one and you’ll find it and good. I hope you do; I’d read that paper.” Her voice was proud and tired at once, the way it always was, the voice of someone who’d had this site explained back to her by more confident outsiders than she could count. “But you keep asking *how do I make the metal flow*. You haven’t once asked *who was standing there when it flowed*.”

Rani went still. This was new. This was not in any paper.

“When they poured,” Lucía said, “it was not three people and a furnace having a bad afternoon. The pour was the *end* of something. The quarrying, the dragging, the dressing of the face — months. Years. People who were related to each other, who owed each other, who’d buried each other’s parents under stones like these.” She touched the cold edge of the H-block nearest her, light, the way you’d touch a sleeping animal. “When the bronze went in, the joint it made was not the only joint being made. You understand? The cramp held two stones. The day held the people who held the two stones.” She looked at the dog’s-leg cramp in Naila’s hand without contempt, which was its own gift. “You’re trying to make the object. They were making the *binding*. The object is what’s left of it.”

Specs, who interrupted everyone, did not interrupt.

“Your process and your outcome,” Lucía said. “You keep treating them like the outcome is the point and the process is the cost. Out here the process *was* the point. The cramp was the receipt.” She straightened her jacket. “So. If your cramp comes out a dog’s leg because three strangers were cold and arguing, that’s — what’s the word — that’s *honest*, like she said.” A nod at Naila. “It’s the right shape for what you actually did.”

The wind turned the pages again. Out past the rope a man had come with a small bundle and was setting something down at the base of a fallen block, careful, unhurried, an offering laid on a working Tuesday. The place was not a ruin to him. Rani watched and did not photograph it and did not write it down, because some things you let stay where they were laid.

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He’d been there long enough that Rani had stopped registering him, the way you stop registering the weather.

Jakobus. Off at the edge of the work, where he always was, never holding more of the moment than the rest of them, never the lead — he’d come up the plateau from somewhere, he was always *up from somewhere*, off his own ground, and he’d spent two days doing nothing but lying near the cramps with a notebook, mapping which joints held and which had failed, drawing lines between them that Rani didn’t understand yet. He had the dog’s-leg cramp in his hand now. She hadn’t seen him take it from Naila. He held it the way Lucía held the

stones — turning it slowly, reading the slump and the bubble and the honest cold-poured wrongness of it, not as a failure, just as a made thing, a thing a person had decided to make in a particular shape for a particular reason.

He looked at the cramp. Then he looked out at the long ruined platform, the H-blocks in their broken rows, the green seam of old bronze still holding two stones together against everything the centuries had thrown — earthquake, conquest, treasure-hunters, the diamond saws of certain men who already knew the answer.

He turned the cramp over one more time.

“What’s it for,” Jakobus said.

He said it to no one. He said it the way you’d hand something across a table. And then he set the dog’s-leg cramp down in the grass and stepped back, out of the centre, and looked at Lucía — handing her the question and the moment and the camera all at once — and waited, with the rest of them, to be told.



## Chapter 16 — The Keeper's Story

The Lens had photographed three wars and a famine, and he had learned that the worst light to work in is the light that flatters. It hides the bones of a thing. Up here, on the plateau, there was no flattery in the light at all. It came off the snow on the Cordillera flat and cold and exact, and it found every seam in every stone, and it found the seams in his own hands where the dry air had cracked them open.

He kept the camera at his chest. He had been told, by Jakobus, in that maddening sideways way the man had, that there would be a moment to lower it. He did not believe him. He had been promised that moment in eleven countries.

Lucía Quispe came across the platform without hurrying. She never hurried. The wind pulled at her jacket and she let it. She had a paper cup of something steaming and she carried it the way a person carries a thing they have carried up a hill before, knowing exactly when it would be cool enough to drink.

"You've been photographing the wrong stones," she said.

"There's a right stone?"

"There's a stone everyone photographs." She tipped her chin at the famous block, the one with the H-cuts arrayed like a typesetter's tray spilled by a giant — those crisp, geometric recesses that the foundation's drone had circled for two days. "And there's a stone people leave alone."

She walked. He followed, because following is most of his job, and because Naila and Rani and Specs were already down on their knees

at the far joint where the morning's failure was still cooling in the mold.

He had watched that failure through the viewfinder. It is easier through glass. Naila had poured arsenical bronze — the right alloy, the published alloy, Rani had read the metallurgical paper aloud twice — down into a channel they had cut into two andesite blocks across three freezing days with hammerstones and patience and a great deal of swearing in four languages. The metal had gone in bright. It had stopped flowing about two-thirds along. The thin cold air had pulled the heat out of it before it could fill the form, and where the official cramps in the ground sat flush and tight and clean, a razor-blade refused at the seam, theirs had cooled into a thing with a void in its belly and a lip standing proud where it should have lain flush. Specs had said it looked like a chocolate bar somebody had sat on. Nobody had laughed, which is how you know a joke is accurate.

"Your bronze is shy," Lucía said, watching Naila pry at the mold.

"It froze," Naila said. Not looking up. Naila did not look up at people; she looked at the work, and the work told her the truth about people obliquely, the way still water tells you about wind. "Preheat wasn't enough. The stone's a heat sink. We're at four thousand metres, the metal's losing thermal energy faster than—"

"I know what altitude does," Lucía said, mild. "My grandmother smelted at this altitude. So did hers."

That stopped the channel of Naila's talk the way a stone stops water. The Lens caught it: the small recalibration in Naila's shoulders, the way she set the cooling cramp down with a sudden care, as if it had become someone's. He did not press the shutter. It would have been theft.

Lucía crouched by the in-situ cramp three blocks over — a real one, two thousand years gone into the groove and still holding — and laid two fingers against it without touching the metal, the way you hover a hand over a sleeping animal.

"You think this is a clamp," she said.

"It is a clamp," Specs said, who could not help it. "Functionally. It's a tie. It pins block to block across the seam, takes the shear when the

ground moves — and the ground moves here, you're on a seismic—”

“Yes.” Lucía did not argue with the engineering. She never argued with the engineering, which was how the Lens knew she had something the engineering didn't reach. “It is a tie. The earthquake comes, the cramp holds, the wall does not walk apart. That is true. Write it down, your historian likes things written down.”

Rani, who had been transcribing, went very still over the notebook, caught between being teased and being seen, and decided — the Lens watched her decide — that she had been seen, and that it was all right.

“But you keep asking *how*,” Lucía went on. “How did they pour it. How did the metal flow. How did they get the alloy so even.” She stood, knees cracking, and looked out at the platform, the scatter of immense red sandstone, the andesite hauled — the consensus says — from a quarry the better part of ninety kilometres off, across this plain, by people whose names are gone. “How is a good question. It is not the first question.”

“What's the first question,” the Lens said.

She looked at him then. Properly. At the camera against his chest, and then up past it, at the man behind it, which most people never bothered to do, because the camera was easier to look at than he was.

“You first,” she said. “Why do you hold that like a shield?”

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Here is the official story, and it deserves to be told straight, with the respect it has earned the hard way:

The Tiwanaku built this. Human beings, ancestral to the Aymara who walk this ground today and leave coca and small bright offerings in the lee of these very blocks. Sometime after 500 of the common era, with stone hammers and bronze tools and abrasive grit and templates and a discipline that beggars the imagination, they pecked and ground these faces flat enough to embarrass a machinist. No steel in the ground. No iron. No laser, no power tool, no fallen-from-the-sky tutor — the material record is exactly, boringly, *gloriously* what you would expect from this people in this place at this time,

and that is the marvel, not its absence.

And to bind the great stones against the shaking earth, they did a thing that is rarer and harder than the cutting: they cut channels — I-shaped, some of them, some like a double-headed key — across the joints, and they poured *molten bronze into the stone itself*, arsenical bronze, smelted and carried and tipped glowing into a form scribed in rock, here, at four thousand metres, where the Lens's own breath came short walking from the truck. And it set. And it held. And some of it holds still.

That is the official story. It has no aliens in it and it does not need them. It is, on its face, enough.

It is also, the crew had now proven with their own ruined chocolate-bar cramp cooling in the dirt, *enormously* hard. The hole is real and it is small and it will not close: nobody has published the whole of it — the smelt, the carry, the pour, the cool, the fit, all of it, at full scale, at this altitude, with these means — and made it come out clean. The fit-up the Tiwanaku achieved as routine is a thing modern hands, with modern advantages, have not cleanly repeated and written down. The metal in the ground is exactly what it should be. The skill that placed it so well is the part that won't sit still on the page.

*If it's so easy, the fish would say, surfacing in the cold lake of it, why is your cramp shaped like furniture an elephant slept on?*

---

"I'll lower it when there's something I'm allowed to see," the Lens said.

Lucía nodded, as if he'd passed something. She did not say what.

She took them — not the famous block, not the drone's darling — to a row at the platform's edge where the cramps ran in a line, joint after joint after joint, marching toward the lake's far blue rim. There were small things tucked at the bases. A twist of red wool. Three coca leaves under a pebble, fresh enough that the wind hadn't taken them. A child's plastic ring, the kind that comes from a market for a coin.

"My nephew," Lucía said, of the ring, without embarrassment. "He leaves it sometimes and takes it back. He's negotiating."

Specs opened his mouth. Closed it. The Lens saw the impatience in



him, the *with-whom* on his tongue, and saw him swallow it, which for Specs was a feat of strength roughly equal to lifting the andesite.

"These cramps," Lucía said, "are not all the same quality. Did your makers notice?"

"Yeah." Naila, low. "Some are tighter than others. Better fill, less void. I figured — varying skill, different crews, different days."

"Different days," Lucía agreed. "Now look where the best ones are."

They looked. The Lens, who reads composition the way Rani reads footnotes, saw it before the others said it — saw the way the cleanest cramps, the tightest, the ones whose metal had clearly flowed full and faithful, clustered. Not at random. They ran along the line that the offerings ran along. The line, Lucía did not have to say and so did not say, that the eye walked toward the lake, toward the place the stones had begun, toward water.

Jakobus was there. He had come up the row without anyone marking it, the way he did, off his own ground and at home in nobody's, hands in his coat. He crouched at one of the good cramps and looked at it a long time and did not touch it.

"This is where they tried hardest," he said. To no one. To the stone. "What is it for — the *best* one. Not the wall. The wall is the others too." He stood, slow. "This one's for being seen. By something."

He looked at Lucía. And then he did the thing he always did, the only thing the Lens had ever entirely trusted about him: he stepped back. Out of the frame. He gave it to her with the small backward motion of a man handing over a tool he knows is not his to use.

Lucía took it.

"You pour bronze into stone," she said, "you are doing a thing that cannot be undone. The metal and the rock — before, they are two. After, they are one, and you cannot make them two again without breaking both. My grandmother had a word." She said the word. The Lens did not catch all of it; the air took the edges. "It means *the joining that costs*. You do not make that join lightly. You do not make it for a stone you do not care about. You make it for the seam that *matters* — where the people will stand, where the water is meant to be read, where the dead are spoken to. The cramp is a clamp, yes.

Write that. It is also a promise poured hot, that will not let go, that *survives the shaking* —” she set her hand flat now on the old metal, finally, gently, ” — the way a thing you have decided to keep survives the shaking.”

She was quiet. The wind worked. Far off, a market seller's call came up thin across the plain.

“Your cramp failed,” she said to Naila, “because you poured it where it didn't matter to anyone. You poured it to prove a point to a foundation. The metal knows.” She let that sit, and then her mouth tilted, dry, kind, tired. “I am teasing you. The metal does not know. It is the altitude and your preheat, your engineer is right. But also.” She looked at the void-bellied cramp cooling in the dirt where Naila had laid it down so carefully. “Also.”

And the Lens raised the camera — and stopped. Hand on the body, thumb on nothing. Because to photograph her hand on the two-thousand-year-old metal, the live offering and the dead pour and the negotiating nephew's plastic ring all in one frame, would be to take it. To make it his. To close it into a rectangle and carry it off the plateau and sell the maybe for the price of a striking image, and the maybe was not his to sell.

He lowered the camera.

Jakobus, three blocks down, off the frame entirely, was watching the lake, but the Lens had the strong sense the old man had seen the camera come down and had been, in some unhurried way, waiting for it.

“I'm not going to shoot this,” the Lens said.

“No,” Lucía agreed. “You're not.” She picked up her cooled cup, drank it off, the timing exact, a woman who knew her own hill. “But you can stay for it. There's a difference between solving a thing and being let near it. Your foundation friends” — she did not sneer; she was past sneering, she was somewhere on the far side of it that the Lens recognized from the worst nights of his old life — “they want to solve it. Take it apart, put it in a paper, put their name where the makers' names are gone. They will get a clamp. A tie. The shear-load and the alloy fraction. All true. Write it down.”

“And miss the line,” Rani said. Quiet. Not looking up. The footnote

at last finding its source.

“And miss the line.” Lucía looked at each of them, and the Lens understood that this was the choosing — that she was deciding, right now, in the flat cold light, exactly how much of her grandmother’s word to set down in this strangers’ notebook and how much to keep in the wind where it had always lived. That the giving was hers. That the withholding was also hers, and was not a wound, and was not a puzzle, but a door she stood in with her hand on the frame.

She made her choice.

“Tomorrow,” she said, “at first light, before your foundation gets out of bed, there is a joining. Not a reconstruction. Not for your cameras. My family makes an offering at the good cramp before the season turns. You may come.” She looked at the Lens, and the corner of her mouth went up one last time, dry as the plateau. “You may bring the shield. You may not raise it.”

She walked off down the line of the best cramps, toward the lake, her cup empty in her hand, and did not look back to see if they were following, because that, too, was up to them.



## Chapter 17 — The Function Revealed

The wind came off the Cordillera the way it always did at that hour — thin, white-edged, smelling faintly of snow that hadn't fallen yet. Naila had her hands jammed in her armpits and her eyes on the groove.

It was a small thing, the groove. A shallow channel cut into the top face of a red sandstone block, T-shaped, no longer than her forearm, mirrored by an identical channel in the block beside it so that together they made a clean dovetailed cavity. Once, molten bronze had been poured into that cavity. The metal had cooled into a cramp — a U or an I of arsenical bronze, locking the two stones along the seam.

The bronze was long gone here, looted or corroded or carried off. But the negative remained. The shape of the absence. She'd been staring at the shape of the absence for three days.

"You keep looking at the empty one," Lucía said.

Naila hadn't heard her come up. Lucía did that — arrived in your peripheral vision like she'd always been standing there, hands behind her back, jacket zipped to the chin against a cold she clearly didn't feel the way the rest of them did.

"It's the cleanest channel on the platform," Naila said. "The geometry's perfect. If I could measure the surviving cramps against the channels that lost theirs, I could back-calculate the shrinkage. Bronze contracts as it cools. If they knew that — if they cut the channel to allow for it —"

"They knew it," Lucía said.

“Then I want to know how they knew it. How many failures it took. Whether they had a—”

“Naila.” Not unkind. “Come walk.”

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Jakobus was already at the far end of the row, crouched, not touching anything. He had a way of crouching that made him look like part of the masonry — weathered, jointed, settled into the plateau as if he’d been mortared there. He’d been on the ground since before any of them arrived, and in three days Naila had heard him say maybe forty words. He stood when they reached him, knees cracking, and gestured down the line of H-blocks with two fingers, the way you’d point out birds to someone who didn’t want to spook them.

“Read it,” he said. To Lucía. Not to the crew. Always to Lucía.

She walked them along the seam.

“Look at the order,” she said. “Not the blocks — the joins. Where the cramps survived, and where they didn’t.” She tapped the air above one channel, then the next, then the next, skipping two. “The looters took the easy ones. The ones at the edges, the ones a man could reach with a bar. But here—” she stopped at a join near the centre of the platform, where two andesite faces met so flat that Specs had spent an afternoon failing to slide a razor blade between them, and the colour going up his neck every minute he failed. “Here the cramp is still in the stone. You know why?”

“Hard to reach,” Specs said.

“Hard to want,” Lucía said.

Rani had been hanging back, eyes on the middle distance the way they always were, but their head came up at that. “Offerings,” they said. “There are offerings at this join. Coca. There’s — that’s a recent offering.” Not a question. Rani didn’t ask questions when the source was right in front of them.

“Every week,” Lucía said. “Sometimes my cousin. Sometimes me.” She let that sit. “The looters are not stupid men. They are poor men with families who do not want the thing that happens if you take a cramp from a join that is still —” she paused, choosing the English the way you’d choose a tool from a roll, “— still being used.”

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Here is the official story, and it is a good one, and it does not need rescuing.

The Tiwanaku — ancestors of the Aymara who walk this ground today, who leave the coca, who custody the stones — were master builders. Between roughly 500 and 950 of the common era they raised a ceremonial centre on the cold roof of the world, and at Puma Punku they did something with stone that still stops engineers mid-sentence. They cut andesite, a volcanic rock so hard it laughs at copper, into faces flat enough to embarrass a machine shop. They fit blocks together with interlocking notches and mortise-and-tenon precision. And along the most critical seams they poured liquid bronze into channels carved to receive it, so that the metal froze into clamps that held megaton stones in a grip that has outlasted a thousand years of frost and earthquake.

No lasers. No off-worlders. The tool marks are stone tool marks and bronze tool marks, and there is no steel anywhere in the ground because nobody here was making steel yet, and that is not a scandal, that is just the eighth century.

Arsenical bronze, the metallurgists confirm. Poured molten. Into stone. At twelve and a half thousand feet, where water boils tepid and a fire fights for every degree it can claw out of the thin air.

Hold that. We'll come back to it.

---

"The cramp does two things," Jakobus said.

It was the most consecutive words Naila had heard from him. He held up one finger, then two, but slowly, like a man who'd learned that the second thing is the one people forget.

"One. It locks the stone." He turned to Lucía, deferring, and she picked it up without a beat.

"The plateau moves," she said. "We are on the Altiplano. The earth here shakes — not always large, but always. A wall of fitted stone with no binding will rack and walk apart in a generation. A wall bound with metal cramps will flex and hold. The bronze is softer than the andesite. When the ground moves, the metal gives a little and pulls

the stone back. It is —” she made a small absorbing gesture with one hand. “A shock absorber. They built earthquakes into the design. They expected the ground to betray them, and they answered before it did.”

Specs made a noise that meant he’d just understood something he wished he’d understood three days ago. “It’s a damper. The cramp’s a damper. We were modelling it as a static fastener and it’s a *damper*.”

“Two,” Jakobus said.

Lucía was quiet for a moment. The wind moved the dry grass. Somewhere past the fence a market woman was calling a price, and a child laughed, and the laugh carried the way sound carries when there’s no tree for a kilometre to break it.

“Two,” she said, “is the join you cannot see by measuring it.”

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She walked them to the centre. To the join with the coca.

“When my grandmother taught me this place,” Lucía said, “she did not start with the blocks. She started with the word for joining. In Aymara we have more than one. There is the joining of two things that were always separate — like tying a rope. And there is the joining of two things that were always meant to be one — like a bone that has healed.” She looked at the cramp still locked in its channel, the green-grey gleam of old bronze in the carved stone. “This is the second kind. The metal is not tying the stones together. The metal is showing that the stones were always one stone, and the people were always one people, and the act of pouring — the fire, the mould, the men who carried the charcoal up from the lake — that act *is* the binding. Not the result. The making.”

Rani had a notebook out and wasn’t writing. That was how Naila knew it had landed.

“You pour the bronze,” Lucía went on, “and the whole community sees the fire. The bronze costs. Arsenic costs — and it poisons the men who smelt it, a little, every time. Charcoal costs. The mould-cutters cost. To put a cramp in a stone is to spend the village. So the cramp is also a record. It says: this seam mattered enough that we burned



ourselves a little to hold it. You can read which joins the ancestors thought were sacred by reading which joins they were willing to bleed for.”

“A ledger,” Naila said. The word came out before she could check it. “It’s a ledger of what they decided was worth binding.”

Lucía’s mouth did something that wasn’t quite a smile. “Your friend the fixer asked me, the first day, *what is it for*. As if a thing has one *for*.” She glanced at Jakobus, who had stepped back, folding himself out of the conversation the way he did, handing the floor over and going still. “It is for holding the stone. It is for surviving the earth. It is for spending the village in public so that everyone knows what is sacred. It is a key. It locks the building and it locks the people to the building and it locks this week’s coca to my grandmother’s hands. All of these. At once. You want me to choose one. I will not choose one.”

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The fish, who has been quiet, surfaces here to say the obvious thing: a damper, a ledger, a sacred mortgage, a shock-absorber, a sign — *very nice, very moving, and we still cannot pour the metal*.

Because here is the potato, and it will not go away no matter how beautifully anyone reads the stone.

Specs’s rig was set up forty metres off, behind the rope line, on a tarp weighted with rebar against the wind. The crucible. The little forced-air furnace they’d dragged up in pieces. The mould — Naila’s mould, a channel cut into a sandstone offcut to match the geometry of the real channels as near as her calipers could make it. They had the alloy right. Rani had chased the composition through three papers and a museum email chain until the arsenic content was within tolerance of the published in-situ analyses. They had the geometry right. They had every advantage the eighth century did not: thermocouples, a blower, a gas line, a man whose entire job was watching the pyrometer.

And the bronze came out a potato.

It had come out a potato four times. Up here the furnace couldn’t hold temperature the way it would at sea level; the air was too lean to feed the fire, and the metal poured sluggish, cooling on the lip

of the channel before it reached the bottom, freezing in mid-flow so that the cramp set as a cold, porous, half-formed thing — surface like cauliflower, voids you could see daylight through, the fit loose where the real cramps gripped like a closed fist. Twice the cold mould had shocked the metal and cracked the stone. The good in-situ cramps were dense, sound, tight, consistent — join after join after join, century after century, with no foundry, no pyrometer, no blower, no gas.

Naila had the data in front of her and the data said: *they did this routinely, at scale, at altitude, and nobody alive has shown — published, full-scale, in this air — exactly how.* Not whether it could be done by human hands. It plainly could; here are the hands' work, in the ground. *How.* The whole workflow — the smelt that holds temperature in lean air, the transport of molten or near-molten metal across an open ceremonial platform in a wind that strips heat off everything, the pour that fills a deep channel before it freezes, the cooling that grips without cracking the stone — that chain, end to end, at the size and consistency the builders managed, is not a thing you can pull off a shelf. It is a hole. A real one, with a citation, and it does not lead anywhere except deeper into the craft.

"We could pre-heat the mould," Specs said. He was already pulling at the tarp. "If we cradle the stone, get the channel up near pour temp before we tip the crucible, we lose less to the shock — Naila, if we wrap the offcut and run the torch over it for twenty minutes—"

"That's the move," Naila said. Her hands were already itching toward the crucible. The closure was right there. One more variable. Pre-heat the channel, the metal flows, the fit comes good, and she would have *done* it, and she could write the line that closed the gap, and she felt the want of it in her teeth.

"It is getting dark," Lucía said.

It was getting dark. The snow on the Cordillera had gone pink, then grey. The market woman had stopped calling. The cold was the kind that meant business now, the plateau pulling its breath in for the night.

Naila looked at the cold cauliflower lump of their last attempt, sitting in the cracked stone. She looked at the join with the coca on it, where the real cramp had held for eleven hundred years and somebody's cousin would come on Sunday and leave more leaves. She looked at

Jakobus, who was watching her, not the crucible — watching her, with an expression she couldn't read except that it wasn't disappointment.

She put her hand on the crucible.

It was still warm. They could light the furnace. They could run it half the night and maybe, with the pre-heat, maybe get one sound cramp out of it before the gas ran low. One. Their first. After eleven hundred years and four tries and every instrument the eighth century never had.

"Tomorrow," she said.

Specs stopped pulling at the tarp.

"We pre-heat tomorrow," Naila said. "In the light. With everyone watching." She looked at Lucía. "If that's all right. To do it where it can be seen."

Something passed over Lucía's face — quick, gone. She didn't answer in words. She bent, and from her jacket pocket took a small twist of green, coca leaves folded in paper, and set them on the rope-line stone beside their cracked failed mould, the way you'd leave something at a join that was, perhaps, still being used.

The crucible cooled under Naila's hand. She left it where it was. The bronze stayed in the ground unpoured, and the channel stayed empty, and the snow went dark on the Cordillera, and nobody up there closed anything that night.



# Chapter 18 — The Wink and the Maybe

Specs had the thing balanced on both palms like a dead bird.

It was supposed to be a bronze cramp. It was, technically, bronze, and it was, technically, cramp-shaped, in the way a sock left in a hot car is technically a sock. One arm of the I had come out fat and blistered; the other had pulled away from the sandstone groove as it cooled, leaving a gap you could read a paperback through. Where the metal had touched the stone too fast, the channel had spalled — a little crescent of red sandstone popped clean off, sitting now in the grass like a chipped tooth.

“Behold,” Specs said. “The lost super-civilization.”

Naila didn’t laugh, which was its own kind of laughing, for her. She crouched over the failed pour with her face very close to it, the way she did, reading the cold flow the way other people read print. “It rolled,” she said. “See? It started to skin over here, then the back of the pour caught up and shoved through. That’s the leak.”

“At three-eight-five-oh metres,” Specs said, “everything skins over. I skin over. I have a permanent altitude headache shaped like a regret.”

The wind came off the plateau the way it always did, all day, a flat hand pressing on the grass. Behind them the real blocks held their morning shadows — the H-blocks in their ruined row, faces so square the eye kept trying to find a flaw and skidding off, finding none. Snow on the Cordillera. The blue lip of the lake. Children somewhere near the gate, shrieking at a dog. A vendor’s radio doing battle with the

wind and losing.

The crew's whole setup looked, in front of all that, like a science fair caught in a hurricane.

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Here is the official story, and it is a good one, and Specs would fight you for it.

The people of Tiwanaku — ancestors of the Aymara who walked this ground right now, hauling marigolds and coca to leave at the stones — cut and dressed andesite and red sandstone with hammerstones and bronze tools and patience that the modern mind genuinely cannot hold in one hand. They pecked. They ground. They tested flatness against templates and against each other and against the low raking sun. And in places, to lock block to block against the slow grind of earthquakes, they carved channels into the stone in the shape of an I or a U, and they poured molten arsenical bronze straight into those channels, and let it set, and so stitched the megaliths together with seams of metal.

No lasers. No saucers. The tool marks are tool marks. The alloy in the cramps is arsenical bronze, exactly the bronze these people are known to have made, smelted from ores you can still point at. Protzen and Nair walked this site for years and wrote the book — the actual book — on how the architecture goes together, and nowhere in it does anyone need to come down from the stars. The marvel is human. That's the better story. That's always the better story.

Specs believed it. Specs would put it on a T-shirt.

Specs also could not make the cramp work.

---

"Run me the day again," Rani said. They were sitting cross-legged on a tarp, not looking at anyone, a notebook open against the wind with two rocks holding the pages. Rani's voice was flat and careful, which meant they were chasing something.

Specs counted on fingers. "Built the charcoal fire. Got the crucible up — barely, the air's so thin the fire sulked. Got the alloy melting. Carried it" — they mimed the agony — "to the groove, which we'd preheated, badly, with the second fire. Poured. It rolled. We cried."

“How long from off-the-fire to in-the-groove?”

“Forty seconds? At a run.”

“And it had already skinned,” Naila said, not a question.

“It was forming a crust before it left the crucible,” Specs said. “Up here the metal’s losing heat the second it stops being shouted at.”

Rani turned a page. “So you’d want the mould hotter. You’d want the channel itself near-glowing, so the bronze doesn’t quench against cold stone.”

“With what,” Naila said. “We had two fires. They had — what. Lucía?”

---

Lucía Quispe had been standing back by the gate for a while, arms folded into her jacket, watching the crew torture metal with the expression of a woman watching tourists try to ride a llama. She came over now, unhurried, and looked at the potato in Specs’s hands for exactly as long as it deserved, which was not very long.

“You poured it like you were afraid of it,” she said.

“I respect it,” Specs said. “Respect and fear are adjacent.”

The corner of her mouth moved. She crouched — easily, the altitude was hers and not theirs — and laid a finger near the spalled crescent without touching. “You see this? You cooked your channel from one side. So one wall took the heat and grew, the other stayed cold and held. When the metal came in, the cold side cracked it off.” She looked at the gap where the cramp had pulled away. “And here you ran out of bronze, or out of nerve. Hard to know which, from outside.”

“Both,” Specs said. “Honestly. Both.”

Naila had gone very still, in the good way, the locked-on way. “You’d have to heat the channel evenly. All around. Long enough that the stone holds the heat itself. Then pour, and the stone keeps the metal liquid while it fills.” She looked up at the row of H-blocks like she could see the bronze flowing into them, a thousand-odd years gone. “That’s a fire built right against the stone. That’s hours.”

“That’s a fire built *for* the stone,” Lucía said.

That landed somewhere. Specs watched Naila's face do the thing where the engineering and the something-else met and didn't quite fit and she let them not fit, which was new, which the trip had been doing to her, slowly.

---

So they walked the real ones.

This is the part Specs would have paid for. Lucía took them along the row and let them put their hands where the cramps had been, into the I-shaped sockets cut into the stone — and the sockets were *clean*. Not roughed-out and filled. Cut. The molten bronze had gone into a channel that fit it the way a key fits, the cooled metal that survived showing a fillet so neat against the stone wall that you could not, with a fingernail, find the seam.

"Razor blade," Naila murmured. It was a thing she said now, since the joint faces. You could not slip a razor blade between two of these stones. She'd tried. She'd brought a razor blade specifically to be humbled by, and she had been.

Specs got down and looked into one of the empty cramp sockets, where the metal had long since been robbed out, and what stayed with them was not the size or even the precision. It was the *casualness* of it. The channel turned a small interior corner — a deliberate undercut, so the bronze would grip and not just sit. Whoever cut that corner cut a lot of them. It was not a miracle made once and worshipped. It was a *detail*, repeated, like a stitch a tailor does ten thousand times and stops seeing.

"They did this *constantly*," Specs said. "This is — this is muscle memory. This is a Tuesday."

"This is a Tuesday for them," Rani agreed, very quietly, writing it down.

And here is the hole, and Specs would also fight you about this, in the other direction, because being pro-craft does not mean lying about what's been done.

Nobody has published it. The whole chain — smelt the arsenical bronze to the right alloy, get it molten at this altitude where fires sulk, transport it to a preheated stone channel, pour it before it skins, fill cleanly, cool without cracking the stone, and get that razor-blade



fit — at full scale, the way it's in the ground here, has not been experimentally reproduced and written up in the modern literature. The cramps are real. The alloy's been studied. The fit is right there to touch. But the *workflow*, end to end, repeatable, at 3,850 metres — that's a thing people keep meaning to do and a thing that, as far as anyone at this tarp could cite, nobody's cleanly done.

Specs found that funnier than the potato. "So we're not failing at a solved problem," they said. "We're failing at an *open* problem. That's — okay, that's a much better failure."

"It is the best kind," Lucía said. "You learned where you are."

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The fish — Specs had named the project mascot a fish, a little dead carved fish-thing scratched into one of the blocks, and Specs had decided the fish heckled — the fish would have said: *you came down here with a degree and a propane torch and you got beaten by guys who didn't have the wheel*. And the fish would not have been wrong, and the fish would not have meant it as a put-down, because the guys without the wheel were geniuses, and that was the whole point, and the fish liked the guys.

They lined up the potato beside a fragment that still held a foot of original cast bronze in its socket. Side by side. The crew's blistered, rolled, leaking, spalled, fat-armed thing, and the thousand-year-old seam that had outlived empires.

Everyone looked at it.

"In our defence," Specs began.

"There is no defence," Naila said, and she was smiling, finally, the rare one. "Look at it. *Look* at it. It's beautiful and we can't do it."

"That's not a confession."

"It absolutely is."

"It's an *invitation*," Specs said, and meant it, and the wind took the word and threw it at the mountains.

---

Lucía let them sit with that. Then she said the thing that Specs would carry home, the thing that wasn't on the tarp and wasn't in Rani's footnotes and wasn't ever going to be replicated by anybody with a propane torch.

"You keep asking how," she said. "How hot, how fast, how clean. Good questions. I like them. They are not *the* question." She looked down the row, at the marigolds someone had left that morning at the base of the largest fallen block, the petals already going brown in the wind. "When you join two stones with bronze, you are saying: these stay together now. Through the shaking ground. After I die. After my children. You are pouring a promise into the stone." She tipped her head at the empty socket. "That seam was somebody's vow. You are trying to copy a vow with a campfire."

Nobody said anything. Even Specs, who had a comment for everything, did not have a comment for that.

"I'm not telling you it was magic," she added, dry as the air. "It was bronze. Real bronze, real fire, real hands. But you are measuring the wrong thing and calling the gap a mystery. The gap is not a hole in the engineering." She stood. "It is the part that belongs to us. You can have the how. You cannot have the why. The why is why this is still ours."

And she did not explain further, and she did not need to, and Specs felt the whole shape of the trip turn over once and settle — the place was not a box they'd failed to open. It was a house with people in it, and they'd been let into the kitchen, and that was already more than they'd earned.

---

The Land Cruiser came up the access track too clean for the road. Of course it did.

Rourke got out alone, no crew, no camera, which was the first surprising thing. He came up to the tarp and looked at the potato, and to his credit — Specs would grant him this, grudgingly, the way you grant the weather a nice afternoon — he didn't smirk.

He looked at it for a long time. Then he looked at the real socket beside it. Then he looked, longer, at the H-blocks holding their hard shadows in the failing light.

"We did a demo last month," he said. "Diamond blade. Angle grinder. We made a face nearly that flat in an afternoon." He nodded at the blocks. "And then we said: see, they couldn't have done this without something like our tools. Something lost." He put his hands in his pockets. "It went over very well. Lots of views."

"It would," Lucía said. Not unkindly. Almost.

"I watched you all morning," Rourke said. "You tried to do it the hard way and it beat you, and you didn't reach for the saucers." He almost laughed. "You just got *interested*."

"Occupational hazard," Specs said.

Rourke turned to Lucía, and the clean-cut certainty was gone out of his face, and what was left was a man who had built a whole foundation on the wrong question and had just, possibly, seen it. "I cut the stone," he said. "I never asked anybody what the cutting *was*." He hesitated. "Would you — would you teach me? Not the foundation. Me. I'd like to understand it the way you understand it. If you'll have me."

The wind pressed the grass flat. Far off, the dog won whatever it had been fighting the children over and went tearing across the plaza, joyous, ancient stones rising on either side of it like the walls of the world.

Lucía looked at Rourke for a while. She didn't say yes. She didn't say no.

"Come back in the morning," she said. "Bring nothing." And she let that hang there, in the cold and the blue and the maybe, and did not tell him whether it would be enough.



# Chapter 19 — Leaving the Ground

The crates went into the van one bad joint at a time.

Naila had made every one of those joints herself, and she could feel each failure through the wood as she lifted. The casting molds, cracked along the pour-line. The graphite crucible with its scorched lip. The cramp — their cramp, the best of the eleven they'd attempted, the one Specs had photographed from four angles because it was the only one that hadn't split — wrapped in a wool blanket she'd bought off a market woman in Tiwanaku for more than it was worth, because the woman had laughed at her cold hands and Naila hadn't known how else to thank her.

The cramp weighed maybe nine kilos. The real ones in the ground weighed less. The real ones fit.

Theirs didn't, quite. You could see daylight under one shoulder of it where the bronze had pulled away from the channel as it cooled, a meniscus of failure no wider than a fingernail. At the start of all this she'd have called that nothing. A tolerance. A rounding error you'd sand out.

Now she knew what nothing cost.

"You're going to drop it," Specs said, not unkindly, hauling the other end. Their hands were wrapped in tape where the rope burns hadn't healed. Three weeks of dragging stone on reed mats at four thousand metres did that. "You've got the face you get."

"I don't have a face."

"You absolutely have a face. It's the face that precedes a dropped thing."

She set the cramp down in the van without dropping it, which she felt was a fair rebuttal, and stood for a second with her hands on the cold metal of the doorframe. The light was going long across the plateau. That particular Altiplano gold that didn't warm anything, just gilded it — the wind-cut grass laid flat, the snow on the Cordillera taking the colour and holding it, the red sandstone of the platform going the deep rust of dried blood. Beyond the fence line the H-blocks stood in their broken rows, throwing shadows ten metres long, each one carved so square the shadow had corners.

Rani was crouched by the largest of them with a notebook, not looking at it. Rani never looked at the thing directly. She looked just past it, the way you look past a faint star to see it, and her pencil moved.

"I keep doing the arithmetic," Rani said, when Naila came to stand near her — near, not over; you didn't stand over Rani. "The andesite. Ninety kilometres, the nearest source anybody agrees on. A hundred and thirty tonnes for the biggest sandstone, and that's only ten kilometres, but the andesite's harder than anything we brought a tool to. They moved it ninety kilometres." She turned a page. "Nobody's shown how. Sledges, rollers, reed boats across the lake — Protzen lists the candidates. Plausible, all of them. Demonstrated at full scale, none of them." Her pencil stopped. "I find that very rude of the record."

"The record's always rude."

"It's rude here specifically." Rani's jaw worked. "It gives you everything but the verb. We have the quarry. We have the stone. We have the place it ended up. We don't have *moved*. We don't have *poured*. The nouns are all present and the verbs are missing and everyone's very calm about it."

A heckle, if the plateau had been a thing that heckled: *the verbs are where the work is, sweetheart. Anyone can own a noun.*

Naila let that sit. Three weeks ago she'd have filled the gap. Built a rig, run the numbers, forced the verb into existence because an open question felt like a wound that wanted closing. She'd have hated Rani's arithmetic for not resolving.

Now she crouched and watched the shadow of the H-block creep across the grass and didn't try to close anything at all. It was the hardest thing she'd done all month and it didn't leak.

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The Fundación's vehicles were leaving by the lower road.

You could see them from the rise — two white Land Cruisers and the box truck that had carried the lift gantry, the one they'd assembled with such ceremony on day three to demonstrate that the precision here was *impossible*, machine-flat, beyond hand tools. They'd had a diamond saw. They'd had a CNC reference plate flown in from La Paz. They'd had Rourke, clean-jawed and sincere, standing in front of a camera with a feeler gauge, sliding it along an original face and saying *you cannot do this by hand, this tolerance does not occur in nature or in human patience*, and meaning every word of it.

He had meant it. That was the thing Naila kept snagging on. Rourke wasn't a liar. He'd looked at the flattest stone he'd ever seen and his mind had reached for the biggest cause it could hold, and a lost super-civilization was bigger than four hundred Tiwanaku stoneworkers with sand and a long winter, so that's where his mind had gone. Confirmation isn't a crime. It's just gravity. Everything falls toward the story you already love.

The box truck was empty. That was the part Naila let herself enjoy, a little, in the cold.

They'd come for a cramp. A real one, in situ, to take to a lab and a press conference — the arsenical bronze poured into the channel, the alloy that fit too well, the marvel the official story actually rested on. They'd argued it was for *analysis*. Lucía had stood in the gap with her permit in one hand and her phone in the other and a flatness in her voice Naila had not heard from her before, and the thing about Lucía was she didn't raise it. She lowered it. She got quieter and quieter until Rourke was the only one making noise, and a man making noise alone in front of a custodian who belongs to the ground is a man who has already lost and just hasn't been told.

They were leaving without it. The cramps stayed in the stone, where they'd been holding the stone together through eleven hundred years and at least one earthquake that had thrown the platform around like

dropped crockery — and the cramps had held, the joints had held, that was the whole astonishing point, the metal had done its one job across centuries that swallowed empires.

“He’ll spin it,” Specs said, watching the dust of the convoy. “He’ll get back and say they were *denied* access. That’s a better story than we *measured it and it was made by people.*”

“Probably.”

“Denied access by the *establishment.*” Specs did the voice, the warm radio baritone of it. “*What are they hiding at Puma Punku?*”

“Nothing,” Naila said. “That’s what makes it good. They’re hiding nothing. It’s all right there in the grass and we still can’t do it.”

The fish, if there’d been one, would have surfaced just to say: *that’s the trick of an honest place. It’s got no secrets and you still leave robbed.*

---

She found the seam under the cramp again that night, in her head, the way you find a sore tooth with your tongue.

The arsenical bronze. They’d matched the alloy — Rani had the numbers, the published assays, arsenic in the right percentage to lower the melting point and harden the cast. They’d matched it and they still couldn’t pour it. Up here the air was thin enough that the fire fought them, that the bronze went sluggish and skinned over before it reached the far end of the channel, that it pulled away as it cooled because nobody had told it not to. Down at sea level it would have flowed sweet. Up here, at four thousand metres, in the exact place where the stones were, in the exact thin cold air the builders had breathed and worked in every day of their lives — it fought.

They’d known something about this air that wasn’t in any paper. Naila was sure of that and could not prove it and would not pretend to. The full workflow — smelt, carry the melt, pour into stone the wind was already stealing heat from, cool it so it gripped instead of shrinking away — nobody had run that, full scale, here, and published it. The cramps in the ground were a sentence in a language we could read every word of and still not speak aloud.

That was the hole. Not aliens. Not lasers. Not a pour like concrete



— there was no concrete here, no geopolymer, the fringe could keep it. The hole was smaller and meaner and entirely real: *they did this, with these things, in this air, and we have their tools and their metal and their stone and we cannot make it fit.*

She fell asleep with her hand on the wool-wrapped failure and did not dream of solving it.

---

Lucía walked the site at the last of the light.

Naila watched from the rim, because you watched Lucía the way you watched weather — you didn't join it. She had a cloth bundle in the crook of her arm and she moved through the broken rows the way a person moves through a house they grew up in, not looking at the famous things, the H-blocks the foreigners flew in to photograph. She stopped at the unspectacular places. A cramp channel here. A joint there. The places where the stones met.

At one of them she crouched and laid something down. Coca leaves, Naila thought. An offering, set in the seam where two stones had been bound by bronze before her great-grandmother's great-grandmother was born. She'd shown them this, days ago — *the best cramps are where the lines want them, she'd said, not just where the engineering needs them. The joining is the point. You think they bound stone to stone. They bound the people who poured it to the people who'd carved it to the ones who'd come after and leave the leaves.* She'd said it without ceremony, eating a sweet roll. *A cramp is a promise that holds two things together when the ground shakes. We have made a lot of those. The stone is only the loudest one.*

She had out-read them on the first day and every day since. They'd come to crack the craft and she'd handed them, at the end, the thing the craft was *for*, which was never the band's to take and which she gave anyway, the way you give a guest the good chair: because she chose to, and because they'd shown her their failures honestly, eleven cracked cramps in a row, and that had been a kind of offering too.

She didn't look up. The light went off the snow. The wind took the last warmth out of the grass.

Naila stayed where she was and let the mystery stay where it was.

That was the whole stake, she understood now — not the answer. The *permission*. The right to walk away from a thing you couldn't do, leaving it undone, leaving it open, and call that wonder instead of defeat. She'd spent her life forcing closure on every gap because an open question felt like a fault line. Here was a fault line that had held a platform together for eleven centuries precisely because it was allowed to move.

She let it move.

---

Jakobus was by the van when she came down.

He'd been everywhere and nowhere this whole time, the way he was — out at the quarry one day, walking the joinery the next, never in front of a camera, never claiming a sentence. He'd found the pattern none of them had: that the tightest joints sat on the ceremonial sightlines, that the engineering and the meaning had been the same decision, made once, by people who didn't separate the two the way the foreigners needed them separated. He'd handed that to Lucía without a flourish and let her say what it meant, because the saying was hers.

He had something in his fist now. He held it out.

It was a piece of their own bronze. A fragment that had cracked off the third cramp, the disaster one, in the cold-shock when the pour went wrong — a curl of arsenical bronze the size of a thumb, one face mirror-bright where it had kissed the stone mold, the other rough as slag where the air had got to it before it set. Both at once. The shine and the failure, in one piece of metal, the size of a thing you'd carry in a pocket.

"For the road," Jakobus said.

That was all. He didn't tell her what it meant. He didn't say *so you'll remember*, or *so you'll come back*, or any of the captions a smaller man would have hung on it. He put the cracked bronze in her hand, and it was still faintly warm — no, that was her imagination, it had been cold for weeks — and he stepped back, and the moment was hers, and the metal was hers, and the question was hers, and none of it was closed.

She turned it over. The bright face. The ruined one. The line between

where it had fit and where it hadn't.

Behind her, far down the lower road, the white trucks were gone into the dark with their empty box and their better story. Ahead, the platform held its joints in the cold the way it had held them through every winter since before the word *impossible* existed in any of the languages now arguing over it. And in her hand, a thing she had made, that proved nothing except that the easy story was hard, that the verbs were still missing, that they had every noun and could not say *poured*.

She closed her fingers around it.

"For the road," she said, and got in the van.



## Chapter 20 — After the Marvel

The Lens had learned, over twenty years and four continents, that the best frame comes after everyone thinks the day is done. The handshake is staged. The triumph is staged. But the moment a person turns to gather their tools, their face forgets to perform — and that's the photograph.

So when Specs flopped down on a fallen sandstone block and announced, "Well, we failed," the Lens didn't raise the camera. Not yet.

The light was going long across the platform. Late afternoon on the altiplano does something to stone that no studio can fake: theandesite goes blue in its shadows and gold on its faces, and the red sandstone burns like a banked fire. The wind had dropped, which it almost never did, and in the lull you could hear the small things — a child somewhere past the fence, a dog, the brittle tick of the grass cooling.

Their cramp lay on the ground between them. Their attempt. It looked, the Lens thought, like something a god had spat out in disgust. The bronze had pooled wrong on one side, a frozen wave of it spilling the channel's lip, and where it should have gripped both stones in one clean U it had instead bonded to one and shrunk from the other, leaving a gap you could lose a coin in. The channel itself had cracked in the cooling, a hairline running off toward the block's edge like a question nobody wanted asked.

"It's not even a good potato," Naila said. She crouched over it, not touching. Her hands hovered, reading the flow lines the way she read everything — where the metal had wanted to go versus where it had

been told. “Look at the chill marks. We poured at — what, Specs, eleven hundred?”

“Eleven-twenty by the rig.”

“And it skinned over before it reached the back of the channel. At sea level it flows. Up here the air pulls the heat out of it the second it leaves the crucible.” She sat back on her heels. “We knew that. We *accounted* for that. Preheated the stone, insulated the runner. And it still cheated us.”

“The official story,” Rani said quietly, from where they sat with their back against a standing block and a notebook open on their knees, “says they did this hundreds of times.” They were not looking at anyone. They never were. “Same alloy. Arsenical bronze, consistent across the cramps that have been analysed. Same fit. They poured into channels cut in stone that you cannot move once it’s set, so there is no second try. Every one of them, first try.”

“Or they had a method to fix the bad ones,” Specs offered.

“Then we’d find the bad ones. We find the cramps that held.”

The Lens lifted the camera then — not at the people. At the platform itself. The H-blocks ran away in their broken rows, each one cut with that geometry that made your eye keep checking it, sure there must be a flaw, and finding none. The recesses were clean. The right angles were *right*. You could put a level to them — the team had — and watch the bubble refuse to move. Pecked and ground and polished by hand, the archaeologists said, and they were almost certainly correct, and standing inside it the Lens understood why so many sensible people had decided that *almost certainly* wasn’t good enough for them and gone looking for a machine.

Through the viewfinder, the flatness of a sandstone face caught the low sun and held it like water holds the sky. No tool mark visible at this distance. Up close there were marks — there were always marks, the careful argument went, the pecking scars under the polish, the proof of the hand. The Lens had photographed them that week, macro, every one. They were real. They were there.

And still the eye kept checking.

“Rourke’s already filed his version,” Specs said. He thumbed his

phone, then put it away with the particular violence of someone who'd promised himself he'd stop. "Preliminary findings. 'Anomalous precision inconsistent with documented Andean toolkits.' He's not even wrong about the *anomalous* part. He's just decided what fills the hole."

"That's the trick of him," Naila said. "He's not lying. He looked at the same gap we're looking at and he was *brave* enough to name it. We're the cowards who keep saying *we don't know yet*."

"It's not cowardice," Rani said, to the notebook.

"No." Naila stood, brushing red dust from her knees that would not come off. "It's worse. It's honest."

The Lens let the camera drift down to their failed cramp again. Centered it. The cracked channel, the shrunk bronze, the gap. Click. Because *this* was the photograph that told the truth, and the truth was that they had come up here with three thousand years of metallurgical advancement, propane and pyrometers and a published literature, and they had been beaten by a U-shaped piece of metal the size of a hand.

Not beaten by aliens. Beaten by the *gap* — the real one, the small one, the one in every honest paper. No one had done it. Not at scale, not at altitude, not start to finish with the smelt and the transport and the pour and the cooling all answered together in a way you could put on a page and let a stranger repeat. The cramps in the ground had done it. Whoever poured them had known something in their hands that hadn't survived into a sentence.

That was the whole of it. That was the entire marvel and the entire hole, and they fit each other like — well. Like a cramp ought to fit a channel, and theirs did not.

"You're brooding," Specs told the Lens. "I can hear you brooding."

"I'm working."

"You're brooding *with equipment*."

A fish, had there been one, would have observed that the band had hauled crucibles up to four thousand metres to lose an argument with a Bronze Age tradesman, and that the tradesman wasn't even present to gloat. The Lens almost smiled.

Footsteps on the gravel path. The Lens knew the gait before turning — unhurried, sure-footed on ground that had thrown every one of the crew at least once that week. Dr. Quispe came across the platform with her clipboard under her arm and her collar up against the cold that was coming, and she stopped a few feet off and looked at the dead cramp with an expression that the Lens, framing her on instinct, did not photograph. It was too private. It was almost tender.

“You poured today,” she said. It wasn’t a question.

“We poured today,” Naila agreed. “It didn’t take.”

Lucía crouched the way Naila had crouched, and looked the way Naila looked — at the flow, at the chill marks, at the cheated gap. The Lens watched her read it. She read it faster. She read it the way you read your own handwriting.

“You insulated the runner,” she said.

“Yes.”

“And preheated the stone.”

“Yes. To about—”

“It’s not the temperature.” Lucía said it without any unkindness, which was somehow harder to take than if she’d been smug. “You all keep solving the metal. The metal is the easy part. You think the channel is a mould.” She set one finger near the cracked groove, not touching. “You cut it like a mould. Square. Tight. So when the bronze shrinks it has nowhere to pull from and it tears the stone, or it lifts off the wall and you get your gap. The old channels aren’t square at the bottom.” She traced an undercut shape in the air. “They open underneath. The metal locks instead of grips. And the cooling is slow because the stone is *warm all the way down*, not skin-hot from a torch an hour before. You build the heat over days. You don’t bring it. You don’t conquer the cold. You make a place where the cold can’t reach.”

Silence on the platform. The dog had stopped barking.

“That’s not in Protzen,” Rani said, almost to themselves. Almost reverent. “That’s not in any of them.”

“No,” Lucía agreed. “It’s not in the books.” She stood, and brushed



the dust from her own knees, and it came off her cleanly, the Lens noticed, as if the ground knew her. “Some of it I was told. Some of it I worked out standing here for twenty years while people flew in to tell me my ancestors had help. You can have that piece. I’m giving it to you.” A dry beat. “I notice you didn’t ask. You were too busy mourning your potato.”

Specs laughed — the surprised, helpless laugh of someone caught out and glad of it. “Will it work? The undercut?”

“I don’t know,” Lucía said. “I’ve never poured one. I’m a custodian, not a smith.” And there it was, the thing the Lens had come a long way to understand: she didn’t need to have done it to *out-read* it. She lived where the answers lived. The crew had borrowed the question. She’d been raised inside it. “You might still fail. Probably you will. It took them longer than your visa allows.”

The Lens raised the camera then. Lucía against the H-blocks, the gold light dying off the andesite behind her, her chin level and her eyes on the crew with that patient irony that had not bent once all week — not for Rourke’s funding, not for the foundation’s cameras, not for the crew’s good intentions. *Click*. That was the photograph of the chapter. The keeper, on her ground, handing over exactly as much as she chose and not one grain more.

“What’s it *for*, do you think,” said a voice from the path, and the Lens didn’t startle, because Jakobus never quite arrived; he simply turned out to have been there. He was looking at the dead cramp too, hands in his pockets, head tilted. He wasn’t asking the crew. He was asking the air, and the question was for whoever wanted it. “All of it. The undercut, the slow heat, the alloy you have to get right with no second try. A peg would hold the stones. A peg’s easy.” He nudged a pebble with his boot. “They chose the hard bond.”

He didn’t answer himself. He never did. He looked at Lucía, and then he stepped back, off to the side, out of the frame, the way he always faded, and left the floor to the only person on the platform who belonged to it.

Lucía didn’t take the question the way the foundation took things — quickly, to own it. She let it sit. The wind came back, the first cold breath of evening sliding down off the Cordillera, and the snow on the high peaks went pink.

“To hold the stones,” she said. “Yes. And to be *seen* holding them. You can’t pour a cramp by yourself. Someone smelts, someone carries the metal up here while it’s hot — runs it, the way you’d run anything you can’t let cool — someone has the channel ready, someone watches the pour, someone reads it after.” She looked at the rows of blocks. “You bind the stones and you’ve bound everyone who touched the binding. The earthquake comes — they come here, all the time — and the stones move and the bronze flexes and holds. Eight hundred years of that. The cramp is the proof the joining worked. The stone is just what it’s made of.”

She let that be the end of it. She did not say *and so it is a metaphor*; she did not say *and so you understand*. The Lens loved her, briefly and professionally, for declining to caption her own meaning.

The light was almost gone. Lucía checked her watch and the sky in the same glance.

“Pack it up,” she said. “I close the gates at six.”

They packed it up. Specs broke down the rig with that fast clever sadness of a man who’d built something that almost worked. Naila wrapped the dead cramp in a cloth — keeping it, the Lens saw, not to fix but to fail by again, properly this time, with an undercut channel and a fire built over days they didn’t have. Rani filled three pages and didn’t look at anyone and was, the Lens thought, completely happy.

The crew filed out through the gate, and the Lens went last, and turned at the threshold for the frame that ends the day.

Lucía stood inside, where she always stood. She drew the gate to. The iron met the post with a sound the platform threw back off the H-blocks, flat and final. She turned the key. She was not locking the crew out of a puzzle they’d nearly solved. She was closing her own house at the end of her own day, the way she’d closed it the day before and would the day after, long after the visas ran out and the foundation went home to film something else.

Behind her the stones held their joints in the dark. The official story stood — hand tools, templates, patience, genius, no machine, no visitors from elsewhere — and it stood unbroken, and it stood unproven in the one small place that mattered, the place where molten metal met cut stone and did, somehow, exactly what it was told.

The Lens lowered the camera without taking the shot.

Some doors you don't photograph closing. You just let them close.



## Chapter 21 — The Permission of Maybe

The notebooks were spread across two pushed-together tables in the back room of the hostel in Tiwanaku town, and Rani had arranged them in the order that made sense, which was not the order anyone else would have chosen. Field notes by date on the left. Photocopied articles by author on the right. The translated transcripts of the elders' interviews in the middle, weighted down at one corner by a chunk of slag from Specs' first failed pour — ugly, bubbled, the colour of an old bruise.

Rani did not look at the others when they came in. Rani looked at the slag, and then at the photograph clipped above the Protzen citation, and the two things would not reconcile, and that was the whole problem, laid out in front of everyone on a table that smelled of instant coffee.

"Okay," Specs said, dropping into a chair, "I've decided the bronze hates me personally. Just so we're documenting that."

"The bronze," Rani said, "is an arsenical alloy. Copper with somewhere between two and five percent arsenic, depending on which cramp you measure. That's confirmed. Lechtman analysed the in situ ones. It's in the literature." They touched the relevant page without quite tapping it. "It does not hate you. It cooled too fast. That's altitude and the cold mould and a hundred other things."

"Hates me," Specs said, comfortably.

Naila came in last, still scrubbing at her forearm where a fleck of the last pour had caught her. She sat without speaking and pulled

the photograph of the in situ cramp toward her, the real one, the U-shaped bridge of bronze sunk flush into its carved channel between two blocks of red sandstone, and she looked at it the way she looked at everything, like it was a sentence she'd been told was simple and she'd already found the lie in it.

Rani began, because someone had to, and reviewing was the part Rani could hold without strain. Reviewing was just putting true things next to each other and refusing to let any of them go soft.

"The official story," Rani said. "I want to say it straight before we say anything else, because it deserves that, and because we keep almost not saying it." They straightened a page that did not need straightening. "Tiwanaku. The builders were Tiwanaku, ancestral to the Aymara. Main megalithic phase after five hundred of the common era. They cut red sandstone and andesite — the andesite very hard, the sandstone softer but enormous, the biggest blocks over a hundred tons. They shaped the faces flat enough that the visitors' refrain about razor blades is, annoyingly, basically true. They cut interlocking notches, T-grooves, mortise and tenon. And they bound the blocks with cast bronze. Molten metal, poured into channels carved into the stone, locking block to block."

"With hand tools," Naila said. Not arguing. Filing it.

"With pecking stones and abrasives and bronze tools and templates and time," Rani said. "No steel found. No iron. No power tools, no lasers, nothing in the ground that doesn't belong to the place and the period. Protzen and Nair walked the whole thing block by block and the precision sits inside what skilled hands can do. The marvel is human. That's not me being polite to the consensus. That's where the evidence lands."

"Read the next bit," Specs said. "Go on. I like the next bit."

Rani did not need to read it. They knew where it was. "The claims," they said. "Twelve thousand years old — no. Stratigraphy says no, radiocarbon says no, the material culture says no, and it isn't close. Poured like concrete — no geopolymer at Puma Punku, no evidence of it, that's a thing people want to be true that the stone doesn't support. Machine-cut, alien-cut, Atlantean—" They stopped. "All of it sits in the column marked claimed. Labelled. We label it. We don't sneer at it and we don't pretend it earned the same shelf as the dated

layers, because it didn't."

"Rourke's whole show on one cramp," Specs muttered.

"Rourke's whole show," Rani agreed, "on the precision. On the flatness. On 'how could primitive people.' And the answer to 'how could primitive people' is that they weren't primitive, they were Tiwanaku, and the question is the rude part, not the stone."

The Lens was in the doorway. Rani hadn't heard them come. They had the camera down, hanging off one shoulder by the strap, lens cap on — which Rani had learned to read, over these weeks, as a kind of weather. Cap on meant the Lens was in the room and not behind the glass for once. They came and stood at the corner of the table and looked at the spread, all of it, the true things laid next to each other, and said nothing, which from them was a full sentence.

"So that's the story," Naila said. "And it's a good one. It's better than the alien one. It's harder and it's truer and it's got actual people in it." She turned the photograph of the in situ cramp ninety degrees, as if a new angle would help. "So why is there a lump of garbage on Rani's transcripts."

Specs laughed, short. "Because we tried it."

This was the part Rani found difficult to keep flat, so they read it from the notes, where the words held still. "Attempt logs. We cast — Specs, how many."

"Eleven," Specs said. "Eleven cramps. Period-plausible furnace, charcoal, the bellows rig that took four of us to keep going at this altitude because the air's so thin the fire just sulks. Arsenical bronze, alloyed to spec, give or take, off the published analyses. Channels cut into sandstone offcuts — and yeah, fine, we cheated, we had steel chisels for the channels and they still took forever, which tells you something about doing it with bronze and a rock."

"Eleven," Rani said. "Two we abandoned at the furnace because the melt wouldn't run hot enough to be safe to pour. Of the nine poured—" They looked at the slag, at the column of small honest entries in their own handwriting. "Of the nine, the bronze cooled before it reached the bottom of the channel four times. Voids. You could see daylight under the bridge. Three cracked the sandstone — the stone takes the heat unevenly, it spalls, the thermal shock split the groove."

“The pretty one,” Specs said.

“Two looked correct from above,” Rani said. “Both leaked at the contact when we tested the bond. Neither fit flush the way the in situ ones fit flush. You can slide things under our cramps. You cannot slide things under theirs.”

Naila said, very quietly, the thing she’d said the first night and had not stopped meaning since: “I have made a lot of things. I made a worse version of this than people made fifteen hundred years ago at four thousand metres with a fire that didn’t want to burn.” She wasn’t wounded by it. She sounded the way she sounded when a problem refused to close, which Rani knew now was the sound of her happiest. “And I knew the answer going in. I knew the alloy. I knew the shape. I had the photos. I had Lechtman. I had everything Rourke says they couldn’t have had, and I had a furnace I understood, and I made a potato.”

“We made a potato,” Specs said.

“We made nine potatoes and two abandoned potatoes,” Rani said, because it mattered to be exact.

The Lens made a small sound that might have been a laugh.

Rani moved to the page they’d been building toward, the one that had taken three drafts because it was so important not to overreach on it. The thing they were proud of was that it said exactly one true thing and stopped.

“Here’s where we land it,” Rani said. “Not in the claimed column. Not in the consensus column. In the middle. The hole.” They read it the way they’d written it, slowly, watching the words for any place they leaned too far. “Bronze cramps are known at other ancient sites — Egypt, Greece, dowels and cramps and clamps, the joining of stone with metal is old and it’s real and it’s not unique to here. But the full workflow at Puma Punku — smelting the arsenical bronze, getting it to the block, pouring it into a carved stone channel at this altitude, in this cold, and having it come out flush and consistent and tight, cramp after cramp — that exact end-to-end process has not been experimentally reconstructed and published at full scale, in situ, at altitude. Nobody’s done it cleanly and shown the working. The in situ cramps are real. The flat faces are real. And the recipe for making



them the way they were made is, as of the papers we can find, not closed.”

Silence. The good kind.

“That’s it,” Naila said. “That’s the whole thing. That’s the only hole and it’s a real one and it’s small and it’s enormous.”

“It doesn’t say aliens,” Rani said, a little sharply, to no one, to the room, to themselves. “I want that in the record. The hole does not say aliens. The hole does not say lost technology. Every material in that ground is exactly what you’d expect from those people in that place at that time. The arsenic is local. The copper is local. The stone is local. The shapes are theirs. The hole isn’t ‘someone else did it.’ The hole is ‘we, with our advantages, have not yet shown that we can do it the way they did it.’ Those are completely different holes.”

“Write that twice,” Specs said, and meant it.

“It’s written,” Rani said. “It’s on the back page.”

The Lens spoke then, which they did rarely, and when they did it was usually a question shaped like an observation. “Lucía would say we’re still doing it wrong.”

And that landed, because it was true, and because it was the part the table couldn’t hold no matter how Rani arranged it. They had laid out every source. Protzen, Nair, Vranich’s 3D printing paper, Janusek on the precursors, Lechtman on the alloy. They had laid out their own eleven failures. And none of the paper said what Lucía had said, three days ago, at the joining, when she had stood between the blocks with the offering in her hands and let them watch — let them, the gift of it, theirs to give and hers to withhold and she had chosen to give — and had said that the cramp was not only how the stones held. That a cramp is a binding. That you do not pour bronze between two stones the way you weld a bracket; you join them, the way you join people, the way the ancestors are joined to the now, and that the fit was tight because the intention was, and that the engineering and the ceremony were not two things her people did at the same time but one thing with two faces, and that this was the part the foreigners always left out of their workflow.

“She out-read us on the first day,” Rani said. “She read Rourke off the first day too. She read all of us. We’ve got eleven cramps and a

citation list and she has the thing the citation list is about.”

“She let us see it,” Naila said. “She didn’t have to.”

“She didn’t have to,” Rani agreed.

They did not write that part down in the same way. Rani had been careful about that. The ceremony was in the notes as the fact that it had happened, with permission, and the meaning Lucía had given was hers, recorded as hers, attributed, hedged with the line Rani had insisted on: this is what the custodian chose to share, and the rest is hers. The how of the bronze they would put in the book. The why of the binding they would point at, and leave standing where it stood, in her hands, on her ground.

Specs picked up the slag, turned it over, set it down. “So here’s my problem with the official story,” they said, grinning, the old impatience and the new humility wrestling on their face. “It’s right. I think it’s right. I’d bet the whole rig it’s right. There’s no spaceman. There’s no laser. It’s hands and fire and templates and a thousand years of knowing how. And—” they tapped the bruise-coloured lump — “if it’s so easy, if it’s just hands and fire, why can’t we do it?”

Nobody answered. That was the agreement. Rani felt the question sit in the room and not get answered and felt, instead of the itch they usually felt when a thing stayed open, something closer to relief. The relief of being allowed not to close it. They had spent a life being asked for the answer, the one answer, the right answer, look at me and give me the answer — and here was a table full of true things and a question on top of it that they were permitted, encouraged, to leave breathing.

“We don’t answer it,” Rani said. “That’s the point. We did the work so the question would be clean. So somebody reading it can’t say we were lazy or we were credulous. The question’s clean now. We hand it over clean. If it’s so easy, why can’t we do it. And then we don’t say.”

“Maybe,” Naila said.

“Maybe,” Rani said.

The Lens raised the camera then — cap off, a small click of the strap — and Rani thought they meant to shoot the table, the evidence, the

slag. But they turned, framed the doorway, the grey morning beyond it, and let it hang there a moment without pressing anything, just looking through the glass at the light, and then lowered it again without a shot. Whatever they'd seen, they let it stay where it was.

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The bus to El Alto left from the corner by the market, and they had cut it fine. Rani carried the notebooks in a dry-bag against the rain that wasn't falling yet, the eleven failures logged inside, the one clean question on the last page.

Lucía was at the stop, which surprised them, holding a paper twist of something that turned out to be roasted broad beans, which she handed to Naila without ceremony. "For the road," she said. "You'll want them. The road is long and you'll be thinking too hard."

"We left the cramps," Naila said. "The ones that didn't crack. For the school, if they want them. As examples of how not to."

"They'll like that," Lucía said, dry. "Children love a thing an adult made badly." She looked at the dry-bag, at all of it bundled up under Rani's arm. "You'll write it the way you said."

"The how in the book," Rani said. "The why stays with you."

Lucía nodded once, the nod of someone who has heard that promise before and watched it get broken and is choosing, this time, to wait and see. "The bronze," she said, as the bus wheezed up, brakes complaining. "You'll keep wondering about the bronze."

"For years, probably," Specs said.

"Good." She stepped back. "Wonder is better company than certainty. Certainty is what your friend Rourke had, and look how thin it left him." A flicker of something that was almost kindness, almost not. "Go on. You'll miss it."

They climbed up into the smell of diesel and wet wool. Rani took the window. Out on the plain the platform of Puma Punku sat low and broken under the grey, the great red blocks scattered, the H-blocks in their rows, the channels where the bronze had run and held, holding still, holding now, fifteen hundred years of holding. The bus pulled out. The stones slid past and did not give anything up, and the question went with the crew, unanswered, alive, the slag in the bag

and the maybe on the last page, and behind them, getting smaller, a woman who had let them see exactly as much as she chose, standing on her own ground, watching them not solve it.

Naila offered the beans around. Outside, the snow line on the Cordillera came clear for one moment between the clouds, and nobody said what it looked like.

## Chapter 22 — Fact, Debate, and Story: A Register

The fish lived in a jar of meltwater on the windowsill of the field office, and the fish had opinions about footnotes.

Not a real fish. A pressed-tin sardine Specs had pried off the lid of an emergency rations tin somewhere around year one of all this, bent into a permanent skeptical curve, and named — over Rani's objections — *the editor*. It rode in a side pocket of the camera bag and got propped wherever the crew did the thing they were doing now: sitting in a borrowed room with the heater ticking, the Cordillera going pink through the glass, trying to write down what was true.

"This is the part nobody films," Specs said, tipping back on a plastic chair. "The bibliography montage."

"It's the part that matters." Rani had three notebooks open and a laptop balanced on a fourth. They were not looking at anyone, which was how Rani looked at you when they meant it most. "If we put a claim in the world, we put the source under it. Otherwise we're just the Fundación with better manners."

The fish, on the windowsill, said nothing, which from the fish was a kind of agreement.

So. The register. Play it straight, the way Lucía had taught them to play everything straight — with respect first, and the wink kept in your pocket until the respect was paid.

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**THE OFFICIAL STORY, AND WHO SAYS SO.**

Puma Punku is part of the Tiwanaku complex, near the southern shore of Lake Titicaca, on the Bolivian altiplano. The platform you walk is ruined, scattered, half-buried, and — this is the thing the postcards never get — *small*, in footprint, for how large it sits in the imagination. The blocks are not small. Red sandstone up to roughly a hundred and thirty metric tons. Andesite, the hard volcanic stuff, cut into the famous H-blocks with their flat faces and crisp interior angles, arrayed in rows like a sentence in a script no living person reads.

The builders were the Tiwanaku culture, ancestral to the Aymara who walk the site this morning with offerings and folding tables and children who treat a thirteen-hundred-year-old monolith as a perfectly good place to leave a backpack. The main megalithic phase goes after 500 CE — call it 500 to 950 — with some carbon dates from organic material in the fill running earlier. Not twelve thousand years. The stratigraphy says so; the radiocarbon says so; the material culture in the ground says so, every potsherd of it agreeing with every other potsherd about what century it is.

For the construction, the consensus is hand tools, templates, abrasives, patience, and genius. Pecking and grinding and polishing, repeated past the threshold of what feels reasonable, which is exactly the threshold the marvel lives on. For the joinery: interlocking notches, T-shaped grooves, mortise-and-tenon. And the cramps — the bronze cramps, U- and I-shaped, poured molten into channels carved to receive them, locking block to block. Metallurgical study identifies the alloy as arsenical bronze.

Rani read the citations aloud in their flat happy cadence: Protzen and Nair, *The Stones of Tiahuanaco*, 2013 — the book that walks the toolmarks one face at a time. Vranich, *Heritage Science*, 2018, on reconstructing the architecture, on what 3D printing the blocks teaches you about how they were meant to fit. Janusek, *Journal of Archaeological Research*, 2008, on the culture and its precursors.

“Put their names big,” Naila said from the floor, where she was sitting against the wall with her hands wrapped around a tin cup. “The people who actually did the looking. We’re tourists in their decades.”

“Vranich 3D-printed the blocks,” Specs said, with the particular reverence Specs reserved for someone who’d built a thing. “Printed them

to model the construction sequence. That's — that's a guy who got tired of arguing and made the argument out of plastic."

The fish approved. You could tell.

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### THE POTATO, FOR THE RECORD.

You should know, before the wink, exactly how badly the crew failed, because the failure is not a confession — it's data.

Naila wrote it in her own hand, because she would not let anyone soften it: *We tried to cast an arsenical-bronze cramp at altitude, full intent, period-plausible method, into a carved sandstone channel. The bronze chilled against the cold stone before it filled. The first pour left a void you could lay a finger in. The second cracked the channel wall with thermal shock. The fit, where we got fit, leaked. We were three experienced makers with a modern crucible, modern thermocouples, modern fuel, and the answers to every question the Tiwanaku metallurgist had to discover for himself. We made a potato.*

She underlined *potato* once. The crew had a rule about not underlining twice.

"It flowed wrong," she said, to the room, to the fish, to the snow going purple now on the range. "I keep thinking flow. The metal didn't want to run. Thin air, cold stone, a pouring temperature you're holding by feel because the wind keeps stealing your fire. And theirs ran. Theirs ran and filled and held, hundreds of times, consistent, across a whole platform, and survived the earthquakes that knocked the blocks loose around them."

"Preheat the channel," Specs offered, for the four hundredth time, gently. It had become a kind of prayer between them.

"We did. We don't know how *they* did. That's the whole —" Naila stopped. Caught herself reaching for closure, the way she always reached. Set it down. "That's the hole."

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### THE HOLE, CITED, MEASURED COLD.

Here is the one genuinely unresolved, checkable thing, and the crew agreed it would go in the back matter in its own box with its own

border, so no one could mistake it for the fringe and no one could pretend it away.

The cramps are *real*. Found in situ. Alloyed, as said, arsenical bronze. Channels carved to take them, visible in the stone. None of that is in dispute, and none of that needs aliens. Cast-metal cramps to bind masonry are known elsewhere in the ancient world — Egypt, Greece — and nobody thinks Karnak was built by anyone but Egyptians.

But: the full workflow — smelting at this altitude, transporting the melt or the means to make it, pouring into carved stone channels, achieving consistent alloy and consistent tight fit, at *full scale*, in *these conditions* — has not, in the published modern literature, been experimentally reconstructed and duplicated end to end. The thing the crew tried and failed to do has, as far as Rani could find across three days and four notebooks, not been cleanly done and written up by anyone.

“Not ‘lost technology,’” Rani said, to be very clear, looking at the table. “Lost *workflow*. Or never-fully-recovered workflow. There’s a difference and we are going to make the difference enormous and visible because the Fundación will collapse it the instant we look away.”

Naila nodded. “And the big andesite. Quarry maybe ninety kilometres off. Sledges, rollers, reed mats, water — all plausible, none of them shown in the ground as *the* method. Plausible isn’t proven. We say plausible. We don’t say solved.”

“We don’t say solved,” Rani repeated, and wrote it.

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### THE FRINGE, LABELLED, NOT SNEERED.

This part Naila wanted handled with a flat hand, no contempt, because contempt is just another way of not looking.

That the faces are “machine-flat” and therefore prove lost technology: not supported. Pecking, grinding, polishing get you there, with time. The crew had felt the time in their own shoulders.

That the site is twelve thousand years old: not supported by stratigraphy, radiocarbon, or material culture. Every line of evidence sits in the same millennium, and they sit there agreeing.



That the blocks were poured like concrete — geopolymer: no evidence of it at Puma Punku.

That only aliens or Atlanteans could have built it: this one Rani wrote out, then sat looking at the words.

“It’s a sad theory,” Rani said finally. “When you actually read it. It’s people loving the place so much they can’t believe humans made it. They mean it as a compliment. They just put the compliment at the wrong address.”

The fish, for once, did not heckle. Even the editor knew when a line had landed.

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The door opened on the cold and Lucía came in stamping her boots, a thermos under one arm, dust of the platform on her trousers. She’d been out walking the rows at the hour she liked, after the tour groups and before the dark. She took in the spread of notebooks, the laptop, the tin fish on the sill, and her mouth did the dry thing it did.

“You’re writing the part where you admit you couldn’t do it.”

“In some detail,” Naila said.

“Good.” Lucía set the thermos down, didn’t sit. “Read me the hole.”

Rani read it — the box, the border, the smelting and pouring and fit, *not reconstructed end to end in the published literature*. Read it carefully, the way you’d read someone their own words back.

Lucía listened with her head tilted. When it was done she was quiet long enough that Specs started to fidget, and then she said:

“It’s correct. As far as it goes.” A beat. “But you’ve written it like the answer is in a journal nobody’s published yet. Like one good field season closes it.”

“Isn’t that —” Specs began.

“The cramp isn’t only the pour.” She picked up Naila’s tin cup, the cold one, turned it in her hands like she was reading the metal’s intention. “You keep measuring the flow of the bronze. My grandmother’s people measured the flow of something else. You join two stones with a cramp, yes — and you join the people who carried them, and

the season you poured in, and the ones poured for. Where do you think the best cramps are? Not the strongest blocks. The blocks in the lines that mattered. The ones you'd stand a person beside, on a day that counted."

Naila looked at her notes. The places Jakobus had quietly marked, weeks back, before any of them understood why — the cramps that fit best clustered not where the engineering was hardest but where the offerings still got left.

"So when you write your box," Lucía went on, mild as the heater's tick, "and you say nobody's reconstructed the workflow — say which workflow. You haven't reconstructed the *pour*. We never lost the *binding*. People bring it up here every week. You filmed them doing it. The thing your literature hasn't closed and the thing my grandmother could close are not the same thing, and if you let them collapse into one, you've done the Fundación's trick from the other side." She handed Naila back the cup. "Write *that* difference enormous and visible."

The room held still. Outside, the wind found a gap somewhere and made the glass hum.

"Yeah," Naila said, very quietly. "Yeah. That's the correction."

Rani was already typing it, lips moving, getting it exact.

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### **A GAZETTEER, BECAUSE PLACES ARE PEOPLE.**

The crew kept one more list, and Lucía read it before it went in, with a pencil, and changed three things.

The platform: ruined, monumental, scattered with sandstone and andesite. The H-blocks in their rows. The T-channels where the cramps once gleamed and, in places, gleam still. Faces so flat the old test — slide a razor blade into the joint — fails on the good ones. Wind-cut grass. Thin air that punishes a fast walk. Snow on the Cordillera, the lake's blue lip to the north.

The town of Tiwanaku, and the rim villages, and the reed islands out on the lake where the sandstone began its journey. The road between, which Jakobus had walked end to end and described to no one, only nodded over.

The people: the Aymara families who belong here in the present tense — not “descendants of the builders” filed under history, but *here, this morning, with bread and a baby and a reason*. The market women. The custodians. The children using a god-block for a goalpost, which Lucía said was the most accurate possible use of it, because it meant the place was still alive enough to be ordinary.

Dr. Lucía Quispe, custodian, who out-read three specialists in their own specialty and didn’t make a thing of it.

And, off in the corner of the list, in Naila’s hand: *Jakobus — present. Asked what it was for. Did not stay to be thanked.*

He’d already gone, of course. He always went before the writing. The road, never the lead.

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### **THE WINK, KEPT IN THE POCKET UNTIL NOW.**

So here is the draught the crew left the reader standing in, and they argued about the wording until the heater ran low.

If the official story is right — and the crew believe it, on the evidence, with their whole chests — then why couldn’t three good makers with every modern advantage cleanly repeat the pour? If the fringe is right, why is every gram of metal in the ground exactly the arsenical bronze you’d expect from this place and this century, and not one atom of anything you wouldn’t?

What’s missing isn’t a spaceship. It’s a step. A knack. A piece of the *how* that lived in hands and never got written down, that the Tiwanaku metallurgist knew the way Lucía’s grandmother knew the binding — by doing it until it was no longer a question.

Specs reached over and turned the tin fish to face the window, toward the dark and the snow and the lake.

“Box it up?” Rani asked.

“Box it up,” Naila said. “And don’t answer it. We don’t get to answer it. That’s the whole point of building the box.”

Rani’s cursor blinked at the bottom of the file. The last line of the register, the one the back matter would close on, the one the fish had been waiting all evening to hear and would never admit it.

Rani typed it, and read it once, soft, to the room:

*Maybe is the gift.*



*Klaus*

Custos, non Conditor